

Light, Water, and Melatonin: The Synergistic Regulation of Phase Separation in Dementia

光、水与褪黑素：痴呆症中相分离的协同调节

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Abstract 摘要

The swift rise in acceptance of molecular principles defining phase separation by a broad array of scientific disciplines is shadowed by increasing discoveries linking phase separation to pathological aggregations associated with numerous neurodegenerative disorders, including Alzheimer's disease, that contribute to dementia. Phase separation is powered by multivalent macromolecular interactions. Importantly, the release of water molecules from protein hydration shells into bulk creates entropic gains that promote phase separation and the subsequent generation of insoluble cytotoxic aggregates that drive healthy brain cells into diseased states. Higher viscosity in interfacial waters and limited hydration in interiors of biomolecular condensates facilitate phase separation. Light, water, and melatonin constitute an ancient synergy that ensures adequate protein hydration to prevent aberrant phase separation. The 670 nm visible red wavelength found in sunlight and employed in photobiomodulation reduces interfacial and mitochondrial matrix viscosity to enhance ATP production via increasing ATP synthase motor efficiency. Melatonin is a potent antioxidant that lowers viscosity to increase ATP by scavenging excess reactive oxygen species and free radicals. Reduced viscosity by light and melatonin elevates the availability of free water molecules that allow melatonin to adopt favorable conformations that enhance intrinsic features, including binding interactions with adenosine that reinforces the adenosine moiety effect of ATP responsible for preventing water removal that causes hydrophobic collapse and aggregation in phase separation. Precise recalibration of interspecies melatonin dosages that account for differences in metabolic rates and bioavailability will ensure the efficacious reinstatement of the once-powerful ancient synergy between light, water, and melatonin in a

modern world.

分子原理定义相分离的现象已被广泛学科迅速接受，但与此同时，越来越多的发现将相分离与包括阿尔茨海默病在内的多种导致痴呆的神经退行性疾病相关的病理性聚集联系起来。相分离由多价大分子相互作用驱动。重要的是，水分子从蛋白质水合壳层释放到体相中产生的熵增促进了相分离，并随后产生不溶性细胞毒性聚集体，从而驱动健康的脑细胞进入疾病状态。界面水的高粘度和生物分子凝聚体内部有限的水合作用促进了相分离。光、水和褪黑素构成了一种古老的协同作用，确保充足的蛋白质水合作用以防止异常相分离。阳光中存在并应用于光生物调节 (photobiomodulation) 的 670 nm 可见红光波长可降低界面和线粒体基质的粘度，通过提高 ATP 合成酶的马达效率来增强 ATP 的产生。褪黑素是一种强效抗氧化剂，通过清除过量的活性氧和自由基来降低粘度并增加 ATP。光和褪黑素降低的粘度提高了游离水分子的可用性，使褪黑素能够采取有利的构象以增强其内在特性，包括与腺苷的结合相互作用，从而强化 ATP 的腺苷部分效应，该效应负责防止导致相分离中疏水塌陷和聚集的脱水现象。精确重新校准考虑到代谢率和生物利用度差异的跨物种褪黑素剂量，将确保在现代世界中有效恢复光、水和褪黑素之间曾经强大的古老协同作用。

Keywords: melatonin; dementia; amyloid- β ; ATP; adenosine; phase separation; infrared light; hydrogen bonds; viscosity; bioavailability

关键词：褪黑素；痴呆；淀粉样蛋白- β ；ATP；腺苷；相分离；红外光；氢键；黏度；生物利用度

1. Introduction 1. 引言

Dementia is a neurodegenerative condition marked by varying levels of cognitive impairment [1], currently affecting approximately 46.8 million people around the world. It is estimated that 10 million people will develop dementia each year, and without approved pharmaceutical intervention to effectively target underlying causes [2-4], by the year 2050, healthcare spending attributable to dementia is projected to become a significant drain on resources representing 11-17% of total global healthcare spending [5]. Alzheimer's disease (AD) is one of the most common causes of dementia [6,7] followed by vascular dementia (VaD) [8,9]. Together with Lewy body dementias [10] and frontotemporal dementia (FTD) [11], these major neurodegenerative disorders account for approximately 90% of all dementia cases [12]. Dysregulated aggregation of biomolecular condensates formed as a result of multivalent macromolecule interactions may underlie the common molecular mechanisms responsible for the development of all AD and non-AD dementia (nADD) [13-15].

痴呆症是一种以不同程度认知障碍为特征的神经退行性疾病 [1]，目前全球约有 4680 万患者。据估计，每年将有 1000 万人患上痴呆症，由于缺乏能够有效针对潜在病因的获批药物干预措施 [2-4]，到 2050 年，痴呆症相关的医疗支出预计将成为资源的重大负担，占全球医疗总支出的 11-17% [5]。阿尔茨海默病 (AD) 是痴呆症最常见的病因之一 [6,7]，其次是血管性痴呆 (VaD) [8,9]。这些主要的神经退行性疾病，连同路易体痴呆 [10] 和额颞叶痴呆 (FTD) [11]，约占所有痴呆病例的 90% [12]。由多价大分子相互作用形成的生物分子凝聚物发生失调性聚集，可能是导致所有 AD 和非 AD 痴呆 (nADD) 发展的共同分子机制 [13-15]。

An Alzheimer's biomarker study performed within a defined population over a period of 15.7 years (maximum) found the absolute remaining lifetime risk for incident dementia to be significantly associated with elevated amyloid accumulation (hazard ratio 2.11; 95% CI 1.43 – 2.79). Even though 87% of the 4984 participants were diagnosed as cognitively unimpaired at enrollment, higher amyloid accumulation was a significant biomarker correlated with accelerated dementia progression [16]. Vascular risks are generally associated with the progression of VaD [8,9]. However, midlife hypertension and late-life amyloid- β ($A\beta$) deposition were found to be independently associated with increased dementia risk in 298 participants aged 45 – 64 in a study that spanned 30 years. The study was unable to identify evidence of synergy between vascular risk and $A\beta$ deposition on a multiplicative scale in subjects with dementia, implying that unique molecular pathways may be involved in the development of dementia [17].

在一项针对特定人群、持续时间最长达 15.7 年的阿尔茨海默病生物标志物研究中，研究发现发生痴呆的绝对剩余终生风险与淀粉样蛋白积累升高显著相关 (风险比 2.11; 95%，置信区间 1.43 – 2.79)。尽管 4984 名参与者中有 87% 在入组时被诊断为认知功能正常，但较高的淀粉样蛋白积累仍是与痴呆进展加速显著相关的生物标志物 [16]。血管风险通常与血管性痴呆 (VaD) 的进展有关 [8,9]。然而，在一项跨越 30 年、涉及 298 名年龄为 45 – 64 的参与者的研究中，发现中年高血压和晚年淀粉样蛋白- β ($A\beta$) 沉积与痴呆风险增加独立相关。该研究未能发现痴呆受试者的血管风险与 $A\beta$ 沉积在乘法尺度上存在协同作用的证据，这暗示痴呆的发生可能涉及独特的分子途径 [17]。

2. Aberrant Phase Separation Is the Fundamental Molecular Driver behind Dementia

2. 异常相分离是痴呆症背后的根本分子驱动因素

In 2017, Banani et al. defined intracellular biomolecular condensates as cytosolic and nuclear micron-scale compartments not bound by membranes and formed via phase separation driven by multivalent macromolecular interactions [18]. These membraneless organelles (MLOs) are responsible for strategic cellular organization in response to changing environments including endogenous and exogenous stress [14,19-21]. MLOs are ubiquitously utilized not only by all eukaryotes, but also bacteria [22], and viruses which are now recognized as master architects of biomolecular condensates, using phase separation to form viral replication “factories” [23,24]. The disruption of phase separation in key cellular processes results

in diseases including neurodegenerative disorders and cancer [25-29].

2017 年, Banani 等人将细胞内生物分子凝聚物定义为细胞质和细胞核中微米级的室室, 它们不受膜的束缚, 是通过多价大分子相互作用驱动的分相形成的 [18]。这些无膜细胞器 (MLOs) 负责在应对包括内源性和外源性压力在内的环境变化时进行战略性的细胞组织 [14,19-21]。MLOs 不仅被所有真核生物普遍利用, 也被细菌 [22] 和病毒所利用。病毒现在被认为是生物分子凝聚物的大师级构建者, 利用相分离形成病毒复制“工厂” [23,24]。关键细胞过程中相分离的破坏会导致包括神经退行性疾病和癌症在内的疾病 [25-29]。

The forces that drive phase separation encompass simple density transitions in single-component fluid systems [30] to changes in macromolecule saturation levels in binary mixtures achieved via manipulating in vitro macromolecule expression levels, interaction energies, and inclusion of hydrotropes/surfactants [31-34]. Spontaneous or driven phase-separated biomolecular condensates are usually nonstoichiometric assemblies of multiple proteins and nucleic acids [35]. These multivalent macromolecules engage in site-specific interactions that conform to the “stickers-and-spacers” architecture [35-37], forming reversible crosslinks that may involve hydrogen bonds [38-40], ionic strength [41], cation- π , and $\pi-\pi$ interactions [42] that fine-tune percolation thresholds that may further define phase separation processes [43-46].

驱动相分离的力量涵盖了从单组分流体系统中的简单密度转变 [30], 到通过操纵体外大分子表达水平、相互作用能以及添加助溶剂/表面活性剂实现的二元混合物中大分子饱和水平的变化 [31-34]。自发或驱动形成的相分离生物分子凝聚物通常是多种蛋白质和核酸的非化学计量组装体 [35]。这些多价大分子参与符合“贴纸与间隙 (stickers-and-spacers)”架构的位点特异性相互作用 [35-37], 形成可能涉及氢键 [38-40]、离子强度 [41]、阳离子- π 和 $\pi-\pi$ 相互作用 [42] 的可逆交联, 从而微调可能进一步定义相分离过程的渗透阈值 [43-46]。

The significant discovery by Kar et al. that fused in sarcoma (FUS) and other phase-separating RNA-binding proteins in the FET family, namely EWSR1 and TAF15, form reversible clusters of varying sizes in subsaturated solutions where phase separation was not observed [47], highlights the relevance of percolation without phase separation in phase transitions in vivo. The aggregation of FUS, EWSR1, and TAF15 are associated with neurological disorders and the three FET family RNA-binding proteins are widely expressed in most cell types [48,49]. Thus, the detection of FUS percolation clusters formed in subsaturated solutions and clusters that are coupled to phase separation in supersaturated solutions [47] offers additional insight in the aggregation of macromolecules in vivo where saturation concentration that can initiate phase separation has been questioned [50]. In this review, in order to accurately capture the concept that phase separation can be coupled to percolation as well as other phase transitions in vivo including the conversion to fibrillar solids [51,52], the term phase separation is employed in lieu of the more popular nomenclature of liquid-liquid phase separation (LLPS) which restrictively implies only viscous liquids are present in the coexisting phases [53].

Kar 等人的重大发现指出, 肉瘤融合蛋白 (FUS) 以及 FET 家族中的其他相分离 RNA 结合蛋白 (即 EWSR1 和 TAF15), 在未观察到相分离的欠饱和溶液中形成了大小不一的可逆团簇 [47], 这突显了无相分离的渗透 (percolation) 在体内相变中的相关性。FUS、EWSR1 和 TAF15 的聚集与神经系统疾病有关, 且这三种 FET 家族 RNA 结合蛋白在大多数细胞类型中广泛表达 [48,49]。因此, 对欠饱和溶液中形成的 FUS 渗透团簇以及过饱和溶液中与相分离耦合的团簇的检测 [47], 为体内大分子的聚集提供了额外的见解, 而此前关于引发相分离的饱和浓度在体内是否存在一直受到质疑 [50]。在本综述中, 为了准确阐述相分离可以与渗透以及体内其他相变 (包括向纤维状固体的转化 [51,52]) 相耦合的概念, 使用了“相分离”一词, 而非更流行的“液-液相分离 (LLPS)”这一术语, 因为后者限制性地暗示共存相中仅存在粘性液体 [53]。

2.1. Phase Separation of α -Synuclein into Amyloid Fibrils in Dementia

2.1. α -共核蛋白在痴呆症中相分离形成淀粉样纤维的过程

In 1992, Hardy and Higgins proposed that the deposition of amyloid fibrils in AD is the direct cause of cell loss, vascular damage, and dementia [54]. Continued research indicated that the AD disease process may be the result of the dyshomeostasis between the production

1992 年, Hardy 和 Higgins 提出, 阿尔茨海默病 (AD) 中淀粉样蛋白原纤维的沉积是导致细胞丢失、血管损伤和痴呆的直接原因 [54]。后续研究表明, AD 的病理过程可能是由于产生量之间的稳态失衡所致。

and clearance of amyloid β -peptides ($A\beta$) [55,56]. The nomenclature committee of the International Society of Amyloidosis (ISA) defines in vivo amyloid fibrils as extracellular protein fibril deposits associated with 36 human amyloid proteins. Intracellular aggregates such as tau and α -synuclein (α -syn), which are present in all synucleinopathies and are the major component of Lewy bodies associated with Lewy body dementia and Parkinson's disease (PD) [57,58], are excluded from this list [59,60]. However, the hallmark feature of amyloid fibrils is the self-association of soluble amyloid monomeric fibers into insoluble cross- β sheets [61-63], and both α -syn [64-66] and tau [67-69] have been reported to self-assemble into cross- β sheet structures.

以及淀粉样蛋白 β -肽 ($A\beta$) 的清除 [55,56]。国际淀粉样变性学会 (ISA) 命名委员会将体内淀粉样纤维定义为与 36 种人类淀粉样蛋白相关的细胞外蛋白质纤维沉积。存在于所有突触核蛋白病中、且作为与路易体痴呆和帕金森病 (PD) 相关的路易体主要成分的细胞内聚集体, 如 tau 和 α -突触核蛋白 (α -syn) [57,58], 被排除在此名单之外 [59,60]。然而, 淀粉样纤维的标志性特征是可溶性淀粉样单体纤维自组装成不溶性的交叉 β 片层 [61-63], 并且据报道 α -syn [64-66] 和 tau [67-69] 均能自组装成交叉 β 片层结构。

Encoded by the SNCA gene on chromosome 4 [70], α -synuclein (α Syn) comprises 140 amino acids [71] with intrinsically disordered regions prone to fibrillization [72]. The aberrant self-assembly of physiological, soluble α Syn monomers into neurotoxic protein aggregates implicated in PD and other synucleinopathies [73,74] is now attributed to phase separation where macromolecular interactions trigger the irreversible liquid-to-solid transition into amyloid hydrogels containing oligomeric intermediates and cross- β -sheet fibrils [75-79]. The documentation of the conformational evolution of α Syn phase transitions has been successfully captured by solution and solid-state magic-angle spinning (MAS) nuclear magnetic resonance (NMR) spectroscopies [51], and the study and analysis of the material components as well as intermolecular interactions of protein molecules within α Syn condensates during phase separations were performed employing fluorescence recovery after photobleaching (FRAP) and Förster resonance energy transfer (FRET) techniques [80]. Since phase separation is an early event in α Syn aggregation, modulating phase separation and/or interfering with liquid-to-solid phase transitions during α Syn amyloid phase transitions become attractive molecular targets [81].

由 4 号染色体上的 SNCA 基因编码 [70], α -突触核蛋白 (α Syn) 包含 140 个氨基酸 [71], 具有易于纤维化的内在无序区域 [72]。生理性可溶性 α Syn 单体异常自组装成与帕金森病 (PD) 及其他突触核蛋白病相关的神经毒性蛋白聚集体 [73,74], 目前归因于相分离, 其中大分子相互作用触发了不可逆的液固转变, 形成含有寡聚中间体和交叉- β -折叠纤维的淀粉样蛋白水凝胶 [75-79]。 α Syn 相变的构象演变过程已通过溶液和固体魔角旋转 (MAS) 核磁共振 (NMR) 波谱学成功记录 [51], 而对相分离过程中 α Syn 凝聚物内物质成分及蛋白质分子间相互作用的研究与分析, 则是利用光漂白后荧光恢复 (FRAP) 和福斯特共振能量转移 (FRET) 技术完成的 [80]。由于相分离是 α Syn 聚集的早期事件, 因此调节相分离和/或干扰 α Syn 淀粉样变过程中的液固相变已成为极具吸引力的分子靶点 [81]。

Phase transitions from soluble monomeric to insoluble β -sheet fibrils observed in medin employing C-direct detection NMR in combination with structural bioinformatics further supports the concept of phase separation as the common molecular pathway underlying not only AD, but also VaD [82]. Among the 36 amyloid proteins recognized by the ISA in 2018 [59], medin (AMed) is the most common amyloid found in the human body [83,84]-being an internal component of milk fat globule-EGF factor 8 (MFG-E8), also known as lactadherin-that is now associated with vascular $A\beta$ in cerebral amyloid angiopathy (CAA) pathology [85].

在 medin 中观察到的从可溶性单体到不溶性 β -折叠纤维的相变, 通过结合 C-直接检测 NMR 与结构生物信息学, 进一步支持了相分离是不仅作为 AD, 也是作为 VaD [82] 基础的共同分子通路的这一概念。在 2018 年被 ISA 认可的 36 种淀粉样蛋白中 [59], medin (AMed) 是人体内发现的最常见的淀粉样蛋白 [83,84]——它是乳脂肪球-EGF 因子 8 (MFG-E8, 也称为 lactadherin) 的内部成分——目前它与脑淀粉样血管病 (CAA) 病理学中的血管 $A\beta$ 相关 [85]。

Present in blood vessels of most adults over the age of 50, medin is cleaved from lactadherin to form insoluble amyloid aggregates [86,87] that co-localize with vascular $A\beta$ deposits [88] to cause cerebrovascular dysfunction in aging mice and human subjects with VaD [89,90]. Cerebral arteriole medin is regarded as a novel biomarker for AD and VaD [91]. Even though the glycoprotein lactadherin has multiple, important physiological functions [92] including phagocytosis [93], angiogenesis [94], and mucosal repair [95], medin aggregates alter cellular homeostasis, causing microvascular endothelial dysfunction by inducing permeability via the formation of pores in lipid membranes that result in upregulated ionic current flow [96,97], a mechanism not dissimilar to how $A\beta$ peptides form calcium ion channels in lipid bilayer membranes [98,99]. However, the conditions that trigger the cleavage of medin from lactadherin causing medin to self-assemble into pathogenic, insoluble fibrils remain unclear [82,83].

Medin 存在于大多数 50 岁以上成年人的血管中, 它从乳粘蛋白 (lactadherin) 中裂解出来, 形成不溶性淀粉样蛋白聚集体 [86,87], 这些聚集体与血管 $A\beta$ 沉积物共定位 [88], 从而导致衰老小鼠和患有 VaD 的人类受试者出现脑血管功能障碍 [89,90]。脑小动脉 medin 被认为是 AD 和 VaD 的新型生物标志物 [91]。尽管糖蛋白乳粘蛋白具有多种重要的生理功能 [92], 包括吞噬作用 [93]、血管生成 [94] 和黏膜修复 [95], 但 medin 聚集体会改变细胞稳态, 通过在脂质膜中形成孔隙诱导通透性, 导致离子电流流量上调, 从而引起微血管内皮功能障碍 [96,97], 这一机制与 $A\beta$ 肽在脂质双分子层膜中形成钙离子通道的方式并无二致 [98,99]。然而, 触发 medin 从乳粘蛋白中裂解并导致 medin 自组装成致病性不溶性原纤维的条件尚不清楚 [82,83]。

The self-recognizing aggregation of amyloid proteins is not limited to homotypic enrichment in one protein, but often also involves heterotypic interactions in condensates containing up to hundreds of proteins [100,101], and the outcome of amyloid aggregation is modified by these associated heterotypic interactions [102]. Aberrant phase separation resulting in delayed disassembly of stress granules (SGs) causes the formation of non-dynamic SGs that entrap and immobilize TAR DNA-binding protein 43 (TDP-43), rendering the protein insoluble in FTD pathogenesis [15]. Increasing understanding that phase-separating RNA-binding proteins such as FUS [47,103] and those that associate with tau [104] play important modulatory roles in the heterotypic interactions that can promote

淀粉样蛋白的自我识别聚集并不局限于单一蛋白的同源富集, 通常还涉及包含多达数百种蛋白的凝聚体中的异源相互作用 [100,101], 且淀粉样蛋白聚集的结果会受到这些相关的异源相互作用的影响 [102]。导致应激颗粒 (SGs) 解体延迟的异常相分离会引起非动态 SGs 的形成, 这些 SGs 会诱捕并固定 TAR DNA 结合蛋白 43 (TDP-43), 使该蛋白在额颞叶痴呆 (FTD) 的发病机制中变得不可溶 [15]。人们日益认识到, 像 FUS [47,103] 这样发生相分离的 RNA 结合蛋白以及那些与 tau 蛋白结合的蛋白 [104], 在能够促进聚集的异源相互作用中发挥着重要的调节作用。

or suppress amyloid aggregation [102,105-107], warrants exploration of specific conditions that may trigger aberrant phase

separation in RNA-binding proteins.

或抑制淀粉样蛋白聚集 [102,105-107], 这使得探索可能触发 RNA 结合蛋白异常相分离的具体条件变得至关重要。

2.2. The Underappreciated Role of Hydrogen Bonds and Protein Hydration in Phase Separation in Dementia

2.2. 氢键和蛋白质水合作用在痴呆症相分离中被低估的作用

Dementia-related neurodegenerative disorders are often associated with gene mutations that may cause the dysregulation of RNA-binding proteins responsible for the aggregation of pathological amyloid fibrils during phase separation [108]. The mechanisms reported mostly involve dysregulation in the low-complexity domains of proteins such as TDP-43 and FUS [103,105,109-111]. Low-complexity domains (LCDs) are generally regarded as universally disordered; however, LCDs can also adopt stable, structured conformations [112]. Therefore, aberrant phase separation observed in LCDs may involve other factors in addition to the dysregulation of intrinsically disordered regions that are essential in the promotion of phase separation. Intrinsically disordered proteins (IDPs) usually serve as necessary scaffolds that facilitate phase separation of biomolecular condensates [18,113-115] which can be tuned by controlling enthalpy, minimizing entropic costs in phase separation [116-118].

与痴呆相关的神经退行性疾病通常与基因突变有关, 这些突变可能导致 RNA 结合蛋白的失调, 而这些蛋白负责相分离过程中病理性淀粉样原纤维的聚集 [108]。报道的机制主要涉及 TDP-43 和 FUS 等蛋白质低复杂性结构域的失调 [103,105,109-111]。低复杂性结构域 (LCDs) 通常被认为是普遍无序的; 然而, LCDs 也可以采用稳定的结构化构象 [112]。因此, 在 LCDs 中观察到的异常相分离除了涉及对促进相分离至关重要的内在无序区域的失调外, 可能还涉及其他因素。内在无序蛋白 (IDPs) 通常作为必要的支架, 促进生物分子凝聚体的相分离 [18,113-115], 这可以通过控制焓来调节, 从而最大限度地降低相分离中的熵代价 [116-118]。

Intrinsically, phase separation is entropically unfavorable and driven predominantly by enthalpically favored protein interactions [119-121]. In addition to energetically favorable multivalent protein-protein interactions that offset entropic costs, variations in ions and salt concentration, pH, and temperature can result in thermodynamic changes in entropy-enthalpy compensation that regulate phase separation [122-124]. Phase separation in proteins such as Ddx4 [125] and hnRNPA1 [126] exhibiting upper critical solution temperature (UCST) cannot take place above a critical temperature at which the system remains homogeneous, whereas proteins exhibiting lower critical solution temperature (LCST) cannot phase separate below a critical temperature at which the system remains homogeneous [127]. Therefore, increasing temperatures can either stabilize or destabilize biomolecular condensates formed by phase separation [128], and variations in salt concentration and pH levels can further promote or disrupt phase separation [122].

从本质上讲, 相分离在熵增上是不利的, 主要由焓驱动的蛋白质相互作用所驱动 [119-121]。除了能够抵消熵代价的能量有利的多价蛋白质-蛋白质相互作用外, 离子和盐浓度、pH 值以及温度的变化也会导致熵-焓补偿的热力学变化, 从而调节相分离 [122-124]。在具有高临界溶液温度 (UCST) 的蛋白质 (如 Ddx4 [125] 和 hnRNPA1 [126]) 中, 相分离不能在临界温度之上发生, 此时系统保持均质状态; 而具有低临界溶液温度 (LCST) 的蛋白质则不能在临界温度之下发生相分离, 此时系统保持均质状态 [127]。因此, 升高温度既可以稳定也可以破坏通过相分离形成的生物分子凝聚体 [128], 而盐浓度和 pH 水平的变化可以进一步促进或破坏相分离 [122]。

Stress granules (SGs) are phase-separated membraneless organelles that are formed under endogenous and exogenous stress conditions; and persistent formation of stress granules may lead to fibrillization associated with neurodegenerative disorders [126,129]. Adjusting pH levels in solutions tunes both UCSTs and LCSTs that trigger phase separation [130,131]. Alterations in tightly controlled cytosolic pH not only affect the survival of yeast and other organisms, but also determine the material properties of phase-separated stress granule-like condensates that regulate stress responses [132,133]. A reduction in pH in yeast generates reversible condensates that dissolve upon restoration of neutral pH; whereas phase-separated condensates induced by heat in yeast can only be reversed with the help of chaperones [134]. Similarly, in lipid membranes, both pH and salt can increase or decrease critical temperatures that trigger phase separation [124].

应激颗粒 (SGs) 是在内源性和外源性压力条件下形成的相分离无膜细胞器; 应激颗粒的持续形成可能导致与神经退行性疾病相关的纤维化 [126,129]。调节溶液中的 pH 值可以微调触发相分离的上限临界溶解温度 (UCSTs) 和下限临界溶解温度 (LCSTs) [130,131]。严格控制的细胞质 pH 值的改变不仅影响酵母和其他生物的存活, 还决定了调节应激反应的相分离应激颗粒样凝聚体的材料特性 [132,133]。酵母中 pH 值的降低会产生可逆的凝聚体, 这些凝聚体在恢复中性 pH 后会溶解; 而酵母中由热诱导的相分离凝聚体只能在分子伴侣的帮助下恢复 [134]。同样, 在脂质膜中, pH 值和盐浓度都可以提高或降低触发相分离的临界温度 [124]。

In vitro elevation of salt concentrations produces either a dehydrating, salting-out (kosmotropic) effect that induces phase separation [135,136], or a hydrating, salting-in (chaotropic) effect that inhibits phase separation [137-139]. Classic interpretations of the Hofmeister effect where kosmotropic anions that remove water molecules from a protein's hydration shell to reduce protein solubility, increasing potential for aggregation via electrostatic and hydrophobic interactions [140], and chaotropic anions that exhibit the opposite effect of increasing protein solubility, functioning as a hydrotrope preventing phase separation and aggregation [141,142] may not fully account for other relevant conditions including the reversal of the Hofmeister effects in anions and cations [143], or the effect of pH on the aggregation of proteins relative to

their isoelectric points (pI) [138].

体外盐浓度的升高会产生两种效应：一种是脱水、盐析 (kosmotropic) 效应，诱导相分离 [135,136]；另一种是水合、盐溶 (chaotropic) 效应，抑制相分离 [137-139]。对霍夫迈斯特效应 (Hofmeister effect) 的经典解释是：余序 (kosmotropic) 阴离子通过移除蛋白质水合壳层中的水分子来降低蛋白质溶解度，从而通过静电和疏水相互作用增加聚集的可能性 [140]；而乱序 (chaotropic) 阴离子则具有相反的效果，即增加蛋白质溶解度，作为助溶剂防止相分离和聚集 [141,142]。然而，这些解释可能无法完全说明其他相关情况，包括阴离子和阳离子中霍夫迈斯特效应的逆转 [143]，或者 pH 值相对于蛋白质等电点 (pI) 对蛋白质聚集的影响 [138]。

At its pI of 4.7pH, α -syn formed highly ordered, fibrillar structures even at low salt concentrations compared to other conditions due to favorable intermolecular energy interactions that compensated for the lack of salting-out effects in a low-salt environment [144]. The hydrophobic, hydrogen-bonded, B-rich amyloid cores in α -syn are intrinsically disor-

在等电点 4.7pH, α 时，与其他条件相比，即使在低盐浓度下， α -突触核蛋白 (α -syn) 也会形成高度有序的纤维结构，这是由于有利的分子间能量相互作用补偿了低盐环境中盐析效应的缺失 [144]。 α -syn 中疏水的、由氢键结合的、富含 β 结构的淀粉样核心本质上是无序的——

dered and participate in dynamic intermolecular energy interactions during fibril assembly and maturation [145-147]. As such, protein hydration exerts a distinct effect on the pathological aggregation of amyloid fibrils in dementia, as the hottest mutational spots are often located in residues that form protective hydrogen bonds but have lost their native protecting functions resulting in protein misfolding [148].

并在纤维组装和成熟过程中参与动态分子间能量相互作用 [145-147]。因此，蛋白质水合作用对痴呆症中淀粉样纤维的病理性聚集产生显著影响，因为最热门的突变位点通常位于形成保护性氢键的残基中，但这些残基已失去其原有的保护功能，从而导致蛋白质错误折叠 [148]。

2.3. Hydration Water Activates Amyloid Aggregation and Regulates Oligomer Toxicity

2.3. 水合水激活淀粉样蛋白聚集并调节寡聚体毒性

The role of water hydrogen bond networks that hydrate protein surfaces in biomolecular systems is known to be active and dynamic [149-152], but its role in intracellular phase separation is often less understood. Hydrophilic residues are more hydrated than hydrophobic residues. Thus, entropy and enthalpy become the two fundamental thermodynamic driving forces in phase separation that provide the requisite energetically favorable decrease in free energy. Lum, Chandler, and Weeks postulated that the price for minimizing broken hydrogen bonds within interfacial hydration water compared to bulk is an increased enthalpic cost that scales with the surface area of the hydrophobic solute [153]. Therefore, the removal of hydration water into bulk (entropic) leads to increased protein concentration that facilitates enthalpically favored protein-protein interactions resulting in condensate formation [154,155].

在生物分子系统中，水合蛋白质表面的水氢键网络的作用被认为是活跃且动态的 [149-152]，但其在细胞内相分离中的作用往往较少被理解。亲水残基比疏水残基具有更高的水合度。因此，熵和焓成为相分离中两个基本的分子热力学驱动力，提供了必要的、在能量上有利的自由能降低。Lum、Chandler 和 Weeks 假设，与体相水相比，为了尽量减少界面水合水内部氢键断裂所付出的代价是焓值的增加，且该成本随疏水溶质的表面积成比例增加 [153]。因此，水合水向体相的移除（熵驱动）导致蛋白质浓度增加，从而促进了在焓上占优势的蛋白质-蛋白质相互作用，最终导致凝聚体的形成 [154,155]。

In other words, desolvation or the release of water molecules from protein hydration shells into bulk water [156-158] create entropic gains that promote phase separation and fibril aggregation [136,159,160]. Tau proteins that phase separate from salting-out effects via increased salt concentration become dehydrated and mature into irreversible, canonical tau fibrils, whereas tau proteins in reversible condensates formed via electrostatically driven phase separation remain hydrated and do not mature into pathogenic fibrils with restricted water accessibility and increased micro-viscosity [135]. Mutational hotspots with structural defects that affect protein interactions in monomeric states can be regions with an immense propensity to aggregate if the exclusion or removal of water in those regions confer a high thermodynamic benefit [148].

换句话说，去溶剂化或水分子从蛋白质水合壳层释放到体相水中 [156-158] 会产生熵增，从而促进相分离和纤维聚集 [136,159,160]。通过增加盐浓度产生的盐析效应而发生相分离的 Tau 蛋白会脱水并成熟为不可逆的标准 Tau 纤维；而通过静电驱动相分离形成的具有可逆性的凝聚体中的 Tau 蛋白则保持水合状态，不会成熟为具有受限的水可及性和更高微观粘度的致病性纤维 [135]。如果排除或移除某些区域的水分能带来巨大的热力学收益，那么单体状态下影响蛋白质相互作用且具有结构缺陷的突变热点区域，可能成为具有极高聚集倾向的区域 [148]。

In 1959, Walter Kauzmann proposed that hydrophobicity in protein hydration shells drives protein folding where protein hydration accumulates hydrophobic free energy and removing the water molecule from the hydration shell can supply the free energy required to drive protein folding [161]. This hypothesis remained largely controversial [162,163] until support from experimental evidence on protein hydration shells was published. When the original clathrate water hydration shell used by Kauzmann in 1959 was replaced by a dynamic one formed by van der Waals (vdW) attraction [164], it became clear that the structural differences between water molecules in hydration shells and bulk [165] contributed to changes in free energy produced in vdW attraction interactions that favored protein folding [166,167]. Furthermore, the fact that the addition of salt can tune the hydrophobic effect by breaking hydrogen bonds in hydration shells [168] and rearrange the hydrogen-bonding environment in interfacial waters [169], provides additional support for the role of dehydration in the

formation of pathogenic amyloid fibrils.

1959 年, Walter Kauzmann 提出蛋白质水合壳层中的疏水性驱动了蛋白质折叠, 其中蛋白质水合积累了疏水自由能, 而从水合壳层中移除水分子可以提供驱动蛋白质折叠所需的自由能 [161]。这一假设在很大程度上一直存在争议 [162,163], 直到有关蛋白质水合壳层的实验证据发表。当 Kauzmann 在 1959 年使用的原始笼形水水合壳层被范德华 (vdW) 引力形成的动态壳层所取代时 [164], 情况变得明朗: 水合壳层中的水分子与体相水之间的结构差异 [165], 有助于范德华引力相互作用中产生的自由能变化, 从而有利于蛋白质折叠 [166,167]。此外, 添加盐可以通过破坏水合壳层中的氢键 [168] 并重新排列界面水的氢键环境 [169] 来调节疏水效应, 这一事实为脱水在致病性淀粉样纤维形成中的作用提供了额外支持。

Highly sensitive femtosecond time-resolved fluorescence spectroscopy revealed the presence of dynamically distinct, confined interfacial hydration water molecules with severely restrained mobility compared to bulk water [170]. The removal of these confined water molecules in the intrinsically disordered amyloidogenic NAC domain of α -syn changes the rate of intramolecular backbone reconfiguration to facilitate the formation of cytotoxic oligomers [171] via intermolecular associations involving chain desolvation, indicating the entropically favored removal of confined water molecules into bulk water [170]. Early studies found the aggregation of protofilaments from $A\beta_{16-22}$ peptides was due to the hydrophobic collapse of protofilaments caused by water molecules being released [172,173]. Similarly, the aggregation propensity of $A\beta_{1-40}$ was significantly elevated via escalating salt concentrations to enhance salting-out effects, with the implication of heightened protein-protein interaction energy and diminished hydrogen-bond strength [174,175]. Out of 3.45 hydrogen bonds formed by a water molecule, only 2.41 are

高灵敏度飞秒时间分辨荧光光谱揭示了动态截然不同的受限界面水分子的存在, 与体相水相比, 其流动性受到严重限制 [170]。在 α -syn 固有无序的致淀粉样变 NAC 结构域中, 这些受限水分子的移除改变了分子内骨架重构的速率, 从而通过涉及链去溶剂化的分子间缔合促进细胞毒性寡聚体的形成 [171], 这表明受限水分子向体相水的移除在熵增上是有利的 [170]。早期研究发现, $A\beta_{16-22}$ 肽原纤维的聚集是由于水分子释放引起的原纤维疏水塌陷所致 [172,173]。同样, 通过提高盐浓度以增强盐析效应, $A\beta_{1-40}$ 的聚集倾向显著升高, 这暗示了蛋白质-蛋白质相互作用能的增强和氢键强度的减弱 [174,175]。在一个水分子形成的 3.45 个氢键中, 只有 2.41 个是

considered “strong” hydrogen bonds. Per the hydrophobic effect, the ability to form hydrogen bonds directly affects the stability of protein where net stabilization at $1-2\text{ kcal/mol}$ can be provided by each intramolecular hydrogen bond [176].

被视为 “强” 氢键。根据疏水效应, 形成氢键的能力直接影响蛋白质的稳定性, 其中每个分子内氢键可提供 $1-2\text{ kcal/mol}$ 的净稳定作用 [176]。

Limited hydration in the interior of MLOs fosters a favorable environment for liquid-to-solid phase transitions observed in amyloidogenic aggregates that are often preceded by liquid-to-liquid phase separation [79,177]. During α -syn nucleation, limited hydration lowers the desolvation barrier and intermolecular hydrogen bond barrier. Thus, the simple removal of confined water molecules in the hydrophobic amyloid NAC domain in α -syn can easily breach high desolvation barriers that normally prevent aggregation of amyloid fibrils [178-180]. Furthermore, the level of protein hydration determines whether homogeneous or heterogeneous nucleation is selected as the primary aggregation mechanism, which further defines the type of amyloid polymorph generated as well as the cytotoxicity of the α -syn oligomers formed [178]. Unfortunately, reduced hydration may be an inevitable phenomenon associated with aging in the human brain.

MLO 内部有限的水合作用为淀粉样变性聚集体中观察到的液固相变提供了有利环境, 而这种相变通常发生在液-液相分离之后 [79,177]。在 α -syn 成核过程中, 有限的水合作用降低了去溶剂化能垒和分子间氢键能垒。因此, 简单地去除 α -syn 疏水性淀粉样 NAC 结构域中的受限水分子, 即可轻易突破通常阻止淀粉样纤维聚集的高去溶剂化能垒 [178-180]。此外, 蛋白质的水合水平决定了是以均质成核还是异质成核作为主要的聚集机制, 这进一步定义了生成的淀粉样多态性类型以及所形成的 α -syn 寡聚体的细胞毒性 [178]。不幸的是, 水合作用减少可能是与人脑衰老相关的必然现象。

During normal aging, even though total protein content in the normal aging brain can decline by $5-15\%$ between the ages of 30 and 90 years, water-soluble protein content actually increases by $16-48\%$, providing a viable explanation for observations of significantly decreased water content in normal aging brain cells [181,182]. The fact that confined and “bridging” interfacial water molecules have limited mobility and exceptionally slow hydrogen-bond rearrangement compared to bulk water, respectively, [170,183] highlights the importance of the binding dynamics of interfacial hydration water around residues located in IDPs prone to phase separation under conditions of limited mobility and hydration [184,185]. Atomistic MD simulations revealed that during the growth of $A\beta_{9-40}$ fibrils, the collective movement of confined interfacial water with reduced mobility provides the entropic energy for pathogenic fibril formation via the removal of 60-85 water molecules that concurrently supplies a dry binding interface between filament and monomer [186]. Consequently, the ability to manipulate the relative thermodynamics of hydrogen bonds [187] in interfacial water compared to bulk becomes an extremely attractive proposition in the regulation of protein aggregation in dementia.

在正常衰老过程中, 尽管 30 至 90 岁之间正常衰老大脑的总蛋白含量会下降 $5-15\%$, 但水溶性蛋白含量实际上增加了 $16-48\%$, 这为观察到的正常衰老脑细胞中水分显著减少提供了可行的解释 [181,182]。与体相水相比, 受限水和 “桥接” 界面水分子的迁移率有限, 且氢键重排异常缓慢 [170,183], 这凸显了在迁移率和水合作用受限的条件下, 易于发生相分离的 IDPs 残基周围界面水合水的结合动力学的重要性 [184,185]。原子级 MD 模拟显示, 在 $A\beta_{9-40}$ 纤维的生长过程中, 迁移率降低的受限界面水的集体运动通过移除 60-85 个水分子为致病性纤维的形成提供了熵能, 同时在纤维丝和单体之间提供了一个干燥的结合界面 [186]。因此, 与体相水相比, 操纵界面水中氢键的相对热力学能力 [187], 成为调节痴呆症中蛋白质聚集的一个极具吸引力的命题。

2.4. The Synergistic Regulation of Hydrogen Bonds by Light, Water, and Melatonin

2.4. 光、水和褪黑素对氢键的协同调节作用

The International Union of Pure and Applied Chemistry (IUPAC) defined a hydrogen bond as “an attractive interaction between a hydrogen atom from a molecule or a molecular fragment X – H in which X is more electronegative than H , and an atom or a group of atoms in the same or a different molecule, in which there is evidence of bond formation” [188]. Although hydrogen bonding can affect important physicochemical properties including density, refractive index, and conductivity [189], due to a limitation of scope, this review is solely focused on the relevant associations between hydrogen bonds and viscosity [190,191] in the context of protein hydration in phase separation in dementia.

国际纯粹与应用化学联合会（IUPAC）将氢键定义为“分子或分子碎片 X – H（其中 X 的电负性强于 H）中的氢原子与同一分子或不同分子中的原子或原子团之间存在成键证据的吸引相互作用” [188]。虽然氢键会影响包括密度、折射率和电导率在内的重要理化性质 [189]，但由于篇幅限制，本综述仅侧重于痴呆症相位分离中蛋白质水合背景下，氢键与黏度之间的相关联系 [190,191]。

Interfacial water can exhibit viscosity 106 times higher than bulk water [192], and the breaking and forming of hydrogen bonds in water [193] can affect viscosity of interfacial water. Viscosity is measured in units of centipoises (cPs) [194], and viscosity can accurately indicate flow resistance in water and other solvents. Viscosity in interfacial water is increased by hydrophilicity and reduced by hydrophobicity, implying the strength of the hydrogen bond is critical to maintaining the integrity of the viscous phase in interfacial water [192]. Low-level microwaves, and other electric and electromagnetic fields (EMF) can restructure hydrogen bonding [195,196] where weakened or broken hydrogen bonds decrease viscosity [197,198] and the formation of stronger hydrogen bonds increases absolute viscosity [199]. Light is a form of electromagnetic radiation (EMR) [200], and plants are exposed to an entire spectrum of EMR from sunlight. However, plants only absorb visible light but reflect infrared light.

界面水的粘度可比体相水高出 10⁶ 倍 [192]，且水中氢键的断裂与形成 [193] 会影响界面水的粘度。粘度的测量单位为厘泊 (cPs) [194]，它可以准确指示水和其他溶剂中的流动阻力。界面水的粘度随亲水性增加而升高，随疏水性增加而降低，这暗示了氢键强度对于维持界面水粘性相的完整性至关重要 [192]。低强度微波以及其他电场和电磁场 (EMF) 可以重构氢键 [195,196]，其中氢键的减弱或断裂会降低粘度 [197,198]，而更强氢键的形成则会增加绝对粘度 [199]。光是电磁辐射 (EMR) 的一种形式 [200]，植物暴露在来自阳光的全光谱 EMR 中。然而，植物仅吸收可见光，而反射红外光。

During canopy photosynthesis, visible light from the sun is absorbed and utilized while a directly proportional amount of infrared is reflected [201-203]. A tight, linear correlation exists between canopy photosynthesis and correspondingly reflected NIR in all

在冠层光合作用过程中，来自太阳的可见光被吸收并利用，而与之成正比的红外光则被反射 [201-203]。在所有情况下，冠层光合作用与相应的近红外（NIR）反射之间都存在紧密的线性相关性。

types of plants examined, including well-watered crops, wetland vegetation, grasslands, and savannas, with different functions, structure, capacity, and even, soil conditions [204]. Surprisingly, or not, a higher level of greenness or presence of vegetation is associated with reduced risk for AD (20%, odds ratio 0.80 ; 95% CI, 0.75 – 0.85) and non-AD dementia (11%, odds ratio, 0.89 ; 95% CI, 0.82-0.96) [205]. However, subjects with dementia treated with UVB irradiation did not exhibit any of the greenness effect even though plants are exposed to both spectrums in sunlight [206].

研究的植物类型包括水分充足的作物、湿地植被、草地和稀树草原，它们具有不同的功能、结构、容量，甚至土壤条件 [204]。令人惊讶（或并不意外）的是，较高的绿化水平或植被存在与阿尔茨海默病（AD）（ 20%，比值比 0.80； 95% CI, 0.75 – 0.85 ）和非阿尔茨海默病痴呆（ 11%，比值比 0.89； 95% CI, 0.82-0.96）的风险降低相关 [205]。然而，接受 UVB 照射治疗的痴呆受试者并未表现出任何绿化效应，尽管植物在阳光下会同时暴露于这两种光谱 [206]。

Similarly, photobiomodulation employing visible red light (670 nm), non-visible near infrared (NIR, 800 – 1090 nm), and even far infrared (FIR, 3 – 25μ m) show encouraging results in the attenuation of symptoms associated with dementia including a reduction in Aβ deposition, size and number of plaque and fibril formation, clearance of misfolded proteins, increased ATP production and reduced ROS production, improved executive and cognitive functions, processing speed, memory performance, mood, energy, and sleep [207-217] (Table 1). The proposal that red and near-infrared wavelengths may promote melatonin synthesis in mitochondria via the pathway involving nitric oxide and enhanced activity of soluble adenylyl cyclase further bolsters the synergistic relationship between light and melatonin [218,219].

同样，采用可见红光（670 nm）、不可见近红外（NIR， 800 – 1090 nm ）甚至远红外（FIR， 3 – 25μ m ）的光生物调节疗法在缓解痴呆相关症状方面显示出令人鼓舞的结果，包括减少 Aβ 沉积、斑块的大小和数量以及原纤维的形成、清除错误折叠的蛋白质、增加 ATP 产量并减少 ROS 产生、改善执行和认知功能、处理速度、记忆表现、情绪、能量和睡眠 [207-217]（表 1）。关于红光和近红外波长可能通过涉及一氧化氮和增强可溶性腺苷酸环化酶活性的途径促进线粒体中褪黑素合成的提议，进一步增强了光与褪黑素之间的协同关系 [218,219]。

Table 1: Table 1. A sample collection of popular wavelengths employed in photobiomodulation, starting from visible 670 nm to non-visible near- and far-infrared wavelengths, and their effects on various symptoms associated with dementia in

animals and humans.

表 1：表 1. 光生物调节中采用的常用波长样本集（从可见光 670 nm 到不可见近红外和远红外波长），以及它们对动物和人类痴呆相关各种症状的影响。

Wavelength 波长	Model/Cell Line/Device 模型/细胞系/设备	Duration/Intensity 时长/强度	Results 结果	Ref. 参考文献
670 nm	APP/PS1 AD transgenic mice/transcranial LED APP/PS1 AD 转基因小鼠/经颅 LED	90 s (4 Joule /cm ²) / day × 20 over 4 wks 90 秒 (4 焦耳 /cm ²) / 天 × 20 / 4 周	Attenuated cerebellar cortex A β deposition, fibril formation. 小脑皮层 A β 沉积减弱, 原纤维形成。	[207]
670 nm	K3 tau, APP/PS1 AD transgenic mice/transcranial LED K3 tau, APP/PS1 AD 转基因小鼠/经颅 LED	90 s (4 Joule/ cm ²)/day × 20 over 4 wks 90 秒 (4 焦耳/ cm ²)/天 × 20 次, 持续 4 周	Neocortex and hippocampus of K3 and APP/PS1 mice showed reduction in tau/fibril formation and size/number of A β , respectively. K3 和 APP/PS1 小鼠的新皮层和海马体分别表现出 tau 蛋白/纤维形成的减少以及 A β 大小/数量的减少。	[208]
670 nm	C57BL/6, transgenic 2576 mice/transcranial LED C57BL/6, 转基因 2576 小鼠/经颅 LED	90 s (4 Joule/ cm ²)/day × 20 over 4 wks 90 秒 (4 焦耳/ cm ²)/天 × 20 次, 持续 4 周	All mice showed reduced A β oligomer binding at CNS synapses. 所有小鼠的中枢神经系统 (CNS) 突触处均显示出 A β 寡聚体结合减少。	[209]
670 nm	h tau, 3 × TgAD mice/transcranial LED h tau, 3 × TgAD 小鼠/经颅 LED	90 s (4 Joule/ cm ²)/day × 20 over 4 wks 90 秒 (4 焦耳/ cm ²)/天 × 20 次, 持续 4 周	Reduced toxic tau oligomers, improved memory deficits, upregulated clearance of misfolded proteins in both models 在两种模型中均减少了毒性 tau 蛋白寡聚体, 改善了记忆缺陷, 并上调了错误折叠蛋白的清除率	[210]
808 nm	A β -treated microglia cells from health mice/ diode laser A β 处理的健康小鼠小胶质细胞/半导体激光器	5 min (9 Joule /cm ²) 5 min (9 焦耳 /cm ²)	Exceeded control cell ATP production after 24 h by 155%, suppressed ROS production promoting neuronal survival. 24 小时后, ATP 产量超过对照组 155%, 并抑制了活性氧 (ROS) 的产生, 从而促进神经元存活。	[211]
810 nm	8 patients diagnosed with dementia/transcranial+ transnasal LED 8 名被诊断为痴呆症的患者/经颅+经鼻 LED 治疗	20 min (pulsed at 40 Hz at 50% duty cycle), 3 times/wk for 12 consecutive wks 20 分钟 (脉冲频率 40 Hz, 占空比 50%) , 每周 3 次, 连续 12 周	Significant score improvements in ADAS-cog (13.8%) NPI-FS (61.4%) compared to baseline ¹ . 与基线 ¹ 相比, ADAS-cog (13.8%) 和 NPI-FS (61.4%) 的评分较基线有显著改善。	[212]
1060 – 1080 nm	11 patients with dementia/transcranial LED helmet 11 位痴呆症患者/经颅 LED 头盔	6 min (1100 LEDs pulsed at 10 Hz at 50% duty cycle)/day × 28 consecutive days 6 分钟 (1100 个 LED 以 10 Hz 频率、 50% 占空比脉冲照射) / 天, 连续 × 28 天	Improved executive functioning in clock drawing, immediate recall, praxis memory, visual attention, and task switching. 在画钟测试、即时回忆、运用记忆、视觉注意力和任务切换方面的执行功能有所改善。	[213]
1060 – 1080 nm	60 patients with mild to moderate dementia/transcranial LED helmet 60 名轻度至中度痴呆症患者/经颅 LED 头盔	2 × 6 min (23.1 mW/cm ²)/day × 8 consecutive weeks 2 × 6 min (23.1 mW/cm ²)/天 × 连续 8 周	Improved cognitive functions, auditory and verbal learning, processing speed, mood, energy, and sleep. 改善认知功能、听觉和言语学习、处理速度、情绪、精力以及睡眠。	[214]
1060 nm	27 healthy participants aged 45+/transcranial LED helmet 27 名 45 岁及以上的健康受试者/经颅 LED 头盔	2 × 6 min (12 mW/cm ²)/ day × 28 minimum 2 × 6 min (12 mW/cm ²)/ 天 × 最少 28 天	Significant improvements in motor function, memory performance, and processing speed. 在运动功能、记忆表现和处理速度方面有显著改善。	[215]
1040 – 1090 nm	APP/PS1 AD double-transgenic mice/LED irradiation APP/PS1 AD 双转基因小鼠/LED 照射	6 min (15 mW/cm ²)/ day × 55 with a 28 -day suspension after day 40 第 6 min (15 mW/cm ²)/ 天 × 55 , 第 40 天后停用 28 天	Improvement in memory, spatial learning ability, and modest plaque reduction; suspension period indicated treatment effects were transient. 改善了记忆力、空间学习能力, 并适度减少了斑块; 停药期表明治疗效果是暂时的。	[216]
500 nm/800 nm/3 – 25 μ m	APP/PS1 AD double-transgenic mice/LED irradiation	60 min (0.13 mW/cm ²)/day × 45	FIR (3 – 25 μ m) enhanced A β phagocytosis via increased ATP production and attenuated	[217]

	APP/PS1 AD 双转基因小鼠/LED 照射	60 分钟 (0.13 mW/cm ²)/天 ×45	cognitive dysfunction compared to other wavelengths tested. 与测试的其他波长相比, FIR (3 – 25 μ m) 通过增加 ATP 产量增强了 Aβ 的吞噬作用, 并减轻了认知功能障碍。
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¹ ADAS-cog Alzheimer’s Disease Assessment Scale-cognitive; NPI-FS: Neuropsychiatric Inventory frequency severity.

¹ ADAS-cog 阿尔茨海默病评估量表-认知部分；NPI-FS：神经精神问卷频率与严重程度。

The ability to increase adenosine triphosphate (ATP) production in mitochondria is one of the most widely accepted mechanisms behind the effectiveness of photobiomodulation in dementia and other health challenges [211,217,220-222]. The fact that both infrared light and melatonin increase ATP production, and the adenosine moiety of ATP which is structurally similar to melatonin is capable of solubilizing protein aggregation point to the existence of a most unexpected, dynamic relationship between NIR light and melatonin that is inextricably connected to the regulation of hydrogen bonds, viscosity, protein hydration, and protein aggregation (Figure 1). The following section will present what is currently known about molecular mechanisms that drive the synergistic relationships between light, water, and melatonin in the regulation of phase separation of pathological aggregates in dementia. In subsequent discussions, the term light refers to red and near-infrared wavelengths unless otherwise indicated.

在线粒体中增加三磷酸腺苷（ATP）生成的能力，是光生物调节治疗痴呆症及其他健康挑战最被广泛认可的机制之一 [211,217,220-222]。红外光和褪黑素都能增加 ATP 的生成，而 ATP 的腺苷部分在结构上与褪黑素相似，且能够溶解蛋白质聚集，这些事实表明近红外光（NIR）与褪黑素之间存在一种出人意料的动态关系，这种关系与氢键、粘度、蛋白质水合作用以及蛋白质聚集的调节密不可分（图 1）。接下来的章节将介绍目前已知的分子机制，这些机制驱动了光、水和褪黑素在调节痴呆症病理性聚集体相分离过程中的协同关系。在随后的讨论中，除非另有说明，术语“光”均指红光和近红外波长。

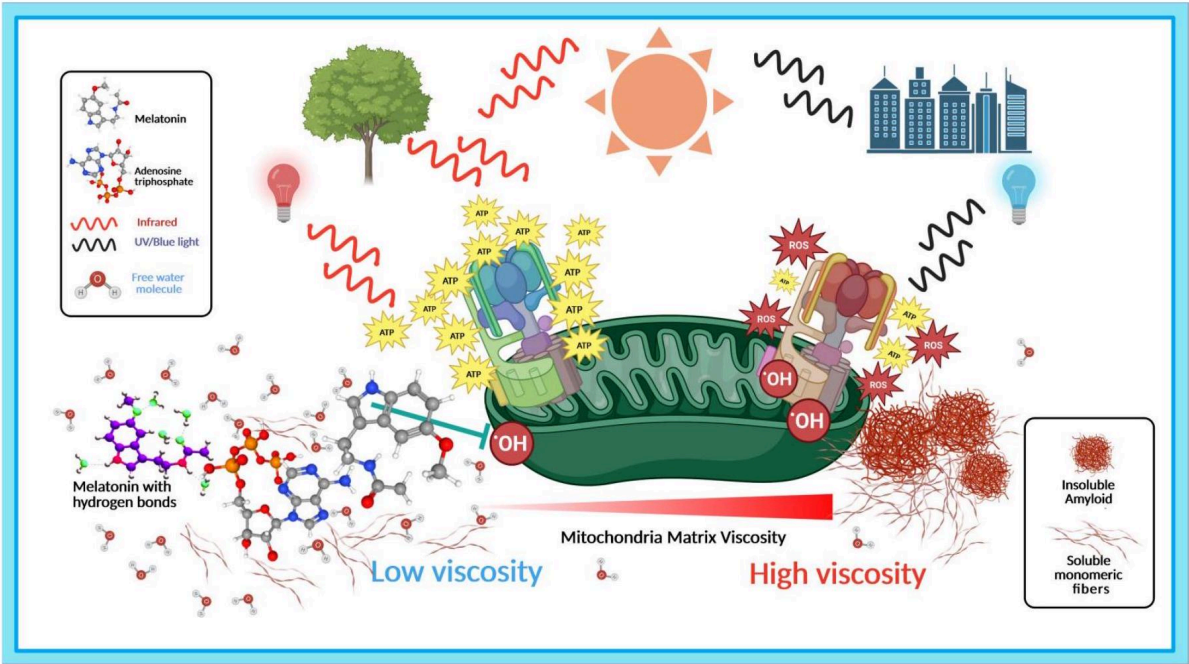


Figure 1: Figure 1. Visible 670 nm red light reduces viscosity in mitochondria interfacial water to increase free water molecules and enhance ATP synthase ability to generate more adenosine triphosphate (ATP). Reactive oxygen species (ROS) increase viscosity and lower ATP synthase efficiency to inhibit ATP production. Melatonin lowers viscosity by scavenging hydroxyl radical (•OH) and ROS. Increased free water molecules from lower viscosity form stronger hydrogen bonds with melatonin to enhance its intrinsic features that include binding interactions with the adenosine moiety of ATP, inhibiting water removal from protein hydration shells that facilitate amyloid fibril aggregation and solubilizing aggregates formed as the result of aberrant phase separation.

图 1：图 1。670 nm 的可见红光降低了线粒体界面水的粘度，从而增加游离水分子，并增强 ATP 合成酶产生更多三磷酸腺苷（ATP）的能力。活性氧（ROS）会增加粘度并降低 ATP 合成酶效率，从而抑制 ATP 的产生。褪黑素通过清除羟基自由基（•OH）和 ROS 来降低粘度。粘度降低带来的游离水分子增多，会与褪黑素形成更强的氢键，从而增强其固有特性，包括：与 ATP 的腺苷部分产生结合相互作用；抑制蛋白质水合壳层的水分流失（该过程会促进淀粉样原纤维聚集）；以及溶解因异常相分离而形成的聚集体。

3. Light, Water, and Melatonin: Ancient Synergies in a Modern World

3. 光、水与褪黑素：现代世界中的古老协同作用

The synergistic relationship between melatonin, water, and light may have originated billions of years ago when primitive unicellular organisms depended on this effective and precise synergy to modulate phase separation to control protein aggregation and associated biological effects. The efficacy of this synergy also provides a credible explanation for the immensely successful and rapid distribution of melatonin via horizontal gene transfer [223]. The discovery of the serotonin N-acetyltransferase (SNAT) gene responsible for the synthesis of essential melatonin substrate N-acetylserotonin (NAS) in archaea [224,225] firmly establishes the quintessential role played by melatonin in early primitive organisms that

褪黑素、水和光之间的协同关系可能起源于数十亿年前，当时原始单细胞生物依赖这种有效且精确的协同作用来调节相分离，从而控制蛋白质聚集及相关的生物学效应。这种协同作用的功效也为褪黑素通过水平基因转移实现极其成功且快速的分布提供了一个可靠的解释 [223]。在古菌中发现负责合成关键褪黑素底物 N-乙酰血清素 (NAS) 的 5-羟色胺 N-乙酰转移酶 (SNAT) 基因 [224,225]，有力地确立了褪黑素在早期原始生物中所发挥的核心作用，这些生物

use phase separation as the fundamental driver for relevant biochemical and biophysical processes to support metabolism, replication, and survival [226-232].

利用相分离作为相关生化和生物物理过程的基础驱动力，以支持代谢、复制和生存 [226-232]。

Melatonin (N-acetyl-5-methoxytryptamine) was first isolated from bovine pineal gland in 1958 [233]. Since then, revelations from the study of melatonin led to a continuously expanding list of appellations that aim to describe its impressive yet often pleiotropic and contradictory behaviors. Melatonin is known as a hormone, an antioxidant, an anticancer agent, an antiviral, an autocoid, a chronobiotic, a hypnotic, an anxiolytic, a glycolytic, a sleep aid, a universal panacea, a biological modifier, and even a Higgs boson [234]. These nomenclatures are excellent illustrations of some of the broad-based metabolic effects achieved by melatonin as it regulates fundamental phase separation processes in living organisms. The role of melatonin in the regulation of phase separation in the context of neurodegenerative disorders, cancer multidrug resistance, and viral phase separation are clearly defined in several in-depth reviews [230,235,236]. Due to a limitation of scope, the reader may review these extensive discussions for a better understanding of molecular mechanisms employed by melatonin in the regulation of phase separation under different biological contexts. This review will focus on the presentation of known, relevant molecular mechanisms that facilitate and enhance the synergistic relationship between light, water, and melatonin in the regulation of phase separation in dementia.

褪黑素 (N-乙酰基-5-甲氧基色胺) 于 1958 年首次从牛松果体中分离出来 [233]。自那时起，对褪黑素的研究揭示了其令人印象深刻但往往具有多效性和矛盾性的行为，导致其称谓列表不断扩大。褪黑素被称为激素、抗氧化剂、抗癌剂、抗病毒药物、自体活性物质、生物节律调节剂、催眠药、抗焦虑药、糖酵解剂、助眠剂、万灵药、生物调节剂，甚至被称为“希格斯玻色子” [234]。这些命名极好地说明了褪黑素在调节生物体基本相分离过程时所实现的广泛代谢效应。褪黑素在神经退行性疾病、癌症多药耐药性和病毒相分离背景下调节相分离的作用，在几篇深入的综述中已有明确定义 [230,235,236]。由于篇幅限制，读者可以查阅这些广泛的讨论，以更好地理解褪黑素在不同生物背景下调节相分离所采用的分子机制。本综述将重点介绍已知的相关分子机制，这些机制促进并增强了光、水和褪黑素在调节痴呆症相分离过程中的协同关系。

3.1. Light, Water, and Melatonin: Viscous Relationships with Hydrogen Bonds

3.1. 光、水与褪黑素：氢键构成的黏性关系

Water molecules confined in interfacial hydration water exhibit severely restrained mobility compared to bulk water [170]. The mobility of these water molecules is reduced by interfacial viscosities as high as 106 times that of bulk [192,237]. However, the viscosity of water constrained in extremely narrow spaces such as the interior of carbon nanotubes increases and decreases with increased and decreased diameters, respectively [238,239]. In carbon nanotubes with diameters below 20Å, water stops behaving like bulk water with different boiling points, self-diffusion coefficient, and viscosity [238,240-242]. Even the mobility of water molecules in ultra-confined spaces is enhanced by reduced viscosity [239,243] which is facilitated by a reduction in hydrogen bonds.

受限于界面水合层中的水分子与体相水相比，其流动性受到严重抑制 [170]。这些水分子的流动性因界面粘度降低，该粘度可高达体相水的 106 倍 [192,237]。然而，限制在极窄空间（如碳纳米管内部）中的水的粘度会分别随着直径的增加和减少而增加和减少 [238,239]。在直径低于 20Å 的碳纳米管中，水的行为不再像体相水，其沸点、自扩散系数和粘度均发生改变 [238,240-242]。甚至在超限空间中，水分子的流动性会因粘度降低而增强 [239,243]，而这种粘度降低是由氢键减少所促成的。

In general, viscosity is increased by stronger intermolecular interactions that form more hydrogen bonds in water molecules [238]. During phase separation, the variation in internal micro-viscosity between tau droplets formed via homotypic and heterotypic associations can be as much as a 7-fold increase [244]. Systematic reductions in droplet micro-viscosity during biological aging may imply continuously evolving intermolecular interactions that shift droplet equilibrium, modifying aggregation potential that favor pathological outcomes [14,245,246]. Therefore, novel properties such as enhanced solubility, diffusion, and electron transfer in specially treated water molecules with lower viscosity and reduced/broken hydrogen bonds [247] may have distinctive effects on the modulation of aberrant protein aggregation in

dementia.

通常情况下，粘度会随着分子间相互作用的增强而增加，这种作用在水分子中形成了更多的氢键 [238]。在相分离过程中，通过同型和异型缔合形成的 tau 蛋白液滴，其内部微观粘度的差异可高达 7 倍 [244]。生物老化过程中液滴微观粘度的系统性降低，可能暗示了分子间相互作用在不断演变，从而改变了液滴的平衡，并修正了有利于病理结果的聚集潜力 [14,245,246]。因此，经过特殊处理、具有较低粘度且氢键减少或断裂的水分子所表现出的增强溶解性、扩散性和电子转移等新特性 [247]，可能对调节痴呆症中异常的蛋白质聚集具有独特的影响。

Hydrogen bonds (HBs) can be reduced/broken by hot electron transfer when plain, deionized bulk water is allowed to flow through gold nanoparticles under resonant illumination. The water-known as plasmon-activated water (PAW)-created by this method exhibits features conspicuously different from bulk even at room temperature [247]. The reduced intermolecular hydrogen bonds in water molecules not only decrease viscosity, but also impart a higher degree of freedom in interaction that allows the formation of stronger intermolecular hydrogen bonding with hydrophilic solutes while enhancing the solubility of hydrophobic solutes [247,248]. Essentially, the elevated interactions with other molecules via increased free water molecules in PAW enhance the intrinsic activities of these molecules. Melatonin is known to dissolve poorly in water [249]; however, melatonin is able to form stronger hydrogen bonds in PAW resulting in enhancement of solubility between $\sim 120\%$ [248] to $\sim 150\%$ [250].

当普通的去离子大体量水在共振照射下流经金纳米颗粒时，热电子转移可以减少或打破氢键（HBs）。通过这种方法产生的水——被称为等离激元活化水（PAW）——即使在室温下也表现出与普通水显著不同的特征 [247]。水分子间氢键的减少不仅降低了粘度，还赋予了更高层次的相互作用自由度，使其能够与亲水性溶质形成更强的分子间氢键，同时增强疏水性溶质的溶解度 [247,248]。从本质上讲，PAW 中增加的游离水分子提升了与其他分子的相互作用，从而增强了这些分子的内在活性。众所周知，褪黑素在水中的溶解性较差 [249]；然而，褪黑素能够在 PAW 中形成更强的氢键，导致其溶解度提高至 $\sim 120\%$ [248] 到 $\sim 150\%$ [250] 之间。

3.1.1. PAW Modulates Melatonin Hydrogen Bonding and Conformation

3.1.1. PAW 调节褪黑素的氢键作用与构象

Melatonin has five distinct hydrogen bonding sites for water, forming up to five hydrogen bonds with water molecules simultaneously at varying strengths. Two of these hydrogen molecules from two water molecules can even reside indefinitely when they are coordinated with the O of the amide group due to the high degree of stability between the H-bond as indicated by Helmholtz free energy [251,252]. For melatonin, water can either be a H-bond donor or acceptor, depending on the site it is attached to. However, even one single water molecule attached to melatonin can change its conformational preference by modulating the relative energies of the conformations and the heights of the barriers that separate conformations, where strong H-bonds can produce substantial electronic frequency shifts. Furthermore, the relative abundance of the conformations can also be regulated by H-bonds, implying that preferential binding between specific sites and water molecules can produce conformational clusters with populations as high as 10 times over other species [252]. In bulk water, melatonin forms the strongest H-bond with its carbonyl O group, stabilizing its tendency to self-aggregate resulting in low solubility [253].

褪黑素具有五个不同的水氢键结合位点，可同时与水分子形成多达五个强度不一的氢键。其中，来自两个水分子的两个氢原子在与酰胺基团的氧原子配位时，甚至可以无限期停留，这是由于亥姆霍兹自由能所显示的氢键高度稳定性 [251,252]。对于褪黑素而言，水既可以是氢键供体，也可以是受体，这取决于其结合的位点。然而，即使是单个水分子与褪黑素结合，也能通过调节构象的相对能量以及分隔构象的能垒高度来改变其构象偏好，其中强氢键会产生显著的电子频率位移。此外，构象的相对丰度也可以通过氢键进行调节，这意味着特定位点与水分子之间的优先结合可以产生构象簇，其种群数量可高达其他物种的 10 倍 [252]。在体相水中，褪黑素与其羰基氧（O）基团形成最强的氢键，稳定了其自聚集倾向，从而导致溶解度较低 [253]。

Melatonin prepared in PAW compared to bulk deionized water exhibited enhanced clearance of hydroxyl radical at 11.9% vs. 6.69%, respectively; its antiviral potency against dengue virus in infected human hepatocarcinoma cells is also enhanced, reducing infectivity by 14.7% vs. 20.6% in bulk [250]. Male Wistar rats subjected to chronic sleep deprivation (CSD) using the disc-on-water methodology [254] and treated with 10mg/kg melatonin via intraperitoneal (IP) injection dissolved in PAW exhibited significantly better results in all parameters detected, including hepatic function and metabolic activity, than control (no treatment), CSD only, and CSD + melatonin dissolved in bulk deionized water groups [250]. It is plausible that when melatonin is dissolved in PAW, the intrinsic anti-inflammatory properties of PAW may also be responsible for molecular mechanisms that support/enhance melatonin's antiviral and antioxidative features. Indeed, APP/PS1 transgenic AD mice treated with PAW showed improved memory function and reduced amyloid burden, potentially via anti-inflammatory and anti-oxidative effects, compared to age-matched wild-type controls [255,256]. There is no doubt that the anti-oxidative properties of PAW enhance melatonin's intrinsic activities. However, the molecular

mechanism involved is an unexpected, viscous one.

与散装去离子水相比，在 PAW 中制备的褪黑素表现出增强的羟自由基清除能力，分别为 11.9% 和 6.69%；其在受感染的人肝癌细胞中对抗登革病毒的抗病毒效力也得到了增强，与散装水中的 14.7% 相比，感染率降低了 20.6% [250]。采用盘上水法 [254] 进行慢性睡眠剥夺 (CSD) 的雄性 Wistar 大鼠，通过腹腔 (IP) 注射溶解在 PAW 中的 10mg/kg 褪黑素进行治疗，在所有检测参数 (包括肝功能和代谢活性) 上均表现出显著优于对照组 (不治疗)、仅 CSD 组以及 CSD + 溶解在散装去离子水中的褪黑素组的结果 [250]。一种合理的解释是，当褪黑素溶解在 PAW 中时，PAW 固有的抗炎特性可能也是支持/增强褪黑素抗病毒和抗氧化特性的分子机制的原因。事实上，与同龄野生型对照组相比，接受 PAW 治疗的 APP/PS1 转基因 AD 小鼠表现出记忆功能改善和淀粉样蛋白负荷减少，这可能是通过抗炎和抗氧化作用实现的 [255,256]。毫无疑问，PAW 的抗氧化特性增强了褪黑素的内在活性。然而，所涉及的分子机制是一个意想不到且复杂的 (viscous) 过程。

3.1.2. Reactive Oxygen Species Increase Viscosity

3.1.2. 活性氧增加粘度

Hydrophilicity enhances viscosity in interfacial water at values up to ~ 106 times that of bulk due to an increase in ordering and hydrogen-bond dynamics [192]. The negative polarity of reactive oxygen species (ROS) is able to increase hydrophilicity and elevate viscosity. When the oxygen atom of one of the most reactive ROS hydroxyl radical ($\bullet\text{OH}$) becomes highly negative and acts as a hydrogen bond acceptor, it can lower the reaction barrier stabilizing $\bullet\text{OH}$ bonding to water during the polar transition state. Thus, water and viscosity of water can modulate and stabilize the highly reactive $\bullet\text{OH}$ [257]. In bulk water, $\bullet\text{OH}$ forms three stable hydrogen bonds and a weaker hemibond with surrounding water molecules comprising its solvation shell [258].

由于有序度和氢键动力学的增加，亲水性使界面水的粘度增强，其数值最高可达体相水的 ~ 106 倍 [192]。活性氧 (ROS) 的负电性能够增加亲水性并提高粘度。当最具活性的活性氧之一——羟基自由基 ($\bullet\text{OH}$) 的氧原子变得高度带负电并作为氢键受体时，它可以降低反应能垒，在极性过渡态期间稳定与水的 $\bullet\text{OH}$ 键合。因此，水和水的粘度可以调节并稳定高活性的 $\bullet\text{OH}$ [257]。在体相水中， $\bullet\text{OH}$ 与构成其溶剂化壳层的周围水分子形成三个稳定的氢键和一个较弱的半键 [258]。

In mitochondria, $\bullet\text{OH}$ is derived from superoxide radicals produced as a result of a one-electron reduction of oxygen (O_2) from electron leakage during mitochondrial electron transport [259,260]. Simply stated, the presence of excess, unneutralized ROS can significantly elevate viscosity in these essential energy-producing organelles, negatively impacting mitochondrial functions and ATP production associated with pathological A β aggregation [261]. Hydrogen peroxide (H_2O_2)-a ubiquitous ROS with classical intracellular signaling functions at lower physiological levels [262]-is also produced in mitochondria from electrons leaked during mitochondrial electron transport activities [259]. Similar to $\bullet\text{OH}$, H_2O_2 accumulation can increase matrix viscosity in mitochondria [263,264]. Furthermore, an NIR emissive fluorescent probe with a large Stokes shift detected significantly elevated viscosity and H_2O_2 levels in brain mitochondria of APP/PS1 transgenic AD mice compared to normal BALB/c mice [265].

在线粒体中， $\bullet\text{OH}$ 衍生自超氧自由基，而超氧自由基是线粒体电子传递过程中电子泄漏导致氧气 (O_2) 发生单电子还原而产生的 [259,260]。简单来说，过量且未被中和的 ROS 的存在会显著提高这些重要能量产生细胞器的粘度，从而对线粒体功能和与病理性 A β 聚集相关的 ATP 产生造成负面影响 [261]。过氧化氢 (H_2O_2) 是一种普遍存在的 ROS，在较低的生理水平下具有经典的细胞内信号传导功能 [262]，它也是在线粒体电子传递活动期间由泄漏的电子在线粒体中产生的 [259]。与 $\bullet\text{OH}$ 、 H_2O_2 类似，其积累会增加线粒体基质的粘度 [263,264]。此外，一种具有大斯托克斯位移的近红外发射荧光探针检测到，与正常的 BALB/c 小鼠相比，APP/PS1 转基因 AD 小鼠脑线粒体中的粘度和 H_2O_2 水平显著升高 [265]。

3.1.3. Reduction in Viscosity and Hydrogen Bonds Enhance Melatonin ROS Scavenging

3.1.3. 粘度降低与氢键减少增强褪黑素对活性氧 (ROS) 的清除能力

Melatonin is known for its ability to scavenge $\bullet\text{OH}$ and other free radicals [266-270] where one molecule of melatonin can scavenge two $\bullet\text{OH}$ radicals to produce the stable cyclic 3-hydroxymelatonin (3-OHM) metabolite [266]. However, the addition of only one water molecule that provides an H-bonding relay pathway significantly lowered the energy barrier in the tautomerization step to enhance the scavenging potential by melatonin [271]. The fact that melatonin prepared in PAW exhibit 78% increased effectiveness in $\bullet\text{OH}$ scavenging compared to bulk (11.9% vs. 6.69%) [250] implies that melatonin may adopt more favorable conformations that enhance its intrinsic activities as a result of stronger H-bonds formed in water with reduced viscosity and H-bonds compared to bulk.

褪黑素以清除 $\bullet\text{OH}$ 和其他自由基的能力而闻名 [266-270]，其中一个褪黑素分子可以清除两个 $\bullet\text{OH}$ 自由基，从而产生稳定的环状 3-羟基褪黑素 (3-OHM) 代谢物 [266]。然而，仅加入一个提供氢键中继途径的水分子，就能显著降低互变异构步骤中的能垒，从而增强褪黑素的清除潜力 [271]。在 PAW 中制备的褪黑素与体相相比，在清除 $\bullet\text{OH}$ 方面表现出 78% 的效率提升 (11.9% 对比 6.69%) [250]，这一事实暗示褪黑素可能会采取更有利的构象，从而增强其内在活性，这是由于在粘度和氢键均低于体相的水中形成了更强的氢键所致。

In the context of aberrant protein aggregation in dementia, the signature reduction in viscosity and H-bonds in PAW inadvertently accentuates an unconventional but relevant perspective on the viscous relationships between light, melatonin, and ROS that surprisingly, or not, converge on the synthesis of ATP in mitochondria. In response to conditions that reduce

ATP, budding yeast conserves energy by increasing cytosolic viscosity to slow cellular processes by reducing protein diffusion rates. Additionally, increased viscosity modulates phase separation, impeding the formation of stress granules and inducing aberrant phase separation to form aggregates that were not present in cells that could not elevate viscosity [272].

在痴呆症相关的异常蛋白质聚集背景下，等离子体活化水（PAW）中粘度和氢键的显著降低，无意中强调了一个非常规但相关的视角，即光、褪黑素与活性氧（ROS）之间的粘性关系，而这些因素出人意料（或意料之中）地汇聚于线粒体中 ATP 的合成。为了应对 ATP 减少的情况，酿酒酵母通过增加细胞质粘度来降低蛋白质扩散速率，从而减缓细胞进程以节省能量。此外，增加的粘度会调节相分离，阻碍应激颗粒的形成，并诱导异常相分离以形成聚集体，而这些聚集体在无法提高粘度的细胞中并不存在 [272]。

3.2. Light, Melatonin, and Viscosity in the Elevation of ATP Synthesis

3.2. 光、褪黑素及黏度在提升 ATP 合成中的作用

Mitochondrial matrix exists mostly as interfacial water due to the density of proteins, and matrix water exhibits similar restrained mobility as interfacial water. Consequently, matrix water is significantly more viscous than cytoplasm [273]. The viscosity of the mitochondrial matrix is correlated with the respiratory state of the organelle that can affect not only signal transduction, but also how mitochondrial networks are organized. The abnormal elevation of viscosity in mitochondria results in dysregulation in metabolite diffusion that can cause aberrant phase separation resulting in malignancies associated with fatty liver, diabetes, atherosclerosis, accelerated aging, cancer, AD, and other neurodegenerative disorders. Therefore, the accurate detection and determination of mitochondrial viscosity can facilitate the understanding of molecular mechanisms behind various diseases associated with mitochondrial dysfunctions [265,274-294]. A fluorescent probe that can detect mitochondrial viscosity fluctuations was used for the first time in the successful, early diagnosis of liver and kidney injury in animal models [295], while other probes are employed to effectively distinguish normal cells from cancerous cells with distinct, elevated viscosity [296-299].

由于蛋白质密度极高，线粒体基质主要以界面水的形式存在，且基质水的流动性表现出与界面水相似的受限特征。因此，线粒体基质的粘度显著高于细胞质 [273]。线粒体基质的粘度与细胞器的呼吸状态相关，这不仅会影响信号转导，还会影响线粒体网络的组织方式。线粒体粘度的异常升高会导致代谢物扩散失调，从而引起异常的相分离，导致与脂肪肝、糖尿病、动脉粥样硬化、加速衰老、癌症、阿尔茨海默病（AD）及其他神经退行性疾病相关的恶性病变。因此，准确检测和测定线粒体粘度有助于理解与线粒体功能障碍相关的各种疾病背后的分子机制 [265, 274-294]。一种能够检测线粒体粘度波动的荧光探针首次被成功用于动物模型肝肾损伤的早期诊断 [295]，而其他探针则被用于有效区分正常细胞与具有明显高粘度的癌细胞 [296-299]。

In HeLa cells, the average viscosity of mitochondria is determined to be $\sim 62.8\text{cP}$ [300], in stark contrast to the $2.04 \pm 0.49\text{cP}$ obtained for HeLa nucleoplasm viscosity which is already higher than that in HeLa cytoplasm [301]. Furthermore, treatment with pharmaceuticals, such as monensin and nystatin, can further drive matrix viscosity up to 90.5 and 109 cP, respectively [300,302]. Mitochondria of HeLa cells under oxidative stress generate a tremendous amount of ROS [303]. Therefore, ROS such as $\cdot\text{OH}$, which is naturally produced during mitochondrial respiration, and excess oxidative stress, can potentially increase matrix viscosity from its hydrogen-bonding interactions with water molecules. Dual-targeting fluorescent probes are developed to easily identify viscosity changes in mitochondria in the presence of specific free radical species [304]. Mechanistically, increased viscosity in the matrix can result in the lower production of ATP catalyzed by the ATP synthase.

在 HeLa 细胞中，线粒体的平均粘度被测定为 $\sim 62.8\text{cP}$ [300]，这与 HeLa 核质粘度所得的 $2.04 \pm 0.49\text{cP}$ 形成鲜明对比，而核质粘度已经高于 HeLa 细胞质的粘度 [301]。此外，使用莫能菌素（monensin）和制霉菌素（nystatin）等药物处理可进一步将基质粘度分别推高至 90.5 和 109 cP [300,302]。处于氧化应激下的 HeLa 细胞线粒体会产生大量的 ROS [303]。因此，像 $\cdot\text{OH}$ 这样在线粒体呼吸过程中自然产生的活性氧（ROS）以及过度的氧化应激，可能会通过与水分子的氢键相互作用增加基质粘度。目前已开发出双靶向荧光探针，以便在存在特定自由基物种的情况下轻松识别线粒体中的粘度变化 [304]。从机制上讲，基质粘度的增加会导致 ATP 合酶催化的 ATP 产量降低。

3.2.1. Efficiency of ATP Synthase Is Modulated by Viscosity

3.2.1. ATP 合成酶的效率受黏度调节

The mitochondrial ATP synthase ($F_0 F_1$) is a rotary motor enzyme with a proton-driven F_0 motor that is embedded in the inner mitochondrial membrane and is connected to the ATP-driven F_1 motor that protrudes into the mitochondrial matrix [305-307]. The higher viscosity of the medium can slow down the rotation of the F_1 motor to reduce ATP

线粒体 ATP 合酶 ($F_0 F_1$) 是一种旋转马达酶，其具有嵌入线粒体内膜的质子驱动型 F_0 马达，并与突出到线粒体基质中的 ATP 驱动型 F_1 马达相连 [305-307]。介质的较高粘度会减慢 F_1 马达的旋转，从而减少 ATP

synthesis not only in mitochondria [308] but also chloroplasts [309]. While ATPase turnover rates are more effective when detected by probes designed with lower viscous drag [310], viscous drag can dramatically slow the rate of rotation to 3% of

the enzyme turnover rate in *Escherichia coli* [311].

合成不仅发生在线粒体 [308] 中，也发生在叶绿体 [309] 中。虽然使用低粘滞阻力设计的探针检测时，ATP 酶的周转率更为有效 [310]，但粘滞阻力会显著降低旋转速率，使其仅为大肠杆菌 [311] 中酶周转率的 3%。

Nonetheless, 100% efficiency of the F_1 rotor can theoretically be achieved if the 120° power strokes rotate at a constant angular velocity [312]. However, power stroke and dwell duration are easily modified by viscosity. Viscous loads applied to the ATP F_1 motor of *E. coli* can cause the increase in the duration of the 120° power stroke that is correlated to a 20-fold increase in the length of the dwell. Thus, the power stroke velocity is limited by the viscous load on the motor, and consequently, increases in transition time are the direct result of increases in viscosity and not from inhibition of the ATPase by other means [313]. A deeper analysis of viscosity sensitivity showed that viscous drag on rotations of the γ -subunit in the F_1 motor [314] can cause variations of more than 5000 -fold by using a variety of rotation probes [315].

尽管如此，如果 120° 做功冲程以恒定角速度旋转，理论上可以实现 F_1 转子的 100% 效率 [312]。然而，做功冲程和停顿持续时间很容易受粘度影响。施加在大肠杆菌 ATP F_1 马达上的粘性负载会导致 120° 做功冲程持续时间的增加，这与停顿长度增加 20 倍相关。因此，做功冲程速度受限于马达上的粘性负载，因此，过渡时间的增加是粘度增加的直接结果，而不是由于其他手段对 ATPase 的抑制 [313]。对粘度敏感性的更深入分析表明，通过使用各种旋转探针 [315]， F_1 马达中 γ 亚基旋转产生的粘性阻力 [314] 可引起超过 5000 倍的变化。

3.2.2. The 670 nm Wavelength Elevates ATP Production in Mitochondria

3.2.2. 670 nm 波长提升线粒体中的 ATP 产量

The benefits of photobiomodulation employing the 670 nm wavelength for dementia and other neurodegenerative disorders are extensively documented (Table 1). Even though improved ATP production and reduced ROS production are associated with the use of 670 nm irradiation, the exact mechanism responsible for these effects remains controversial. Experimental works employing 670 nm report the reduction in inflammation via increased expression of cytochrome C oxidase (COX) in an age-related macular degeneration mouse model [316]; increased COX expression and ATPase activities in Wistar rats exposed to suppressive effects of fluorescent light [317]; significantly elevated ATP production in aging mouse retina via increased COX expression [318]; and the restoration of neuronal ATP and prevention of apoptosis induced by potassium cyanide—an irreversible inhibitor of COX [319]. Therefore, benefits from photobiomodulation, especially the enhancement of ATP production and mitochondrial function, are generally believed to be associated with the involvement of COX via increased COX expression and activities.

使用 670 nm 波长的光生物调节疗法对痴呆症和其他神经退行性疾病的益处已有广泛记录（表 1）。尽管 670 nm 照射与 ATP 产量的提高和活性氧（ROS）产量的减少有关，但产生这些效应的确切机制仍存在争议。采用 670 nm 的实验研究报告称：在年龄相关性黄斑变性小鼠模型中，通过增加细胞色素 C 氧化酶（COX）的表达来减轻炎症 [316]；在暴露于荧光抑制效应的 Wistar 大鼠中，COX 表达和 ATPase 活性增加 [317]；在衰老小鼠视网膜中，通过增加 COX 表达显著提高 ATP 产量 [318]；以及恢复神经元 ATP 并预防由氰化钾（一种不可逆的 COX 抑制剂）诱导的细胞凋亡 [319]。因此，光生物调节的益处，特别是 ATP 产量和线粒体功能的增强，通常被认为与 COX 的参与（通过增加 COX 的表达和活性）有关。

COX, or complex IV [320], is the fourth enzyme that catalyzes the transfer of electrons from ferricytochrome C to oxygen in the mitochondrial electron-transport complexes, and COX is highly susceptible to inactivation by oxidative damage induced by ROS including $\cdot\text{OH}$ and 4-hydroxynonenal (HNE), a major lipid peroxidation product [321-325]. Even though COX is viewed as the primary photoacceptor, molecular mechanism that elucidates the association of irradiation by 710 – 790 nm and 650 – 680 nm wavelengths with reduced and oxidized states of COX, respectively, remain elusive [326]. Furthermore, experimental work that combined nanoindentation and 670 nm laser irradiation to modulate viscosities of interfacial water supports the proposal that lower viscosity in mitochondria is the real driver behind photobiomodulation propelling enhanced ATP synthesis, and not increased COX expression and activities [327-329]. However, if reduced viscosity from light irradiation is responsible for increased ATP synthesis via increased power stroke velocity producing more efficient F_1 motor rotations, then this proposal should be inclusive of COX involvement also.

COX，即复合体 IV [320]，是线粒体电子传递链中的第四种酶，负责催化电子从铁细胞色素 C 转移到氧气。COX 极易受到活性氧（ROS）诱导的氧化损伤而失活，这些活性氧包括 $\cdot\text{OH}$ 和 4-羟基壬烯醛（HNE，一种主要的脂质过氧化产物）[321-325]。尽管 COX 被视为主要的受光体，但阐明 710 – 790 nm 和 650 – 680 nm 波长的照射分别与 COX 的还原态和氧化态之间关联的分子机制仍不清楚 [326]。此外，结合纳米压痕技术和 670 nm 激光照射来调节界面水粘度的实验工作支持了这样一种观点：线粒体中较低的粘度才是光生物调节作用推动 ATP 合成增强的真正驱动力，而非 COX 表达和活性的增加 [327-329]。然而，如果光照降低粘度是通过提高动力冲程速度、产生更高效的 F_1 马达旋转来负责增加 ATP 合成，那么这一提议也应将 COX 的参与纳入其中。

3.2.3. Viscosity Modulates COX Activities in Mitochondria

3.2.3. 黏度调节线粒体中的 COX 活性

In 1987, the main activity of COX—the oxidation of ferricytochrome C by COX—was demonstrated to be viscosity-dependent at both high and low ionic strengths [330]. While laser flash photolysis revealed a dramatic decrease in the rate of intramolecular electron transfer (IET) between the heme and molybdenum centers of chicken liver sulfite oxidase when solution viscosity was increased [331]. The evidence supporting the enhancement of ATP synthesis via light irradiation is solid [332,333], and it is also not unreasonable to propose that the reduction in mitochondrial matrix viscosity by light or ROS scavenging can increase ATP production, and increased ATP is associated with clearance of pathogenic aggregates from aberrant phase separation.

1987 年，细胞色素 c 氧化酶 (COX) 的主要活性——即 COX 对铁细胞色素 C 的氧化作用——被证明在高离子强度和低离子强度下均具有粘度依赖性 [330]。而激光闪光光解技术显示，当溶液粘度增加时，鸡肝亚硫酸盐氧化酶的血红素中心与钼中心之间的分子内电子传递 (IET) 速率会大幅下降 [331]。支持通过光照射增强 ATP 合成的证据是充分的 [332,333]，因此，提出通过光照或活性氧 (ROS) 清除来降低线粒体基质粘度可以增加 ATP 产量也是合理的，而 ATP 的增加与清除异常相分离产生的致病性聚集体有关。

Thus, the ability to clear $A\beta$ aggregation by the antioxidant epigallocatechin-3-gallate (EGCG) may be the result of upregulated COX activities and ATP production from reduced ROS and matrix viscosity [334,335]. In human neuroblastoma (SH-EP) cells, 670 nm irradiation dramatically elevated ATP levels by 20% which was subsequently diminished after irradiation-associated clearance of $A\beta_{42}$ aggregation. Both 670 nm irradiation and EGCG were independently able to reduce $A\beta_{42}$ aggregation at the expense of ATP consumption compared to controls. However, the combined, complementary treatment produced even better results in the clearance of amyloid aggregates compared to controls [334].

因此，抗氧化剂表没食子儿茶素-3-gallate (EGCG) 清除 $A\beta$ 聚集的能力，可能是由于活性氧 (ROS) 和基质粘度降低导致 COX 活性上调及 ATP 产量增加的结果 [334,335]。在人神经母细胞瘤 (SH-EP) 细胞中，670 nm 照射显著提高了 20% 的 ATP 水平，随后在照射相关的 $A\beta_{42}$ 聚集清除后，ATP 水平有所下降。与对照组相比，670 nm 照射和 EGCG 均能独立减少 $A\beta_{42}$ 聚集，但会消耗 ATP。然而，与对照组相比，这种联合互补治疗在清除淀粉样蛋白聚集体方面产生了更好的效果 [334]。

3.3. Melatonin Prevents and Disaggregates Aberrant Protein Aggregation in Dementia via Association with ATP

3.3. 褪黑素通过与 ATP 结合预防并解聚痴呆症中的异常蛋白质聚集

Melatonin, a mitochondria-targeted molecule [336] that is known for being a potent ROS scavenger [266-271], promotes ATP synthesis via the elevation of COX expression and activities. Melatonin administered to aged rats at 10mg/kg per day in drinking water prevented the 30% age-related decline in COX activity while abolishing concomitant elevation of H_2O_2 in brain mitochondria of aged rats compared to controls [337]. Melatonin administered orally at 10mg/kg/day for 17 weeks to male Zucker diabetic fatty (ZDF) rats restored the 25% decline in renal mitochondrial COX activity and attenuated other mitochondrial dysfunctions including diminished ATP production compared to lean controls [338]. Melatonin administered in drinking water at the same amount to ZDF rats reversed the 76% decline in brown adipose tissue mitochondria COX activity by 35% while increasing COX activity by a staggering 31% in normal, lean controls [339].

褪黑素是一种靶向线粒体的分子 [336]，以其强大的活性氧 (ROS) 清除能力而闻名 [266-271]，它通过提高 COX 的表达和活性来促进 ATP 合成。与对照组相比，每天在饮用水中给予老年大鼠 10mg/kg 的褪黑素，可防止其脑线粒体中 30% 与年龄相关的 COX 活性下降，同时消除伴随的 H_2O_2 升高 [337]。对雄性 Zucker 糖尿病肥胖 (ZDF) 大鼠口服给予褪黑素 10mg/kg/天，持续 17 周，与消瘦对照组相比，恢复了其肾线粒体 COX 活性的 25% 下降，并减轻了包括 ATP 产量减少在内的其他线粒体功能障碍 [338]。对 ZDF 大鼠在饮用水中给予相同剂量的褪黑素，使棕色脂肪组织线粒体 COX 活性恢复了 76%，同时使正常消瘦对照组的 COX 活性惊人地增加了 35% [339]。

3.3.1. Melatonin Elevates ATP Production via Modulation of COX and Viscosity

3.3.1. 褪黑素通过调节 COX 和黏度提高 ATP 产量

Even though cyanide (CN^-)—a highly cytotoxic molecule that inhibits COX to suppress mitochondrial respiration and ATP production, and elevates ROS by modulating antioxidant defense—is proposed to be a novel mammalian gasotransmitter that can stimulate COX activity and enhance cellular bioenergetics at low endogenous nanomolar levels, at levels beyond $10\mu M$, CN^- remains exceedingly toxic [340-345]. As a result, sophisticated dual-response sensors and probes are used to detect fluctuations in mitochondria viscosity in the presence of varying levels of cyanide in living cells [278,346].

尽管氰化物 (CN^-)——一种具有高度细胞毒性的分子，通过抑制 COX 来抑制线粒体呼吸和 ATP 产生，并通过调节抗氧化防御来升高 ROS——被认为是一种新型哺乳动物气体递质，在低内源性纳摩尔水平下可刺激 COX 活性并增强细胞生物能量，但当水平超过 $10\mu M$, CN^- 时，它仍然具有极高的毒性 [340-345]。因此，复杂的双响应传感器和探针被用于检测活细胞在不同氰化物水平下线粒体粘度的波动 [278,346]。

Not unexpectedly, the in vitro study of rat brain mitochondria treated with 5 μ M potassium cyanide revealed that 50% inhibition of COX activity was nearly entirely counteracted by treatment with 100 μ M melatonin in a dose-dependent manner compared to control; while COX activity in rat liver mitochondria under same treatment conditions achieved 30% higher efficiency than control. However, at 100 μ M cyanide exposure, even 5 mM of melatonin was unable to reverse the 100% inactivation of COX [347]. In vivo administration of melatonin at 10mg/kg (IP) significantly elevated COX activity in rat brain and liver mitochondria in a time-dependent manner while reversing COX activity inhibition and preventing mitochondrial damage and oxidative stress induced by ruthenium red treatment at 60 μ g/kg (IP) [348]. Ruthenium complexes can also increase viscosity and induce cell apoptosis via ROS-mediated mitochondrial pathways [349,350].

不出所料，一项针对用 5 μ M 氰化钾处理的大鼠脑线粒体的体外研究显示，与对照组相比，使用 100 μ M 褪黑素处理以剂量依赖的方式几乎完全抵消了 50% 的 COX 活性抑制；而在相同处理条件下，大鼠肝脏线粒体中的 COX 活性比对照组实现了 30% 的更高效率。然而，在 100 μ M 氰化物暴露下，即使是 5 mM 的褪黑素也无法逆转 100% 的 COX 失活 [347]。体内给予 10mg/kg (IP) 的褪黑素能以时间依赖的方式显著提高大鼠脑和肝脏线粒体中的 COX 活性，同时逆转 COX 活性抑制，并防止由 60 μ g/kg (IP) 钌红处理诱导的线粒体损伤和氧化应激 [348]。钌配合物还可能增加粘度，并通过 ROS 介导的线粒体途径诱导细胞凋亡 [349,350]。

3.3.2. Fibril Disaggregation by Melatonin Is Dose-Dependent

3.3.2. 褪黑素诱导的纤维解聚具有剂量依赖性

Melatonin is intensively studied and extensively reviewed as a likely ideal therapeutic molecule for AD and other neurodegenerative disorders [351-355]. A novel understanding of melatonin regulation of biomolecular condensate phase separation in neurodegenerative disorders [230] provides additional relevant molecular mechanisms behind reported observation including the inhibition, destabilization, reduction, and delay of α -Syn and A β fibril aggregation. Melatonin not only increased survival rates in transgenic AD mice, but also reversed A β -induced synaptic disorder, memory deficit, neurodegeneration, as well as phosphorylation of tau in wild-type mice injected with A β peptides [356-366] (Table 2). However, inconsistent results were observed when there were discrepancies in dosage and timing/duration of administration [363].

褪黑素作为一种可能的阿尔茨海默病（AD）及其他神经退行性疾病的理想治疗分子，已得到深入研究和广泛综述 [351-355]。关于褪黑素调节神经退行性疾病中生物分子凝聚物相分离的新见解 [230]，为已报道的观察结果提供了额外的相关分子机制，包括对 α -Syn 和 A β 纤维聚集的抑制、去稳定化、减少和延迟。褪黑素不仅提高了转基因 AD 小鼠的存活率，还逆转了注射 A β 肽的野生型小鼠中由 A β 诱导的突触功能障碍、记忆缺失、神经变性以及 tau 蛋白磷酸化 [356-366]（表 2）。然而，当剂量和给药时间/持续时间出现差异时，观察到的结果并不一致 [363]。

Table 2: Table 2. In vitro and in vivo studies that reveal important relationships between melatonin dosage, timing, and administration that produced different results for symptoms associated with dementia.

表 2：表 2。揭示了褪黑素剂量、时机和给药方式之间重要关系的体外和体内研究，这些研究对痴呆相关症状产生了不同的结果。

Melatonin Dosage/Duration 褪黑素剂量/持续时间	Study Design 研究设计	Results 结果	Ref. 参考文献
25μM, 250μM, 2.5mM	In vitro α-Synuclein peptide aggregation 体外 α -α-突触核蛋白肽聚集	Blocked α-Syn fibril formation and destabilized preformed fibrils in a dose- and time-dependent manner; increased viability of primary mixed neurons treated with α-Syn to ~ 97% in a time-dependent manner. 以剂量和时间依赖性方式阻断了 α -Syn 纤维的形成，并使预形成的纤维不稳定；以时间依赖性方式将经 α -Syn 处理的原始混合神经元的存活率提高至 ~ 97% 。	[356]
10mg/kg (IP) ×5/ day for 2 days, then ×2/ day for 5 days 10mg/kg (IP) ×5/ 天, 持续 2 天, 然后 ×2/ 天, 持续 5 天	Arsenite-induced oxidative injury in substantia nigra of adult male rats 亚砷酸盐诱导的成年雄性大鼠黑质氧化损伤	Attenuated arsenite-induced α-Syn aggregation, lipid peroxidation, and glutathione depletion. 减轻了亚砷酸盐诱导的 α -Syn 聚集、脂质过氧化和谷胱甘肽耗竭。	[357]
Melatonin dissolved in 2 mM ammonium acetate 溶于 2 mM 乙酸铵的褪黑素	Aβ peptide (1-40) β-sheet/fibril formation Aβ 肽 (1-40) β β-折叠/纤维形成	Melatonin alone delayed fibril formation from 仅褪黑素就延迟了纤维形成，其源自	[359]
1 μM melatonin 1 mM 褪黑素	Aβ1-40 peptide, profibrillogenic apoE4/apoE Aβ1 -40 多肽, 促纤维化 apoE4/apoE	24 h up to 72 h . Combined with either apoE4 or apoE3, inhibition remained effective at termination of experiment. 24 小时至 72 小时。无论是与 apoE4 还是 apoE3 结合，抑制作用在实验结束时依然有效。	[360]
2mg/mL in drinking water starting at age 4 months until euthanasia 饮用水中的 2mg/mL , 从 4 个月大开始直至处死	Transgenic Tg2576 AD mice, terminated at 4 months 1 wk or 15.5 months 转基因 Tg2576 AD 小鼠, 在 4 个月 1 周或 15.5 个月时处死	The brains of animals treated with melatonin terminated at 15.5 months exhibited dramatic decline in oligomeric Aβ40 together with a significant increase in soluble monomeric Aβ40, and a decreasing trend in Aβ42 compared to untreated mice at same age. Melatonin prolonged survival rates of 15.5 -month mice to levels attained by non-transgenic mice. 41 survivals) compared to untreated (13 deaths/31 survivals). 在 15.5 个月时停止褪黑素治疗的动物大脑显示，与同龄未治疗的小鼠相比，寡聚体 Aβ40 剧烈下降，同时可溶性单体 Aβ40 显著增加，且 Aβ42 呈下降趋势。褪黑素将 15.5 个月大小鼠的存活率延长至非转基因小鼠的水平。与未治疗组（13 例死亡/31 例存活）相比，褪黑素治疗组（41 例存活）。	[361]
0.5mg/mL in drinking water starting at age 4 months 饮用水中的 0.5mg/mL , 从 4 个月大开始	Transgenic Tg2576 AD mice 转基因 Tg2576 AD 小鼠	Striking reductions in Aβ levels in brain tissues of treated mice at 8, 9.5, 11, and 15.5 months. 在 8, 9.5, 11 和 15.5 个月时，治疗组小鼠脑组织中的 Aβ 水平显著降低。	[362]
16μ g/mL in drinking water starting at age 14 months 饮用水中 16μ g/mL 的含量, 从 14 个月大开始	Transgenic Tg2576 AD mice 转基因 Tg2576 AD 小鼠	Melatonin treatment failed to reduce brain Aβ levels or even oxidative damage. 褪黑素治疗未能降低大脑 Aβ 水平，甚至未能减轻氧化损伤。	[363]
40-ppm (w/w) in pelleted minimal basal diet 40-ppm (w/w) 颗粒状最低限度基础饮食	Male B6C3F1 mice aged 6, 12, and 27 months 6、12 和 27 月龄的雄性 B6C3F1 小鼠	tissues: 57% in Aβ40 and 73% in Aβ42; increased melatonin levels in cerebral cortex in all 3 treated age groups (12 > 6 > 27mos) compared to untreated. 组织: Aβ40 中的 57% 和 Aβ42 中的 73% ; 与未治疗组相比, 所有 3 个治疗年龄组 (12 > 6 > 27mos) 的大脑皮层褪黑素水平都有所增加。	[364]
10mg/kg in drinking water from day 7 after tauopathy induction to day 28 at termination 从诱导 tau 蛋白病后的第 7 天到结束时的第 28 天, 在饮用水中加入 10mg/kg	Male wild-type C57BL/6N mice (8 wks old) injected with Aβ1 – 42 peptide 注射了 Aβ β1 – 42 肽段的雄性野生型 C57BL/6N 小鼠 (8 周龄)	Melatonin treatment reversed Aβ1 - 42 - induced synaptic disorder, memory deficit, and prevented Aβ1 – 42-induced apoptosis, neurodegeneration, and tau phosphorylation. Increased ROS and tau hyperphosphorylation starting at day 7 precedes cognitive decline; melatonin-treated animals showed reduced memory impairment, tau hyperphosphorylation,	[365]

		ROS, and neuroinflammation. 褪黑素治疗逆转了 A β 1 - 42 诱导的突触障碍、记忆缺失，并预防了 A β 1 - 42 诱导的细胞凋亡、神经变性和 tau 蛋白磷酸化。从第 7 天开始增加的活性氧（ROS）和 tau 蛋白过度磷酸化先于认知能力下降；经褪黑素治疗的动物表现出记忆受损、tau 蛋白过度磷酸化、ROS 和神经炎症的减轻。	
10 μ mol/L	Ex vivo brain slices from 3-month-old SD rats exposed to okadaic acid to induce tau hyperphosphorylation 来自 3 月龄 SD 大鼠的离体脑切片，经冈田酸处理以诱导 tau 蛋白过度磷酸化	Melatonin reduced tau hyperphosphorylation and ROS to control levels in OA-treated brain slices. 褪黑素将 OA 处理的脑切片中的 tau 蛋白过度磷酸化和 ROS 降低至对照水平。	[366]
100 μ M – 5000 μ M	Aggregation/disaggregation of repeat domain Tau (K18wt) 重复结构域 Tau (K18wt) 的聚合/解聚	Pre-formed tau fibril disaggregation was dose-dependent: 14% with 100 μ M, 54% with 5000 μ M. 预形成的 tau 蛋白纤维解聚呈剂量依赖性：14% 配合 100 μ M, 54% 以及 5000 μ M。	[367]
200 – 5000 μ M	Aggregation/Disaggregation full-length tau (hTau40wt) 聚合/解聚全长 tau 蛋白 (hTau40wt)	Tau treated with 200 μ M melatonin showed no change in morphology compared to controls; 5000 μ M melatonin treatment did not prevent aggregation but disaggregated tau fibrils into broken filaments. 用 200 μ M 褪黑素处理的 Tau 蛋白与对照组相比，在形态上没有变化；5000 μ M 褪黑素处理未能阻止聚集，但使 Tau 蛋白纤维解聚成断裂的细丝。	[368]

Similarly, in vitro studies on tau fibril aggregation and disaggregation in the presence of melatonin at varying strengths found disaggregation effects to be dose-dependent

同样，关于在不同浓度的褪黑素存在下，tau 蛋白纤维聚合与解聚的体外研究发现，其解聚效应具有剂量依赖性 where 100 μ M led to 14% disaggregation while 5000 μ M disaggregated ~ 54% of preformed repeat domain tau [367] (Table 2). However, 200 μ M melatonin treatment in full-length tau aggregates failed to produce morphological changes, and 5000 μ M treatment could not prevent aggregation but was able to disaggregate tau fibrils into small, broken filaments [368] (Table 2).

其中 100 μ M 导致了 14% 的解聚，而 5000 μ M 使 ~ 54% 的预形成重复结构域 tau 蛋白发生了解聚 [367]（表 2）。然而，200 μ M 褪黑素处理全长 tau 蛋白聚集体未能产生形态学变化，且 5000 μ M 处理无法阻止聚集，但能够将 tau 蛋白原纤维解聚成细小的断裂丝状体 [368]（表 2）。

3.3.3. Melatonin Hydrogen Bonding May Modulate Salt Bridge Formation in Aggregates

3.3.3. 褪黑素氢键可能调节聚集体中盐桥的形成

The general consensus on the disaggregation mechanism employed by melatonin is the disruption of salt bridge formation or the reduction of hydrophobic interaction between proteins [358,359] (Table 2). Salt bridges formed between tau proteins can strengthen and stabilize the core of the paired helical filaments which enhances aggregation [367,369]. Both hydrogen bonds and salt bridges provide favorable free energy during protein-protein binding. Therefore, unfulfilled hydrogen bonds or isolated charges without forming salt bridges can destabilize binding due to the desolvation effect [370,371].

关于褪黑素所采用的解聚机制，普遍共识是破坏盐桥的形成或减少蛋白质之间的疏水相互作用 [358,359]（表 2）。tau 蛋白之间形成的盐桥可以增强并稳定双股螺旋丝（paired helical filaments）的核心，从而促进聚集 [367,369]。氢键和盐桥在蛋白质与蛋白质结合过程中都能提供有利的自由能。因此，由于去溶剂化效应，未满足的氢键或未形成盐桥的孤立电荷会降低结合的稳定性 [370,371]。

During A β oligomerization, the prerequisite expulsion of water molecules from protein hydration shells facilitates the formation of salt bridges [372]. In general, weaker hydrogen bonds are formed in interfacial regions due to the restrictive translational and rotational freedom constraints in interfacial regions. Consequently, more water molecules are required in interfacial regions for bridging hydrogen bond networks across protein interfaces [370]. When interacting with melatonin, water can act as both a H -bond donor to the amide carbonyl, methoxy oxygen, or indole π clouds and a H-bond acceptor from the amide NH and indole NH groups [252]. Therefore, the ability to form π hydrogen bonds [373] potentially allows melatonin to prevent the formation of salt bridges that impede intramolecular tau filament aggregation [374]. However, in vitro studies produced results that did not fully support in vivo and ex vivo work on melatonin and tau

hyperphosphorylation [367,368] (Table 2).

在 $A\beta$ 寡聚过程中，蛋白质水合壳层中水分子的预先排出促进了盐桥的形成 [372]。通常，由于界面区域平移和旋转自由度的限制，界面区域形成的氢键较弱。因此，界面区域需要更多的水分子来跨越蛋白质界面桥接氢键网络 [370]。当与褪黑素相互作用时，水既可以作为酰胺羰基、甲氧基氧或吲哚 π 云的氢键供体，也可以作为酰胺 NH 和吲哚 NH 基团的氢键受体 [252]。因此，形成 π 氢键的能力 [373] 可能使褪黑素能够阻止阻碍 tau 蛋白丝分子内聚集的盐桥形成 [374]。然而，体外研究产生的结果并未完全支持关于褪黑素和 tau 蛋白过度磷酸化的体内及离体研究工作 [367,368] (表 2)。

3.3.4. Hyperphosphorylation Reduces Water Hydration during Fibril Aggregation

3.3.4. 过度磷酸化在纤维聚集过程中减少水分合作用

Hyperphosphorylation of tau is a reversible physiological process, but abnormal hyperphosphorylation in neurodegenerative disorders including AD is resistant to dephosphorylation and proteolysis [375-378]. It is believed that the cytotoxicity of $A\beta$ is tau-dependent where tau and $A\beta$ together drive healthy neurons into diseased states and that both $A\beta$ and tau toxicity reinforce each other via a feedback loop [379-382]. The oligomerization of tau fibrils resulting in the formation of pathological tau aggregates is thermodynamically facilitated by hyperphosphorylation of tau proteins [383]. Hydrophobically driven phase separation which leads to the removal of water molecules from protein hydration shells is the predominant interaction that amplifies hydrophobic attractions that cause hyperphosphorylation of tau and fibrillization [135,384]. Recall tau proteins that phase separate due to salting-out effects mature into pathogenic, irreversible, canonical tau fibrils with restricted water accessibility and increased micro-viscosity [135] (see Section 2.3).

Tau 的过度磷酸化是一个可逆的生理过程，但在包括 AD 在内的神经退行性疾病中，异常的过度磷酸化对去磷酸化和蛋白水解具有抵抗性 [375-378]。据信 $A\beta$ 的细胞毒性是 Tau 依赖性的，其中 Tau 和 $A\beta$ 共同驱动健康神经元进入疾病状态，并且 $A\beta$ 和 Tau 的毒性通过反馈回路相互增强 [379-382]。Tau 蛋白的过度磷酸化在热力学上促进了 Tau 原纤维的寡聚化，从而导致病理性 Tau 聚集体的形成 [383]。疏水驱动的相分离导致水分子从蛋白质水合壳层中移除，这是放大疏水吸引力的主要相互作用，进而引起 Tau 的过度磷酸化和原纤维化 [135,384]。回想一下，由于盐析效应而发生相分离的 Tau 蛋白会成熟为致病性的、不可逆的、典型的 Tau 原纤维，其水分可及性受限且微观粘度增加 [135] (见第 2.3 节)。

Furthermore, hyperphosphorylation can generate conformation changes critical for in vitro phase separation of full-length tau which precedes aggregation. Hyperphosphorylation shifts the equilibrium between soluble and phase-separated tau to favor the droplet state, enhancing maturation that initiates pathological aggregation [385]. Consequently, the ability to form hydrogen bonds to maintain protein solubility may determine the level of effectiveness of melatonin treatment in the prevention of tau hyperphosphorylation and subsequent phase separation events that ultimately result in the formation of pathological tau aggregates.

此外，过度磷酸化会诱导构象改变，这对于全长 tau 蛋白在体外发生聚合前的相分离至关重要。过度磷酸化使可溶状态与相分离状态之间的平衡向液滴状态偏移，从而增强了启动病理性聚合的成熟过程 [385]。因此，通过形成氢键来维持蛋白质溶解度的能力，可能决定了褪黑素治疗在预防 tau 蛋白过度磷酸化及随后导致病理性 tau 聚集体形成的相分离事件中的有效程度。

Therefore, in vitro work that showed dose-dependent disaggregation of pre-formed tau fibrils but the inability to prevent aggregation even at high concentrations of $5000\mu\text{M}$ in contrast to in vivo and ex vivo work that reported a significant reduction in tau hyperphosphorylation even after the establishment of tauopathy (Table 2) may simply reflect the absence of ATP that can modulate hydrophobic interactions from hydrogen-bonding

因此，体外研究显示 $5000\mu\text{M}$ 能够呈剂量依赖性地解聚预先形成的 tau 蛋白原纤维，但即使在高浓度下也无法阻止其聚集；与之相反，体内和离体研究则报道称，即使在 tau 病变形成后， $5000\mu\text{M}$ 也能显著降低 tau 蛋白的过度磷酸化 (表 2)。这种差异可能仅仅反映了体外环境中缺乏 ATP，而 ATP 能够调节氢键产生的疏水相互作用

activities. In the context of phase separation in dementia, ATP may be the quintessential lynchpin that brings light, water, and melatonin together in a dynamic and effective synergy. After all, the regulation of aberrant protein aggregation in dementia by light and melatonin is associated with molecular mechanisms including reduced viscosity, hydrogen bonding, protein hydration, and elevation of ATP synthesis (Figure 1).

在痴呆症相分离的背景下，ATP 可能是将光、水和褪黑素结合在一起，形成动态且有效协同作用的核心关键。毕竟，光和褪黑素对痴呆症中异常蛋白质聚集的调节，与包括降低粘度、氢键结合、蛋白质水合作用以及提升 ATP 合成在内的分子机制密切相关 (图 1)。

3.4. Light, Water, and Melatonin: The Adenosine Moiety Effect of ATP

3.4. 光、水与褪黑素：ATP 的腺苷部分效应

The ability of ATP to solubilize hydrophobic substances in aqueous solutions at neutral and elevated pH was first reported by Mandl and Neuberger in 1952 [386]. Several decades later, ATP was observed to behave as a hydrotrope, solubilizing and dissolving protein aggregates in *Xenopus* oocyte nucleoli, preventing the aggregation of synthetic $A\beta_{42}$ peptides, and even dissolving preformed tau fibrils [387,388]. However, employing all-atom molecular dynamics (MD) simulations, Kurisaki et al. observed contradictory results where ATP actually did not have any effect on the dissociation of monomers or the decomposition of the $A\beta_{42}$ oligomer. Instead, the hydrophobic adenosine moiety of ATP was reported to dissociate $A\beta_{42}$ monomers via contacts with $A\beta_{42}$ backbone atoms, potentially dissolving the $A\beta_{42}$ oligomer by shifting thermal equilibrium from an onpathway species to an off-pathway species [389].

ATP 在中性和较高 pH 值的条件下，能够增加疏水性物质在水溶液中溶解度的能力，最早由 Mandl 和 Neuberger 于 1952 年报道 [386]。几十年后，人们观察到 ATP 表现出助溶剂 (hydrotrope) 的特性，能够增溶并溶解非洲爪蟾卵母细胞核仁中的蛋白质聚集体，防止合成的 $A\beta_{42}$ 肽聚集，甚至能溶解预先形成的 tau 蛋白纤维 [387,388]。然而，Kurisaki 等人利用全原子分子动力学 (MD) 模拟观察到了矛盾的结果，即 ATP 实际上对单体的解离或 $A\beta_{42}$ 寡聚体的分解没有任何影响。相反，据报道 ATP 的疏水性腺苷部分通过与 $A\beta_{42}$ 骨架原子的接触使 $A\beta_{42}$ 单体解离，可能通过将热力学平衡从正向路径物种转移到非正向路径物种，从而溶解 $A\beta_{42}$ 寡聚体 [389]。

These observations were further clarified by Mehringer et al. demonstrating via MD simulations that ATP did not exhibit classic features of a hydrotrope or displayed chaotropic salting-in effects. In fact, ATP can be considered a kosmotropic anion with salting-out effects as a result of the triphosphate moiety of ATP capable of lowering the solubility of organic compounds in water. The ability of ATP to prevent and dissolve aggregates formed by phase separation observed in earlier works [387,388] is facilitated mainly by the interaction of the aromatic adenosine moiety in ATP with intrinsically disordered proteins, while the highly charged phosphate moiety served to heighten the solubility of the hydrophobic adenosine in ATP [390]. This molecular mechanism clearly explains why AD transgenic mice exhibit significantly reduced ATP production and mitochondrial dysfunction [391]. The adenosine moiety prevents hydrophobic collapse and aggregation by increasing solubility that prevents water removal [173]. As a consequence, the presence of ATP is highly effective in the suppression of $A\beta_{16-22}$ peptide aggregation [392].

这些观察结果由 Mehringer 等人进一步阐明，他们通过 MD 模拟证明，ATP 并不具备水溶助长剂 (hydrotrope) 的典型特征，也不表现出离液序列 (chaotropic) 的盐溶效应。事实上，由于 ATP 的三磷酸部分能够降低有机化合物在水中的溶解度，ATP 可被视为一种具有盐析效应的嗜序 (kosmotropic) 阴离子。在早期研究 [387,388] 中观察到的 ATP 预防和溶解由相分离形成的聚集体的能力，主要是通过 ATP 中的芳香族腺苷部分与内在无序蛋白质的相互作用实现的，而高电荷的磷酸部分则起到了提高 ATP 中疏水性腺苷溶解度的作用 [390]。这一分子机制清楚地解释了为什么 AD 转基因小鼠表现出 ATP 产量显著降低和线粒体功能障碍 [391]。腺苷部分通过增加溶解度来防止水分流失，从而防止疏水塌陷和聚集 [173]。因此，ATP 的存在对于抑制 $A\beta_{16-22}$ 肽聚集非常有效 [392]。

Adenosine is a primordial metabolite [393] that is an integral component of ATP and RNA [394,395]. Not unexpectedly, both ATP and RNA modulate phase separation biphasically where low concentrations enhance phase separation but high concentrations inhibit droplet formation [387,396-399]. MD simulations demonstrate succinctly that the dissolution of FUS by ATP- Mg^{2+} is promoted by solubilization via the adenine moiety and the phosphate moiety served only to enhance the requisite hydration effect [400].

腺苷是一种原始代谢物 [393]，是 ATP 和 RNA 的重要组成部分 [394,395]。不出所料，ATP 和 RNA 均能双向调节相分离，即低浓度促进相分离，而高浓度则抑制液滴形成 [387,396-399]。分子动力学 (MD) 模拟简洁地证明，ATP- Mg^{2+} 对 FUS 的溶解是通过腺嘌呤部分的增溶作用促进的，而磷酸基团仅起到增强必要的补水效应的作用 [400]。

Mechanistically, the adenosine moiety may prevent amyloid fibril formation by interfering with $A\beta$ peptide $\pi - \pi$ stacking [401,402]. Interestingly, the indole ring of tested indole derivatives effectively inhibited the formation of amyloid fibrils in hen egg-white lysozyme induced by low pH and high temperatures via hydrophobic interactions that accelerated disaggregation and destabilized the amyloid fibrillar state [403,404]. Therefore, it is perhaps not an evolutionary coincidence that melatonin not only exhibits structural homology to the adenosine moiety of ATP [401] (Figure 2), but also binds to adenosine via a hydrogen bond [405-408]. Consequently, ATP and melatonin may have been used for billions of years by living organisms to efficiently regulate phase separation in proteins with a high propensity for aggregation [230,401,409].

从机制上讲，腺苷部分可能通过干扰 $A\beta$ 肽的 $\pi - \pi$ 堆积来防止淀粉样原纤维的形成 [401,402]。有趣的是，所测试的吲哚衍生物的吲哚环通过疏水相互作用，有效地抑制了低 pH 值和高温诱导的鸡蛋清溶菌酶中淀粉样原纤维的形成，这种相互作用加速了解聚并破坏了淀粉样原纤维状态的稳定性 [403,404]。因此，褪黑素不仅在结构上与 ATP 的腺苷部分具有同源性 [401] (图 2)，而且还通过氢键与腺苷结合 [405-408]，这或许并非进化上的巧合。因此，ATP 和褪黑素可能已被生物体使用了数十亿年，以有效地调节具有高聚合倾向蛋白质的相分离 [230,401,409]。

Arguably, the absence of ATP, despite the ability of melatonin to disrupt salt bridge formation, may be the reason why in vitro works on melatonin and tau fibril aggregation reported distinctly different results in the inhibition of fibril formation

that could not confirm in vivo and ex vivo observations even at high concentrations of $5000\mu\text{M}$ (Table 2).

可以说，尽管褪黑素能够破坏盐桥的形成，但 ATP 的缺失可能是体外研究中关于褪黑素与 tau 蛋白纤维聚集报告了截然不同的纤维形成抑制结果的原因，这些结果即使在 $5000\mu\text{M}$ 的高浓度下也无法证实体内和离体观察到的现象（表 2）。

Extracellular adenosine is derived from the degradation of ATP and adenosine monophosphate (AMP), whereas hydrolysis of AMP is the main source of intracellular adenosine [410,411]. It is estimated that extracellular adenosine can rise 1000-fold from the

胞外腺苷来源于 ATP 和单磷酸腺苷 (AMP) 的降解，而 AMP 的水解是胞内腺苷的主要来源 [410,411]。据估计，胞外腺苷可升高 1000 倍，从

low nanomolar range of $\sim 20 - 300\text{nM}$ to the low micromolar range as high as $30\mu\text{M}$ under conditions of high physical stress including extreme exercise and high altitude with low ambient oxygen [412]. Neurodegenerative diseases, inflammatory conditions, autoimmune diseases, cancer, diabetes, and cerebral ischaemia are pathological conditions associated with elevated extracellular adenosine [410,413-417].

在剧烈运动和低环境氧气的高海拔等高物理压力条件下，其浓度范围从 $\sim 20 - 300\text{nM}$ 的低纳摩尔水平到高达 $30\mu\text{M}$ 的低微摩尔水平 [412]。神经退行性疾病、炎症状态、自身免疫性疾病、癌症、糖尿病和脑缺血是与细胞外腺苷升高相关的病理状态 [410,413-417]。

Under optimal conditions, the high reserve/maximum capacity of melatonin synthesis in humans theoretically confers enhanced survival fitness as higher melatonin production allows rapid adaptation to unpredicted internal and external stressors [418]. Assuming that melatonin can be bound to adenosine at a 1 : 4 ratio [406-408], $6 - 20\text{nM}$ plasma adenosine in venous blood collected from normal, healthy subjects [419] can theoretically bind to $1.5 - 5\text{nM}$ of plasma melatonin. However, the lower range of 1.5 nM already reflects the highest 1.13 nM median level detected in nocturnal plasma melatonin concentration in children between the ages of 1 - 3 [420], and melatonin production begins to decline at the early age of 20 - 30 to approximately 0.12 nM after the age of 50 [421-423].

在理想条件下，人类褪黑素合成的高储备/最大容量在理论上赋予了更强的生存适应性，因为较高的褪黑素产量能够快速适应不可预测的内部和外部压力源 [418]。假设褪黑素能以 1 : 4 的比例与腺苷结合 [406-408]，那么从正常健康受试者采集的静脉血中 $6 - 20\text{nM}$ 的血浆腺苷 [419]，理论上可以结合 $1.5 - 5\text{nM}$ 的血浆褪黑素。然而， 1.5 nM 的较低范围已经反映了在 1 - 3 儿童夜间血浆褪黑素浓度中检测到的 1.13 nM 最高中位水平 [420]，且褪黑素产量在 20 - 30 的早期阶段就开始下降，到 50 岁以后降至约 0.12 nM [421-423]。

Furthermore, although there are contrary outcomes in some other reports, there may be conditions where endogenous production of melatonin is suppressed by constant exposure to 60 Hz magnetic field [424] and ambient light at night [425,426]. In addition to binding adenosine, melatonin can significantly elevate ATP production in mitochondria [347,348]. Therefore, the adenosine moiety effect of ATP in phase separation is directly affected by how much melatonin is available, and the dosage of melatonin becomes a critical moving target in the study of phase separation regulation in dementia.

此外，尽管在其他一些报告中存在相反的结果，但在某些条件下，持续暴露于 60 Hz 磁场 [424] 和夜间环境光 [425,426] 可能会抑制内源性褪黑素的产生。除了结合腺苷外，褪黑素还能显著提高线粒体中 ATP 的产量 [347,348]。因此，ATP 在相分离中的腺苷部分效应直接受到褪黑素可用量的影响，而褪黑素的剂量成为痴呆症相分离调节研究中一个关键的动态目标。

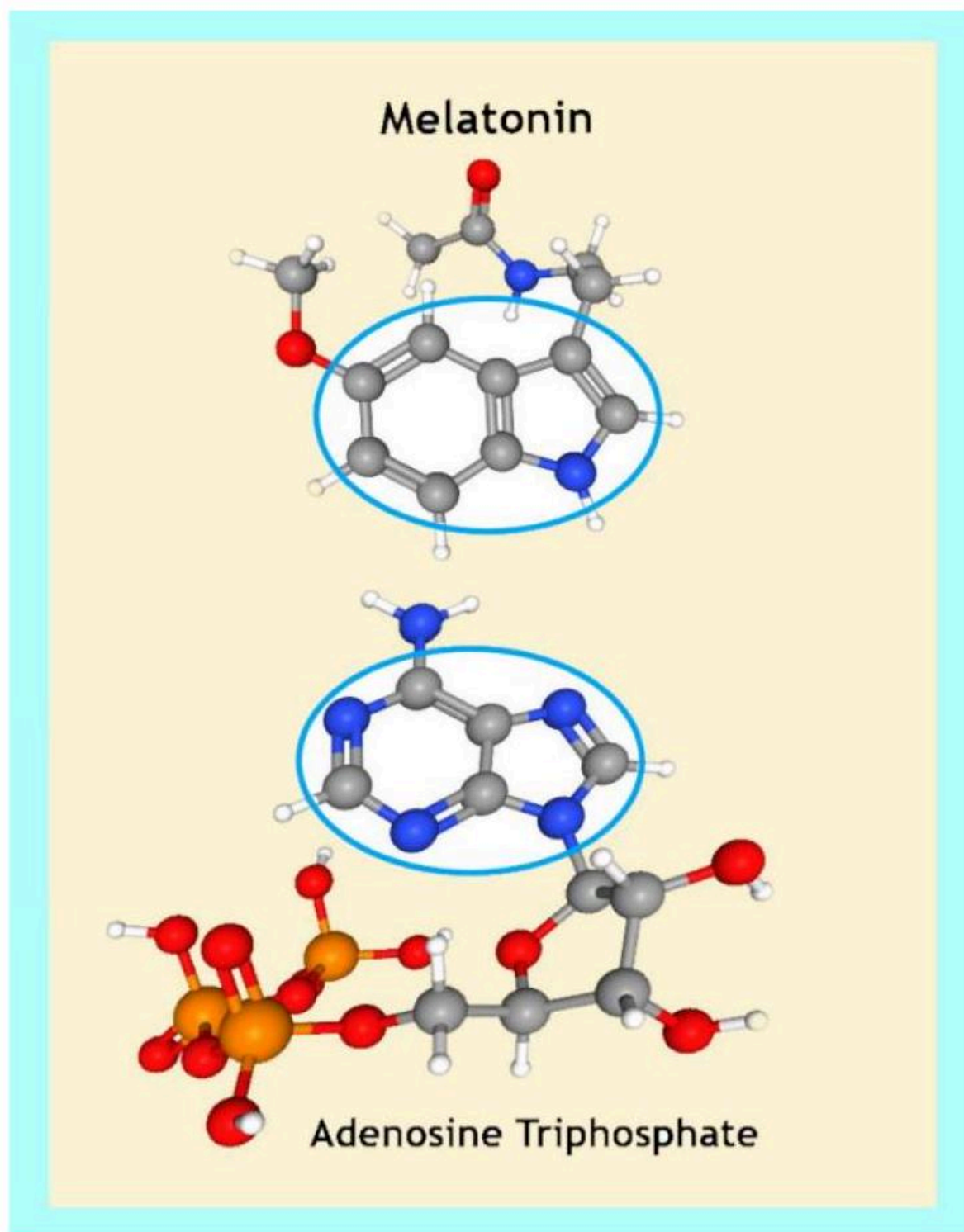


Figure 2: Figure 2. Homologous molecular structures between the electron-rich aromatic indole moiety in the melatonin molecule [427] and the adenosine moiety of ATP [428,429].

图 2：图 2。褪黑素分子中的富电子芳香吲哚部分 [427] 与 ATP 的腺苷部分 [428,429] 之间的同源分子结构。

4. Of Mice and Men: Perfecting the Human Equivalent Dose for Melatonin in the Regulation of Phase Separation in Dementia

4. 《人鼠之间：完善褪黑素在调节痴呆症相分离中的人类等效剂量》

The in vitro and in vivo effects of melatonin in dementia is not only dose-dependent, but may also be time-, and perhaps even age-dependent (Table 2). Three experiments that tested the same strain of transgenic Tg2576 AD mice with 0.016, 0.5, and 2.0mg/mL of melatonin added to the drinking water starting at various ages, produced not only different, but also contradictory results (Table 2). Tg2576 mice are leaner compared to wild types as they age [430-432]. Assuming an average weight of 22.5 g for each animal drinking 3 mL

褪黑素在痴呆症中的体外和体内效应不仅具有剂量依赖性，还可能具有时间依赖性，甚至可能具有年龄依赖性（表 2）。针对同一种转基因 Tg2576 AD 小鼠进行的三项实验，在饮用水中分别添加了 0.016, 0.5 和 2.0mg/mL 的褪黑素，并从不同年龄开始给药，结果不仅不同，甚至相互矛盾（表 2）。随着年龄增长，Tg2576 小鼠比野生型小鼠更瘦 [430-432]。假设每只动物的平均体重为 22.5 g，饮水量为 3 mL

of water per day [361,362], the approximate daily melatonin supplementation would have been 2.13, 66.66, and 266.66mg/kg, respectively.

每天 [361,362] 的饮水量，每日褪黑素的近似补充量分别为 2.13, 66.66 和 266.66mg/kg。

Tg2576 mice receiving $\sim 2.13\text{mg/kg}$ daily starting at age 14 months failed to show any benefit in the reduction in $A\beta$ accumulation in the brain or oxidative stress levels [363]; whereas Tg 2576 mice receiving $\sim 66.66\text{mg/kg}$ daily starting at age 4 months showed a significant reduction in $A\beta$ levels in brain tissues, as well as lowered abnormal nitration of proteins [362]. Importantly, Tg 2576 mice receiving $\sim 266.66\text{mg/kg}$ daily starting at age 4 months produced the most impressive results where the brains of mice terminated at 15.5 months not only exhibited a dramatic decline in oligomeric $A\beta_{40}$, but also a significant increase in soluble monomeric $A\beta_{40}$. A noticeable decreasing trend in $A\beta_{42}$ was observed in treated compared to untreated mice at the same age [361]. When Tg 2576 mice from two separate experiments were administered $\sim 266.66\text{mg/kg}$ melatonin in drinking water daily starting at age 4 months until termination at 15.5 months, survival was significantly increased in treated compared to untreated mice [361,362]. Melatonin treatment at $\sim 266.66\text{mg/kg}$ daily in drinking water was able to reduce mortality in Tg 2576 mice to levels observed in wild-type mice [361] (Table 2). Consequently, the effective translation of melatonin doses between animals and humans becomes the primary consideration when designing the dosage for clinical trials.

从 14 个月大开始每日接受 $\sim 2.13\text{mg/kg}$ 的 Tg2576 小鼠，其大脑中的 $A\beta$ 累积或氧化应激水平并未表现出任何减少的益处 [363]；而从 4 个月大开始每日接受 $\sim 66.66\text{mg/kg}$ 的 Tg 2576 小鼠，其脑组织中的 $A\beta$ 水平显著降低，且蛋白质的异常硝化作用也有所减少 [362]。重要的是，从 4 个月大开始每日接受 $\sim 266.66\text{mg/kg}$ 的 Tg 2576 小鼠产生了最令人印象深刻的结果：在 15.5 个月大时处死的小鼠大脑中，不仅寡聚体 $A\beta_{40}$ 大幅下降，而且可溶性单体 $A\beta_{40}$ 显著增加。与同龄的未处理小鼠相比，处理组小鼠的 $A\beta_{42}$ 呈现出明显的下降趋势 [361]。在两项独立的实验中，当 Tg 2576 小鼠从 4 个月大开始每日通过饮用水摄入 $\sim 266.66\text{mg/kg}$ 褪黑素直至 15.5 个月大处死时，处理组小鼠的存活率较未处理组显著提高 [361,362]。每日通过饮用水进行 $\sim 266.66\text{mg/kg}$ 的褪黑素处理能够将 Tg 2576 小鼠的死亡率降低至与野生型小鼠观察到的水平相当 [361]（表 2）。因此，在设计临床试验剂量时，褪黑素在动物与人类之间的有效剂量转换成为了首要考虑因素。

4.1. Aiming at Moving Targets in Allometric Scaling of Melatonin Interspecies Conversion

4.1. 瞄准褪黑素物种间换算异速生长比例关系中的移动目标

Animals have different metabolic rates. In general, larger animals have lower metabolic rates; therefore, the metabolic rate requires scaling in the conversion of interspecies doses. Allometry broadly describes the study of consequences between body and organ sizes [433,434]. The concept of interspecies allometric scaling was first presented in 1637 by Galileo Galilei [435]. Since that time, various allometric approaches have been proposed and used to determine the most efficacious human equivalent dose (HED) [436-450]. However, the identification of a definitive unified principle that effectively scales and optimizes different energy metabolism systems across animal species remains elusive and highly controversial [441,451-454].

动物具有不同的代谢率。通常情况下，体型较大的动物代谢率较低；因此，在跨物种剂量转换中需要对代谢率进行比例换算。异速生长（Allometry）广泛描述了身体与器官大小之间关系的研究 [433,434]。跨物种异速生长比例换算的概念最早由伽利略·伽利莱（Galileo Galilei）于 1637 年提出 [435]。自那时起，人们提出并使用了各种异速生长方法来确定最有效的人体等效剂量（HED）[436-450]。然而，要确定一个能够有效衡量和优化不同动物物种间能量代谢系统的明确统一原则，目前仍然难以实现且极具争议 [441,451-454]。

In 1880, Rubner first proposed the body surface law that scales metabolic rate with body mass raised to the power of $2/3$ [455]. The seminal work by Kleiber in 1932 led to extensive empirical evidence that supports the metabolic rate in most animals and plants scales to the power of $3/4$ of body mass instead of $2/3$ [454,456]. To date, there is no consensus as to whether $2/3$ or $3/4$ power of body mass should be used as the basal metabolic rate scale in connection with the body mass to determine dose conversion. In general, the exponent of $2/3$ may be more applicable for pharmaceuticals that are eliminated via the kidneys, whereas the exponent of $3/4$ is more suitable for molecules such as melatonin that are cleared by metabolism or via combined metabolism and renal elimination [441,445]. However, once a correction factor (K_m)-a ratio that accounts for the interspecies difference between humans and animals obtained by dividing body weight by body surface area (BSA)

[447,457]-is applied during the conversion of animal to human doses, where

1880 年, Rubner 首次提出了体表面积定律, 该定律认为代谢率与体重的 $2/3$ 次方成比例 [455]。Kleiber 在 1932 年的开创性工作提供了广泛的经验证据, 支持大多数动植物的代谢率与体重的 $3/4$ 次方而非 $2/3$ 次方成比例 [454,456]。迄今为止, 关于在确定剂量换算时应使用体重的 $2/3$ 次方还是 $3/4$ 次方作为基础代谢率标准, 尚未达成共识。通常情况下, $2/3$ 指数可能更适用于通过肾脏消除的药物, 而 $3/4$ 指数则更适用于像褪黑素这样通过代谢或代谢与肾脏联合消除的分子 [441,445]。然而, 一旦在动物剂量向人类剂量换算的过程中应用了修正因子 (K_m)——即通过体重除以体表面积 (BSA) 得出的、用于解释人类与动物之间种属差异的比率 [447,457]——其中

$$\text{HED (mg/kg)} = \text{Animal dose (mg/kg)} \times (\text{animal } K_m / \text{human } K_m)$$

$$\text{HED (mg/kg)} = \text{动物剂量 (mg/kg)} \times (\text{动物 } K_m / \text{人类 } K_m)$$

The results obtained are invariably higher than the $2/3$ metabolic scale exponent of 0.67 (assuming human $K_m = 37.9$).

所得结果总是高于 0.67 的 $2/3$ 代谢标度指数 (假设人类 $K_m = 37.9$)。

For example, to calculate the HED for a 70 kg human, BSA 1.846, K_m 37.9, the metabolic rate exponents for a mouse weighing 0.02 kg, BSA 0.007 (M^2), K_m 2.857, and K_m ratio 13.265; and a rat weighing 0.15 kg, BSA 0.025 (M^2), K_m 6, and K_m ratio 6.317, after adjustments would be 0.683 and 0.700, respectively. For a mini pig weighing 40 kg, BSA 1.14 (M^2), K_m 35.088, and K_m ratio 1.08, the exponent for metabolic scaling for a 70 kg human becomes 0.863.

例如, 计算一个体重 70 kg、BSA 为 1.846, K_m 37.9 的人的 HED, 在对体重 0.02 kg、BSA 为 0.007 (M^2), K_m 2.857、 K_m 比率为 13.265 的小鼠, 以及体重 0.15 kg、BSA 为 0.025 (M^2), K_m 6、 K_m 比率为 6.317 的大鼠进行调整后, 代谢率指数分别为 0.683 和 0.700。对于体重 40 kg、BSA 为 1.14 (M^2)、 K_m 为 35.088、 K_m 比率为 1.08 的迷你猪, 70 kg 人的代谢标度指数变为 0.863。

Thus, the accuracy of HED calculations during interspecies conversion is strictly dependent upon both body weight and BSA. The human BSA can be estimated via the formula $BSA = 1/6 (\text{Weight} \times \text{Height})^{0.5}$ [458], whereas specific animal BSA is often more difficult

因此, 物种间转换过程中 HED 计算的准确性严格取决于体重和体表面积 (BSA)。人类的 BSA 可以通过公式 $BSA = 1/6 (\text{体重} \times \text{身高})^{0.5}$ [458] 来估算, 而特定动物的 BSA 通常更难获得。

to ascertain. Popular manuals containing instructions for interspecies HED conversions rely upon predetermined animal BSAs based upon a mathematical formula ($BSA = k W^{2/3}$) postulated by Meeh in 1879, where BSA is derived from a constant k and volume estimated from body mass that is scaled to the $2/3$ -power. For most small animals including mice, the mean constant k is accepted to be 9.83 [459,460]. However, empirical determination for Meeh constants in mice with different body compositions and shapes revealed a range from 9.822 (normal) to 8.288 (obese) [459]. Therefore, the difference in Meeh constants between measured and calculated values would be important considerations when using mice with altered body composition. Furthermore, modifications for cell porosity per fractal theory indicate that the scaling exponent can vary from 0.694 to 0.83 [450].

难以确定。目前流行的包含物种间 HED 转换指令的手册, 主要依赖于基于 Meeh 在 1879 年提出的数学公式 ($BSA = k W^{2/3}$) 预设的动物 BSA, 其中 BSA 是由常数 k 和根据体重估算的体积 (按 $2/3$ 次方缩放) 推导而来的。对于包括小鼠在内的大多数小动物, 平均常数 k 被公认为 9.83 [459,460]。然而, 对具有不同身体成分和体型的小鼠进行 Meeh 常数的经验测定显示, 其范围从 9.822 (正常) 到 8.288 (肥胖) 不等 [459]。因此, 在使用身体成分发生改变的小鼠时, 测量值与计算值之间 Meeh 常数的差异将是重要的考虑因素。此外, 根据分形理论对细胞多孔性的修正表明, 缩放指数的变化范围可从 0.694 到 0.83 [450]。

As such, the estimated HEDs in this review will be calculated employing metabolic rate scaling using body weight raised to the $3/4$ -power where

因此, 本综述中估算的 HED (人用等效剂量) 将采用代谢率换算方法进行计算, 即使用体重提升至 $3/4$ 次幂, 其中

$$\text{HED (mg/kg)} = \text{animal dose (mg/kg)} \times (\text{WEIGHT [kg]}_{\text{animal}} / \text{WEIGHT [kg]}_{\text{human}})^{(1-0.75)}$$

$$\text{HED (mg/kg)} = \text{动物剂量 (mg/kg)} \times (\text{WEIGHT [kg]}_{\text{animal}} / \text{WEIGHT [kg]}_{\text{human}})^{(1-0.75)}$$

Using this formula, the estimated HED for a rat weighing 0.2 kg with a 10mg/kg melatonin dose will be 2.12mg/kg and 2.4mg/kg for a 100 kg and 60 kg human, respectively. Nevertheless, the often-large differential in interspecies bioavailability and pharmacokinetics that can be modulated by route of administration, dosage, solubility, and formulation must also be taken into account for the accurate determination of an efficacious HED during conversion/scaling processes.

利用该公式，对于体重为 0.2 kg 且褪黑素剂量为 10mg/kg 的大鼠，其估算的人体等效剂量（HED）对于 100 kg 和 60 kg 的成年人分别为 2.12mg/kg 和 2.4mg/kg。尽管如此，在转换/换算过程中，为了准确确定有效的 HED，还必须考虑到物种间生物利用度和药代动力学的巨大差异，这些差异可能会受到给药途径、剂量、溶解度和制剂的影响。

4.2. The Many Faces of Bioavailability in the Interspecies Conversion of Melatonin

4.2. 褪黑素物种间转换中生物利用度的多面性

The bioavailability of a substance is generally accepted as a key pharmacokinetic parameter, expressed as a percentage, that describes the rate and extent the substance becomes available in the general circulation after being delivered from a pharmaceutical form. Absolute bioavailability is a percentage obtained by comparing extravascular administration to intravenous injection (IV) assumed to be 100% available, whereas relative bioavailability compares different routes or formulations without reference to an IV administration [461]. Early work found healthy male subjects who ingested a single 80 mg melatonin (gelatin capsules) displayed varying levels of peak serum melatonin and absorption levels (up to 25 -fold difference) 60 – 150 min after ingestion. This peak level could be extended from ~ 1.5 h to 4 – 6 h when subjects were given one 80 – mg capsule /h over a 3 h period [462]. Melatonin bioavailability in humans, absolute or relative, is delivery-, dose-, solubility-, and formulation-dependent.

物质的生物利用度通常被认为是一个关键的药代动力学参数，以百分比表示，描述了物质从药物剂型中释放后进入全身循环的速率和程度。绝对生物利用度是将血管外给药与假设为 100% 可用的静脉注射（IV）进行比较获得的百分比，而相对生物利用度则是比较不同的给药途径或剂型，而不参考静脉给药 [461]。早期研究发现，摄入单次 80 mg 褪黑素（明胶胶囊）的健康男性受试者在摄入后表现出不同水平的血清褪黑素峰值和吸收水平（差异高达 25 倍）60 – 150 min。当受试者在 3 小时内每隔 /h 服用一粒 80 – mg 胶囊时，该峰值水平可以从 ~ 1.5 h 延长至 4 – 6 h [462]。人体内褪黑素的生物利用度（无论是绝对还是相对）取决于给药方式、剂量、溶解度以及剂型。

4.2.1. Administration Routes Modulate Melatonin Bioavailability

4.2.1. 给药途径调节褪黑素的生物利用度

The bioavailability of melatonin is affected by different routes of administration, where the mean bioavailability for 25 mg of melatonin delivered via intravesical, transdermal, rectal, and vaginal administration in healthy female volunteers were 3.6%, 10.0%, 36.0%, and 97.8%, respectively, compared to IV administration [463]. However, the determination of the bioavailability of oral melatonin may be complicated by melatonin metabolism. In humans, melatonin is mainly cleared by first-pass hepatic metabolism. When the clearance of an IV melatonin dose was combined with plasma concentrations of oral doses from previous data, the calculated oral bioavailability of melatonin was estimated to be 3 – 6% after a 2.5 mg dose, 3 – 76% after an 80 mg dose, but only 9% after a 100 mg dose [464].

褪黑素的生物利用度受不同给药途径的影响。在健康女性志愿者中，通过膀胱内、透皮、直肠和阴道途径给予 25 mg 褪黑素，与静脉注射（IV）相比，其平均生物利用度分别为 3.6%、10.0%、36.0% 和 97.8% [463]。然而，口服褪黑素生物利用度的测定可能会因其代谢过程而变得复杂。在人体中，褪黑素主要通过肝脏首过代谢清除。将静脉注射剂量的清除率与先前数据中口服剂量的血浆浓度相结合，计算得出褪黑素的口服生物利用度在 2.5 mg 剂量后估计为 3 – 6%，在 80 mg 剂量后为 3 – 76%，但在 100 mg 剂量后仅为 9% [464]。

In 2000, DeMuro and coworkers determined the absolute oral bioavailability of 2 and 4 mg melatonin doses (tablets) to be 14.3% ± 7% and 15.9% ± 6%, respectively, compared to IV melatonin (2mg) [465]. Fifteen years later, a systematic review of 22 studies identified from 392 records that tested oral or IV melatonin dosages between 0.3 and 100 mg found the bioavailability of melatonin to be approximately 15% with significant variability between individuals, where critically ill patients often displayed accelerated absorption but compromised elimination [466].

2000 年，DeMuro 及其同事测定，与静脉注射褪黑素（2mg）相比，2 毫克和 4 毫克褪黑素剂量（片剂）的绝对口服生物利用度分别为 14.3% ± 7% 和 15.9% ± 6% [465]。十五年后，一项针对从 392 条记录中筛选出的 22 项研究进行的系统评价（测试的口服或静脉注射褪黑素剂量在 0.3 至 100 毫克之间）发现，褪黑素的生物利用度约为 15%，且个体间存在显著差异，其中危重患者通常表现出吸收加速但消除受阻的情况 [466]。

4.2.2. Melatonin Bioavailability Is Inversely Correlated to 6-Sulfatoxymelatonin

4.2.2. 褪黑素的生物利用度与 6-硫酸氧褪黑素呈负相关

As such, the interpretation of bioavailability may not be straightforward considering the fact that absolute bioavailability of oral melatonin has also been reported at $\sim 3\%$ (10mg gelatin capsule) albeit with considerable variability among the 12 tested healthy male subjects (20 – 40yr old) [467]. Low absolute bioavailability in oral melatonin is often the prominent effect of first-pass hepatic metabolism which produces the major melatonin metabolite 6-sulfatoxymelatonin (6-OHMS). Consequently, low endogenous production of melatonin in the elderly is associated with a significant reduction of 6-OHMS in older test subjects compared to younger ones (82-21 years old) [421]. However, a small study sample found a significant inverse correlation between oral bioavailability and 6-OHMS, where lower oral bioavailability (10%, 12%) was correlated with high plasma 6-OHMS (31%, 14%). Conversely, high bioavailability (56%, 54%) was associated with lower 6-OHMS in plasma (4%, 3%) of healthy male subjects (21 to 32 years old) tested [468].

因此，考虑到口服褪黑素的绝对生物利用度也有报道为 $\sim 3\%$ (10mg 明胶胶囊)，尽管在 12 名受试健康男性 (20 – 40yr 岁) 中存在相当大的差异 [467]，对生物利用度的解读可能并不简单。口服褪黑素的低绝对生物利用度通常是首过肝脏代谢的显著结果，首过代谢产生了主要的褪黑素代谢物 6-硫氧基褪黑素 (6-OHMS)。因此，与较年轻的受试者 (82-21 岁) 相比，老年人内源性褪黑素产量的降低与 6-OHMS 的显著减少有关 [421]。然而，一项小样本研究发现口服生物利用度与 6-OHMS 之间存在显著的负相关，其中较低的口服生物利用度 (10%, 12%) 与高血浆 6-OHMS (31%, 14%) 相关。相反，在接受测试的健康男性受试者 (21 至 32 岁) 中，高生物利用度 (56%, 54%) 与较低的血浆 6-OHMS (4%, 3%) 相关 [468]。

This inverse relationship was also observed in children admitted to an intensive care unit where septic patients who did not survive exhibited nocturnal melatonin levels that were significantly higher than survivors, but total 6-OHMS excretion was dramatically lower in nonsurvivors compared to survivors. Additionally, septic shock patients had higher nocturnal melatonin levels than non-septic patients [469]. Low plasma 6-OHMS is correlated with autism [470,471], and low 6-OHMS excretion level is associated with adults who were lean at birth but obese in adult life, and high excretion rates were associated with opposite observations [472]. Similarly, patients with unstable angina exhibited significantly lower 6-OHMS than healthy controls and no negative correlation with age was observed in coronary patients as opposed to healthy subjects [473]. Therefore, the interpretation of melatonin bioavailability becomes more meaningful when 6-OHMS levels are taken into consideration.

这种反向关系在入住重症监护室的儿童中也有所观察到：未存活的脓毒症患者其夜间褪黑素水平显著高于存活者，但未存活者的总 6-OHMS 排泄量却显著低于存活者。此外，脓毒性休克患者的夜间褪黑素水平高于非脓毒症患者 [469]。低血浆 6-OHMS 与自闭症相关 [470,471]，低 6-OHMS 排泄水平与出生时消瘦但成年后肥胖的成年人相关，而高排泄率则与相反的情况相关 [472]。同样，不稳定型心绞痛患者的 6-OHMS 显著低于健康对照组，且与健康受试者不同，冠心病患者未观察到 6-OHMS 与年龄的负相关性 [473]。因此，当考虑到 6-OHMS 水平时，对褪黑素生物利用度的解读会变得更更有意义。

4.2.3. Animals Show Large Variations in Melatonin Bioavailability

4.2.3. 动物在褪黑素生物利用度上表现出巨大差异

In animals, melatonin bioavailability via different administration routes varies greatly with strain, species, and first pass metabolism after administration. Yeleswaram and coworkers determined the absolute bioavailability of melatonin for a 10mg/kg oral dose compared to IV in male Sprague Dawley (SD) rats to be 53.5%, but more than 100% in dogs and monkeys. However, the oral bioavailability in dogs is dose-dependent, where 1mg/kg resulted in only 16.9% bioavailability. IP injection of melatonin at 10mg/kg in SD rats increased bioavailability to 74.0% compared to oral at 53.5%.

在动物体内，褪黑素经由不同给药途径的生物利用度随品系、物种以及给药后的首过代谢而有很大差异。Yeleswaram 及其同事测定，与静脉注射 (IV) 相比，雄性 Sprague Dawley (SD) 大鼠口服 10mg/kg 褪黑素的绝对生物利用度为 53.5%，但在狗和猴子中超过了 100%。然而，狗的口服生物利用度具有剂量依赖性，其中 1mg/kg 仅导致 16.9% 的生物利用度。在 SD 大鼠中，腹腔注射 (IP) 10mg/kg 褪黑素的生物利用度提高至 74.0%，而口服 53.5% 的生物利用度较低。

In rats, IV administration at half the dose (5mg/kg) achieved 80% bioavailability via IP at 10mg/kg [474]. Rats, regardless of strain and administration, metabolize melatonin completely. SD rats excreted 60 – 70% of radiolabeled melatonin via IV injection as the major metabolite 6-OHMS [475]; and male Wistar rats administered 12.5 – 250 μ g melatonin via IP injection also showed concentration-dependent increases in plasma of melatonin and 6-hydroxymelatonin, which always maintained a constant ratio of 1% of plasma melatonin irrespective of dosage. However, the sulfate conjugate 6-OHMS reached at maximum, ~ 64 -fold elevation of maximum plasma 6-hydroxymelatonin levels

[476].

在大家鼠中，以一半剂量（5mg/kg）进行静脉注射（IV），在10mg/kg时通过腹腔注射（IP）达到了80%的生物利用度[474]。无论品系和给药方式如何，大鼠都能完全代谢褪黑素。SD大鼠通过静脉注射放射性标记的褪黑素后，其主要代谢物6-OHMS排泄量为60—70%[475]；通过腹腔注射给予12.5—250 μ g褪黑素的雄性Wistar大鼠，其血浆中褪黑素和6-羟基褪黑素的浓度也呈剂量依赖性增加，且无论剂量大小，两者始终保持血浆褪黑素1%的恒定比例。然而，硫酸盐结合物6-OHMS达到峰值时，比血浆6-羟基褪黑素的最高水平高出 $\sim$ 64倍[476]。

Similar to rats, female C57BL/6 mice (age 8-10 weeks) administered varying doses of melatonin at 31.25, 62.5, 125, 250, and 500mg/kg showed no difference in the ability to clear and eliminate melatonin; and the concentration of melatonin in the liver and gastrointestinal tracts was higher than other vital organs by 5- to 10-fold, indicating that hepatic first-pass metabolism is also prominent in mice. However, the effect of melatonin in mice is also dose-dependent even at supra-pharmacological concentrations. After exposure to lethal radiation, mice administered 500mg/kg had the highest survival rate (55%) compared to 250 and 125mg/kg (40%) [477].

与小鼠相似，在31.25, 62.5, 125, 250和500mg/kg时间点给予不同剂量褪黑素的雌性C57BL/6小鼠（8-10周龄）在清除和消除褪黑素的能力上没有差异；且肝脏和胃肠道中的褪黑素浓度比其他重要器官高出5到10倍，这表明肝脏首过代谢在小鼠中也十分显著。然而，即使在超药理浓度下，褪黑素在小鼠中的作用也具有剂量依赖性。在遭受致死剂量辐射后，给予500mg/kg的小鼠存活率最高（55%），优于250和125mg/kg (40%)组[477]。

4.2.4. Solubility and Formulation Modulate Melatonin Bioavailability

4.2.4. 溶解度和剂型调节褪黑素的生物利用度

The oral bioavailability of melatonin, at any dose, can be modulated by altering solubility. The oral bioavailability of melatonin in critically ill patients with sepsis was

褪黑素在任何剂量下的口服生物利用度都可以通过改变溶解度来调节。在患有脓毒症的危重患者中，褪黑素的口服生物利用度为

greatly enhanced by the use of solvents, where melatonin dissolved in glycerol achieved a 5-fold increase in relative bioavailability over melatonin in capsules at the same doses (20 or 50 mg) [478]. Similarly, when compared to IV solution (62.5mg/kg dissolved in water), absolute oral bioavailability in mice of an aqueous melatonin suspension at 250mg/kg administered via gavage tube was 29%, whereas 250mg/kg melatonin dissolved in a popular co-solvent polyethylene glycol 400 (PEG400) and administered in the same manner achieved absolute bioavailability of 98.5% [477]. However, PEGs are very hydrophilic molecular crowders that can amplify entropy gain from water-release, causing dehydration that drives phase separation [136]. Hence, the use of PEG as a solvent in applications for the regulation of phase separation must be carefully weighed.

通过使用溶剂，其效果得到了极大增强，其中溶解在甘油中的褪黑素在相同剂量（20或50mg）下，其相对生物利用度比胶囊中的褪黑素提高了5倍[478]。同样，与静脉注射溶液（溶解在水中的62.5mg/kg）相比，通过灌胃管给予的小鼠水性褪黑素悬浮液（剂量为250mg/kg）的绝对口服生物利用度为29%，而溶解在常用助溶剂聚乙二醇400 (PEG400)中并以相同方式给予的250mg/kg褪黑素，其绝对生物利用度达到了98.5%[477]。然而，PEG是极具亲水性的分子拥挤剂，会放大释放水带来的熵增，从而导致驱动相分离的脱水现象[136]。因此，在调节相分离的应用中使用PEG作为溶剂必须经过仔细权衡。

Variations in formulation also affect melatonin bioavailability. In rabbits, intranasal delivery of melatonin encapsulated in starch microspheres achieved absolute bioavailability of 84.07%, whereas intranasal administration via solution produced much lower pharmacokinetics [479]. While intranasal melatonin administration in male Wistar rats via niosomes-bilayer vesicles of nonionic surfactant-based liposomes-achieved absolute bioavailability of 98.7% compared to IV melatonin solution [480]. Therefore, the successful conversion of an animal melatonin dose into an efficacious human equivalent requires equal considerations of metabolic rate scaling, bioavailability as determined by intrinsic differences between species, administration route, as well as solubility and formulation.

制剂配方的差异也会影响褪黑素的生物利用度。在兔子实验中，通过淀粉微球包裹的褪黑素经鼻给药实现了84.07%的绝对生物利用度，而通过溶液经鼻给药产生的药代动力学数值则低得多[479]。而在雄性Wistar大鼠中，通过类脂质体（niosomes，一种基于非离子表面活性剂的双层囊泡脂质体）经鼻给药褪黑素，与静脉注射（IV）褪黑素溶液相比，实现了98.7%的绝对生物利用度[480]。因此，要将动物褪黑素剂量成功转化为有效的人类等效剂量，需要综合考虑代谢率换算、由物种间内在差异决定的生物利用度、给药途径以及溶解度和制剂配方。

4.3. Timing Is Everything in the Dosing of Melatonin for the Regulation of Phase Separation in Dementia

4.3. 褪黑素调节痴呆症中相分离的给药时机至关重要

The daily supra-pharmacological dose of 266.66mg/kg administered in drinking water to transgenic Tg2576 mice from 4 months to 15.5 months not only prevented aggregation of amyloid fibrils but also prolonged survival compared to untreated mice [361,362]. This dose can be converted into a HED using metabolic scaling to the $\frac{3}{4}$ -power, with the assumption of mice and human body weight to be 0.0225 kg and 70 kg, respectively; and oral bioavailability of mice and humans to be 63.75% and 15%, respectively. 63.75% oral bioavailability is a conservative estimation of a 50% enhancement of solubility in water achieved by first dissolving melatonin in hydroxy methyl cyclodextrin before dilution in drinking water to the final concentration of 2mg/mL [361]. Cyclodextrins (CDs) are small carbohydrates that enhance the solubility of molecules and drugs, resulting in higher bioavailability [481,482]. The HED obtained before the bioavailability adjustment is 2499 mg. Without solubility enhancement, the adjusted bioavailability HED dose is 4831 mg. After correcting for a 50% enhancement in bioavailability (calculated based on oral bioavailability data obtained by Choudhary et al. [477]), the final HED is a staggering dose of 10,621mg.

从 4 个月到 15.5 个月，在饮用水中给予转基因 Tg2576 小鼠每日超药理剂量的 266.66mg/kg，不仅防止了淀粉样原纤维的聚集，而且与未治疗的小鼠相比延长了存活时间 [361,362]。假设小鼠和人类的体重分别为 0.0225 kg 和 70 kg，小鼠和人类的口服生物利用度分别为 63.75% 和 15%，利用代谢比例 $\frac{3}{4}$ 次方，可以将该剂量转换为人类等效剂量（HED）。63.75% 的口服生物利用度是一个保守估计，这是基于先将褪黑素溶解在羟甲基环糊精中，然后稀释在饮用水中至最终浓度 2mg/mL，从而使水溶性提高了 50% [361]。环糊精（CDs）是能增强分子和药物溶解度的小分子碳水化合物，从而导致更高的生物利用度 [481,482]。在进行生物利用度调整前获得的 HED 为 2499 mg。在不增强溶解度的情况下，调整后的生物利用度 HED 剂量为 4831 mg。在对生物利用度提高 50% 进行修正后（根据 Choudhary 等人 [477] 获得的口服生物利用度数据计算），最终的 HED 是一个惊人的剂量：10,621mg。

Even though the toxicity of melatonin as defined by LD₅₀ has not been determined in human or rodents, where early studies failed to produce death in mice at 800mg/kg [483], and acute oral toxicity that result in LD₅₀ in rats is reported at concentrations higher than 3200mg/kg (in one single dose [484]) according to the latest Merck safety data sheet on melatonin (Regulation (EC) No. 1907/2006, revised 17 November 2021), without a convincing rationale for a high HED in the context of phase separation in dementia, this extreme supra-pharmacological HED may seem unjustified.

尽管褪黑素在 LD₅₀ 定义下的毒性尚未在人类或啮齿动物中确定，早期研究未能使小鼠在 800mg/kg [483] 剂量下死亡，且根据最新的默克褪黑素安全数据表（法规 (EC) No. 1907/2006，2021 年 11 月 17 日修订），导致大鼠 LD₅₀ 的急性经口毒性报告浓度高于 3200mg/kg（单次给药 [484]），但在痴呆症相分离的背景下，如果没有令人信服的高 HED 合理性依据，这种极端的超药理学 HED 似乎是不合理的。

4.3.1. The Rationale for Frequent Division of Melatonin Doses

4.3.1. 褪黑素频繁分次给药的依据

The Tg 2576 mice drank ~ 3 mL of water containing a total of 6 mg melatonin in a 24 h period. Accordingly, total HED should also be administered in divided doses of 885mg × 12. This hypothetical HED now resembles the HED used by Martin et al. to elevate ATP production via complex I and COX (complex IV) activities in the brain and liver mitochondria of rats [348].

Tg 2576 小鼠在 24 小时内饮用了含有总计 6 mg 褪黑素的 ~ 3 mL 水。因此，总 HED 也应分次服用，剂量为 885mg × 12。这一假设的 HED 现在类似于 Martin 等人用于提高大鼠脑和肝脏线粒体中复合物 I 和 COX（复合物 IV）活性的 HED [348]。

Male Wistar rats with a body weight between 200-230 g were administered 10mg/kg melatonin via IP injection. Respiratory complex activity enhancements were tissue- and time-dependent. Complex I activities in the liver achieved peak levels and returned close

体重在 200-230 g 之间的雄性 Wistar 大鼠通过腹腔注射（IP injection）给予 10mg/kg 褪黑素。呼吸链复合物活性的增强具有组织和时间依赖性。肝脏中的复合物 I 活性达到峰值后恢复至接近

to baseline at ~ 30 and ~ 180 min, respectively; whereas in the brain, peak activity levels were attained at ~ 60 min and returned to baseline at ~ 180 min. COX activities in both the liver and brain became significantly elevated at ~ 30 min, but reached a peak in the liver at ~ 100 min before declining close to baseline at 180 min, whereas brain activities quickly dropped to baseline soon after 120 min [348]. In other words, in the brain, complex I and COX reached peak activity levels at 60 and 30 min, respectively, before returning to baseline at ~ 120 min. Whereas in the liver, complex I and COX activities were both elevated at ~ 30. Complex I steadily declined to close to baseline at ~ 180 min, but COX activities

remained elevated and reached a peak at ~ 100 min before declining close to baseline at ~ 180 min.

分别在 ~ 30 和 ~ 180 min 时恢复至基线水平；而在大脑中，峰值活动水平出现在 ~ 60 min，并在 ~ 180 min 时恢复至基线。肝脏和大脑中的 COX 活性在 ~ 30 min 时均显著升高，但肝脏中的活性在 ~ 100 min 时达到峰值，随后在 180 min 时下降至接近基线，而大脑中的活性在 120 min 后很快降至基线 [348]。换句话说，在大脑中，复合物 I 和 COX 分别在 60 和 30 min 时达到峰值活动水平，随后在 ~ 120 min 时恢复至基线。而在肝脏中，复合物 I 和 COX 活性在 ~ 30 时均有所升高。复合物 I 在 ~ 180 min 时稳步下降至接近基线，但 COX 活性保持升高并在 ~ 100 min 时达到峰值，随后在 ~ 180 min 时下降至接近基线。

The difference in peak and duration of respiratory enzyme activities may reflect the effect of prominent first-pass hepatic metabolism where more melatonin is retained in the liver and gastrointestinal tracts than other vital organs such as the brain [475,477]. Regardless, bioavailability via IP in rats is ~ 74%, or 4.933 -fold higher than oral bioavailability in humans. Therefore, assuming an average body weight of 0.215 kg and 70 kg for rats and humans, respectively, the total daily HED that can effectively maintain peak complex I and COX activity at a sustained level throughout a 24 h period in order to provide adequate ATP and adenosine that can prevent and solubilize aberrant phase separation and aggregation is 9755 mg, adjusted for differences in metabolic rate and bioavailability, assuming average intake of $812.90\text{mg} \times 12$ in a 24 h period. However, the amount quickly doubles to 19, 509.60mg if maximum complex I and COX activities were to be sustained in the brain over a 24 h period based on observations reported by Martin et al. [348].

呼吸酶活性的峰值和持续时间差异可能反映了显著的肝脏首过代谢效应，即与其他重要器官（如大脑）相比，更多的褪黑素被保留在肝脏和胃肠道中 [475,477]。无论如何，大鼠通过腹腔注射（IP）的生物利用度是 ~ 74%，比人类的口服生物利用度高出 4.933 倍。因此，假设大鼠和人类的平均体重分别为 0.215 kg 和 70 kg，为了在 24 小时内将复合体 I 和 COX 活性有效维持在持续的峰值水平，从而提供充足的 ATP 和腺苷以预防并溶解异常的分相和聚集，在调整代谢率和生物利用度差异后，每日总人类等效剂量（HED）为 9755 mg（假设 24 小时内的平均摄入量为 $812.90\text{mg} \times 12$ ）。然而，根据 Martin 等人 [348] 报告的观察结果，如果要在 24 小时内在大脑中维持最大的复合体 I 和 COX 活性，该剂量将迅速翻倍至 19, 509.60mg。

At this point, the estimated daily total HED of 10, 621mg obtained from Tg 2576 mice taking 266.6mg/kg in drinking water becomes quite reasonable and theoretically justifiable. In addition, in vitro work found a strong correlation between ATP and melatonin concentration for disaggregation of fibrils [367,388], where 1 mM of ATP and melatonin both were able to dissolve 20% of aggregates, respectively; 4 mM and 5 mM of ATP and melatonin dissolved 50% and ~ 60% of aggregates, respectively (Figure 3). Therefore, the rationale supporting supra-pharmacological oral melatonin doses to maintain elevated ATP synthesis that prevents aberrant phase separation and aggregation warrants further investigation.

此时，从饮用含有 266.6mg/kg 的水的 Tg 2576 小鼠中获得的每日总 HED 估计值 10, 621mg 变得相当合理，且在理论上是成立的。此外，体外研究发现 ATP 与褪黑素浓度在解聚纤维方面存在强相关性 [367,388]，其中 1 mM 的 ATP 和褪黑素分别能够溶解 20% 的聚集体；4 mM 和 5 mM 的 ATP 和褪黑素则分别溶解了 50% 和 ~ 60% 的聚集体（图 3）。因此，支持使用超药理剂量的口服褪黑素以维持升高的 ATP 合成，从而防止异常相分离和聚集的理论依据值得进一步研究。

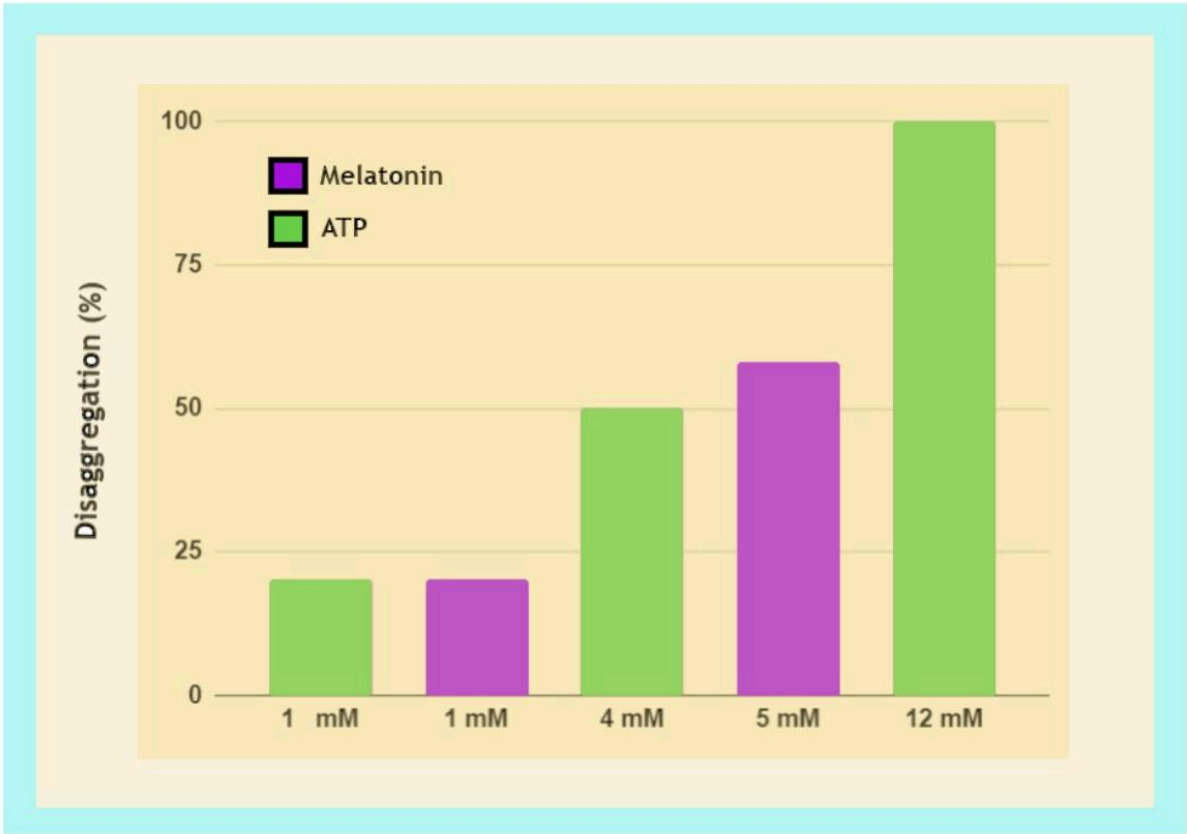


Figure 3: Figure 3. Comparison of disaggregation percentages obtained when different concentrations of melatonin and ATP were added to mediums containing repeat tau aggregates [367] and egg white aggregates [388], respectively.

图 3: 图 3. 在分别含有重复 tau 蛋白聚集体 [367] 和蛋清聚集体 [388] 的培养基中, 添加不同浓度的褪黑素和 ATP 后获得的解聚百分比比较。

4.3.2. The Calculation of HED Estimates Adjusted for Differences in Metabolic Rates, Bioavailability, and Formulation

4.3.2. 针对代谢率、生物利用度和剂型差异进行调整后的 HED 估算值计算

A close examination and comparison of various HED estimates obtained from the different in vivo experiments discussed in Table 2 may provide clarification on melatonin doses required for the effective regulation of phase separation in dementia. Importantly, there is a difference in doses required to obtain similar results in healthy versus diseased, transgenic animal models.

对表 2 中讨论的不同体内实验所获得的各种 HED (人用等效剂量) 估算值进行仔细检查和比较, 可能有助于阐明有效调节痴呆症中相分离所需的褪黑素剂量。重要的是, 在健康动物模型与患病的转基因动物模型中, 获得相似结果所需的剂量存在差异。

Table 3 illustrates how (A) oral melatonin HED for a human weighing 70 kg is converted from animal doses by using metabolic rate scaled to $\frac{3}{4}$ power with body weight ($M_b^{3/4}$); (B) where HED (A) is further adjusted by interspecies bioavailability difference that takes into account both species differentials and administration routes; and © adjusts (A) to reflect enhancements via solubility/formulation as per study design. In the absence of data, where applicable, the average body weight of transgenic Tg 2576 and wild-type mice is assumed to be 0.0225 kg and 0.025 kg, respectively. Daily food intake for mice [364] is assumed to be ~ 4.5 g [432]. The oral bioavailability of melatonin in humans and mice is assumed to be 15% [465,466] and 29% [477], respectively, in the conversion for values in column (B). Bioavailability enhancement via increased solubility is estimated at a conservative 50% increase based on data reported by Choudhary and coworkers [477]. Therefore, values in column © are obtained by multiplying (A) by 4.25.

表 3 说明了: (A) 如何根据与体重 ($M_b^{3/4}$) 成 $\frac{3}{4}$ 次幂比例的代谢率, 将 70 kg 成人的口服褪黑素 HED (人类等效剂量) 从动物剂量转换而来; (B) 在 HED (A) 的基础上, 进一步根据物种间生物利用度差异进行调整, 该差异综合考虑了物种差异和给药途径; 以及 (C) 根据研究设计, 对 (A) 进行调整以反映通过溶解度/剂型改良所实现的增强效果。在缺乏数据的情况下 (如适用), 转基因 Tg 2576 小鼠和野生型小鼠的平均体重分别假定为 0.0225 kg 和 0.025 kg。小鼠的每日摄食量 [364] 假定为 ~ 4.5 g [432]。在计算栏 (B) 的数值时, 人类和小鼠的褪黑素口服生物利用度分别假定为 15% [465,466] 和 29% [477]。根据 Choudhary 及其同事 [477] 报告的数据, 通过提高溶解度实现的生物利用度增强保守估计为增加了 50%。因此, 栏 (C) 的数值是通过将 (A) 乘以 4.25 得出的。

Table 3. Calculations of three HEDs converted from animal doses using different adjustments that account for differences in (A) Metabolic rates by scaling to the $\frac{3}{4}$ -power; (B) Bioavailability; © Bioavailability that is enhanced by solubility and/or formulation.

表 3. 使用不同调整方法将动物剂量转换为三种 HED 的计算结果, 这些调整考虑了以下方面的差异: (A) 通过按 $\frac{3}{4}$ 次方缩放得出的代谢率; (B) 生物利用度; (C) 通过溶解度和/或制剂增强的生物利用度。

Study Design/Total Daily Dose/Duration/Ref.	Results 结果	(A) HED Daily Total (mg/kg) Scaled to $M_b^{3/4}$ (A) HED 每日总量 (mg/kg) 已缩放至 $M_b^{3/4}$	(B) Dose (A) Adjusted by Bioavailability (B) 剂量 (A) 经生物利用度调整	(C) Dose (A) Adjusted by Enhanced Bioavailability (C) 剂量 (A) 经增强生物利用度调整
2mg/mL in drinking water, Tg2576 AD mice/266.66 mg/kg/11.5 mos starting at 4 mos old/[361,362] 2mg/mL 饮水水中, Tg2576 AD 小鼠/266.66 mg/kg/11.5 个月, 从 4 月龄开始/[361,362]	Striking reductions in $A\beta$ aggregates at all ages during treatment; dramatic extension of survival of AD mice to levels similar to wild types. 治疗期间, 各年龄段的 $A\beta$ 聚集 体均显著减少; 阿尔茨海默病 (AD) 小鼠的生存期大幅延 长, 达到与野生型小鼠相似的水 平。	2499mg(35.7mg/kg)	4831mg(69mg/kg)	10, 621mg(151.73mg/kg)
0.5mg/mL in drinking water, Tg2576 AD mice/66.66 饮水水中的 0.5mg/mL , Tg2576 AD 小鼠/66.66	Striking reductions in $A\beta$ levels in brain tissues of treated mice at 8, 9.5, 11, and 15.5 months. 在 8, 9.5, 11 和 15.5 个月时, 治疗组小鼠脑组织中的 $A\beta$ 水平 显著降低。	625mg(8.928mg/kg)	1208mg(17.26mg/kg)	2656mg(37.94mg/kg)
mice/ 2.13mg/kg/10 wks starting at age 14 mos old/[363] 小鼠/ 2.13mg/kg/10 周, 从 14 个月大开始/[363]	Failed to reduce brain $A\beta$ levels, unable to reverse oxidative damage. 未能降低大脑 $A\beta$ 水平, 无法逆 转氧化损伤。	19.96mg(0.285mg/kg)	38.58mg(0.55mg/kg)	84.83mg(1.21mg/kg)
10mg/kg in drinking water, healthy, normal C57BL/6J mice/14 days after tauopathy initiation/[366] 饮水水中 10mg/kg , 健康、正 常的 C57BL/6J 小鼠/tau 病变诱 导后 14 天/[366]	Reduced memory impairment, tau hyperphosphorylation, ROS, and neuroinflammation. 减轻了记忆损伤、tau 蛋白过度 磷酸化、活性氧 (ROS) 以及神 经炎症。	96.23mg(1.375mg/kg)	186.0mg(2.66mg/kg)	408.98mg(5.84mg/kg)
40 ppm in food pellets, healthy, normal B6C3F1 mice/7.2 mg/kg/11 weeks different age groups/[364] 40 ppm 饲料浓度, 健康、正 常的 B6C3F1 小鼠/7.2 mg/kg/11 周龄不同年龄 组/[364]		69.29mg(0.99mg/kg)	133.94mg(1.91mg/kg)	Not applicable 不适用
10mg/kg IP injection, C57BL/6J mice treated with $A\beta_{1-42}$ peptide/daily IP injections for 3 wks/[365] 10mg/kg 腹腔注射, C57BL/6J 小鼠接受 $A\beta_{1-42}$ 多肽治疗/每日腹腔注射, 持续 3 周/[365]	Reversed $A\beta_{1-42}$ -induced synaptic disorder, memory deficit; prevented $A\beta_{1-42}$ -induced apoptosis, neurodegeneration, and tau phosphorylation. 逆转了 $A\beta_{1-42}$ 诱导的突触 功能障碍和记忆缺陷; 预防了 $A\beta_{1-42}$ 诱导的细胞凋亡、神 经退行性变和 tau 蛋白磷酸化。	98.55mg(1.41mg/kg)	486.15mg(6.95mg/kg)	Not applicable 不适用

The selection of a “perfect” HED dose for melatonin under different contexts is ultimately at the sole discretion of the investigator(s) who will determine the “parameters to be scaled, independent variables, and the mathematical relationship used in the scaling process” [438,439]. Therefore, the values presented in Table 3 are intended purely as an informative guide to various potentially effective HEDs for melatonin that can be applied in the regulation of phase separation in dementia under distinct conditions.

被缩放、自变量以及缩放过程中使用的数学关系” [438,439]。因此，表 3 中提供的值纯粹是为了作为参考指南，介绍在不同条件下可应用于调节痴呆症中相分离的各种潜在有效的褪黑素人类等效剂量（HEDs）。

5. Conclusions 5. 结论

Modernization of infrastructure leads to inevitable environmental changes that restrict easy access to the ancient, dynamic synergy between light, water, and melatonin. Individuals who live in densely populated urban areas are affected by the lack of adequate greenness that limits exposure to red and infrared frequencies from sunlight, generously reflected by plants [205]. Furthermore, continuous exposure to low-level microwaves and varying levels of EMF can restructure hydrogen bonding to either decrease or increase intracellular viscosity [195-199]. Even exposure to magnetic fields at 0.5 T causes water molecules to form new hydrogen bonds resulting in larger-sized water clusters that increase viscosity but reduce the

proportion of free water molecules [485]. At the same time, the endogenous production of melatonin may be impacted under some circumstances by constant exposure to 60 Hz magnetic field [424] and ambient light at night [425,426]. In older adults with varying risks for dementia, increased light exposure in the evening results in earlier dim light melatonin onset (DLMO) time. This shift in the circadian phase may disturb rhythmicity that is often associated with dementia [486-488].

基础设施的现代化导致了不可避免的环境变化，限制了人们获取光、水和褪黑素之间古老而动态的协同作用。居住在人口密集的城市地区的人们受到缺乏充足绿化的影响，这限制了他们接触来自阳光的红光和红外线频率，而这些频率会被植物大量反射 [205]。此外，持续暴露于低水平微波和不同水平的电磁场（EMF）会重构氢键，从而降低或增加细胞内粘度 [195-199]。即使暴露在 0.5 T 的磁场中，也会导致水分子形成新的氢键，产生更大尺寸的水团簇，从而增加粘度，但减少了游离水分子的比例 [485]。同时，在某些情况下，持续暴露于 60 Hz 磁场 [424] 和夜间环境光 [425,426] 可能会影响内源性褪黑素的产生。在具有不同痴呆风险的老年人中，傍晚光照强度的增加会导致褪黑素分泌起始（DLMO）时间提前。这种昼夜节律相位的偏移可能会干扰通常与痴呆相关的节律性 [486-488]。

Our brave, new world offers unlimited potential in technological advances in every frontier imaginable but exacts an exorbitant premium on our health by creating intracellular conditions that favor aberrant phase separation resulting in pathological protein aggregations that are associated with a wide range of health challenges, including dementia. The reinstatement of this powerful but lost synergy is a provocative proposal that entails the conditional rescaling of an ancient theme to harmonize with the cacophony of modern influences, restoring, once again, balance in optimum health.

我们这个美丽新世界在每一个想象得到的领域都提供了无限的技术进步潜力，但却以损害我们的健康为代价，索取了过高的溢价。它创造了有利于异常相分离的细胞内环境，导致病理性蛋白质聚集，而这与包括痴呆症在内的广泛健康挑战密切相关。恢复这种强大但已丧失的协同作用是一个具有启发性的提议，它要求对一个古老的主题进行有条件的重新调整，以使其与现代影响的嘈杂声相和谐，从而再次恢复最佳健康状态的平衡。

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Abbreviations 缩略语表

6-OHMS	6-sulfatoxymelatonin 6-羟基褪黑素硫酸酯
A β	amyloid- β β -淀粉样蛋白- β
AD	Alzheimer's disease 阿尔茨海默病
AMP	adenosine monophosphate 腺苷一磷酸
α -syn α - 同步	alpha-synuclein α -突触核蛋白
ATP	adenosine triphosphate 三磷酸腺苷
CP	centipoise 厘泊
COX	cytochrome C oxidase 细胞色素 C 氧化酶
EMF	electromagnetic fields 电磁场
FTD	frontotemporal dementia 额颞叶痴呆
FUS	fused in sarcoma
H ₂ O ₂	hydrogen peroxide 过氧化氢
IDP	intrinsically disordered protein 本质无序蛋白质

IP	intraperitoneal 腹腔内注射
IV	intravenous 静脉注射
LCD	low-complexity domain 低复杂性结构域
MD	molecular dynamics 分子动力学
MLO	membraneless organelle 无膜细胞器
•OH	hydroxyl radical 羟自由基
PD	Parkinson's disease 帕金森病
pI	isoelectric point 等电点
ROS	reactive oxygen species 活性氧
SD	Sprague Dawley
SG	stress granule 应激颗粒
VaD	Vascular dementia 血管性痴呆

References
 参考文献

IP	intraperitoneal
IV	intravenous
LCD	low-complexity domain
MD	molecular dynamics
MLO	membraneless organelle
•OH	hydroxyl radical
PD	Parkinson's disease
pI	isoelectric point
ROS	reactive oxygen species
SD	Sprague Dawley
SG	stress granule
VaD	Vascular dementia

References

1. Qiu, S.; Miller, M.I.; Joshi, P.S.; Lee, J.C.; Xue, C.; Ni, Y.; Wang, Y.; De Anda-Duran, I.; Hwang, P.H.; Cramer, J.A.; et al. Multimodal Deep Learning for Alzheimer's Disease Dementia Assessment. *Nat. Commun.* **2022**, *13*, 3404. [\[CrossRef\]](#)
2. Anstey, K.J.; Peters, R.; Clare, L.; Lautenschlager, N.T.; Dodge, H.H.; Barnes, D.E.; Shahrar, S.; Brodaty, H.; Rees, G. Joining Forces to Prevent Dementia: The International Research Network on Dementia Prevention (IRNDP). *Int. Psychogeriatr.* **2017**, *29*, 1757–1760. [\[CrossRef\]](#)
3. Qiu, C.; Kivipelto, M.; von Strauss, E. Epidemiology of Alzheimer's Disease: Occurrence, Determinants, and Strategies toward Intervention. *Dialogues Clin. Neurosci.* **2009**, *11*, 111–128. [\[CrossRef\]](#)
4. Piscopo, P.; Crestini, A.; Carbone, E.; Rivabene, R.; Ancidoni, A.; Lo Giudice, M.; Corbo, M.; Vanacore, N.; Lacorte, E. A Systematic Review on Drugs for Synaptic Plasticity in the Treatment of Dementia. *Ageing Res. Rev.* **2022**, *81*, 101726. [\[CrossRef\]](#) [\[PubMed\]](#)
5. Velandia, P.P.; Miller-Petrie, M.K.; Chen, C.; Chakrabarti, S.; Chapin, A.; Hay, S.; Tsakalos, G.; Wimo, A.; Dieleman, J.L. Global and Regional Spending on Dementia Care from 2000–2019 and Expected Future Health Spending Scenarios from 2020–2050: An Economic Modelling Exercise. *EClinicalMedicine* **2022**, *45*, 101337. [\[CrossRef\]](#) [\[PubMed\]](#)
6. McKhann, G.M.; Knopman, D.S.; Chertkow, H.; Hyman, B.T.; Jack, C.R.; Kawas, C.H.; Klunk, W.E.; Koroshetz, W.J.; Manly, J.J.; Mayeux, R.; et al. The Diagnosis of Dementia due to Alzheimer's Disease: Recommendations from the National Institute on Aging-Alzheimer's Association Workgroups on Diagnostic Guidelines for Alzheimer's Disease. *Alzheimer's Dement.* **2011**, *7*, 263–269. [\[CrossRef\]](#) [\[PubMed\]](#)
7. Hendrie, H.C. Epidemiology of Dementia and Alzheimer's Disease. *Am. J. Geriatr. Psychiatry* **1998**, *6* (Suppl. 1), S3–S18. [\[CrossRef\]](#)
8. O'Brien, J.T.; Thomas, A. Vascular Dementia. *Lancet* **2015**, *386*, 1698–1706. [\[CrossRef\]](#)
9. Iadecola, C. The Pathobiology of Vascular Dementia. *Neuron* **2013**, *80*, 844–866. [\[CrossRef\]](#)
10. Walker, Z.; Possin, K.L.; Boeve, B.F.; Aarsland, D. Lewy Body Dementias. *Lancet* **2015**, *386*, 1683–1697. [\[CrossRef\]](#)
11. Bang, J.; Spina, S.; Miller, B.L. Frontotemporal Dementia. *Lancet* **2015**, *386*, 1672–1682. [\[CrossRef\]](#)
12. Bolla, L.R.; Filley, C.M.; Palmer, R.M. Dementia DDX. Office Diagnosis of the Four Major Types of Dementia. *Geriatrics* **2000**, *55*, 34–37, 41, 42, 45, 46.
13. Li, M.; Fan, Y.; Li, Q.; Wang, X.; Zhao, L.; Zhu, M. Liquid-Liquid Phase Separation Promotes Protein Aggregation and Its Implications in Ferroptosis in Parkinson's Disease Dementia. *Oxidative Med. Cell. Longev.* **2022**, *2022*, 7165387. [\[CrossRef\]](#)
14. Alberti, S.; Hyman, A.A. Biomolecular Condensates at the Nexus of Cellular Stress, Protein Aggregation Disease and Ageing. *Nat. Rev. Mol. Cell Biol.* **2021**, *22*, 196–213. [\[CrossRef\]](#)
15. Mackenzie, I.R.; Nicholson, A.M.; Sarkar, M.; Messing, J.; Purice, M.D.; Pottier, C.; Annu, K.; Baker, M.; Perkersen, R.B.; Kurti, A.; et al. TIA1 Mutations in Amyotrophic Lateral Sclerosis and Frontotemporal Dementia Promote Phase Separation and Alter Stress Granule Dynamics. *Neuron* **2017**, *95*, 808–816.e9. [\[CrossRef\]](#)
16. Jack, C.R., Jr.; Therneau, T.M.; Lundt, E.S.; Wiste, H.J.; Mielke, M.M.; Knopman, D.S.; Graff-Radford, J.; Lowe, V.J.; Vemuri, P.; Schwarz, C.G.; et al. Long-Term Associations between Amyloid Positron Emission Tomography, Sex, Apolipoprotein E and Incident Dementia and Mortality among Individuals without Dementia: Hazard Ratios and Absolute Risk. *Brain Commun.* **2022**, *4*, fca017. [\[CrossRef\]](#)
17. Gottesman, R.F.; Wu, A.; Coresh, J.; Knopman, D.S.; Jack, C.R., Jr.; Rahmim, A.; Sharrett, A.R.; Spira, A.P.; Wong, D.F.; Wagenknecht, L.E.; et al. Associations of Vascular Risk and Amyloid Burden with Subsequent Dementia. *Ann. Neurol.* **2022**, *92*, 607–619. [\[CrossRef\]](#)
18. Banani, S.F.; Lee, H.O.; Hyman, A.A.; Rosen, M.K. Biomolecular Condensates: Organizers of Cellular Biochemistry. *Nat. Rev. Mol. Cell Biol.* **2017**, *18*, 285–298. [\[CrossRef\]](#) [\[PubMed\]](#)
19. Yoo, H.; Bard, J.A.M.; Pilipenko, E.; Drummond, D.A. Chaperones Directly and Efficiently Disperse Stress-Triggered Biomolecular Condensates. *bioRxiv* **2021**. [\[CrossRef\]](#)
20. Wen, Y.; Ma, J. Phase Separation Drives the Formation of Biomolecular Condensates in the Immune System. *Front. Immunol.* **2022**, *13*, 986589. [\[CrossRef\]](#)
21. Ditlev, J.A.; Case, L.B.; Rosen, M.K. Who's In and Who's Out-Compositional Control of Biomolecular Condensates. *J. Mol. Biol.* **2018**, *430*, 4666–4684. [\[CrossRef\]](#) [\[PubMed\]](#)

22. Yeong, V.; Werth, E.G.; Brown, L.M.; Obermeyer, A.C. Formation of Biomolecular Condensates in Bacteria by Tuning Protein Electrostatics. *ACS Cent. Sci.* **2020**, *6*, 2301–2310. [\[CrossRef\]](#)
23. Sagan, S.M.; Weber, S.C. Let's Phase It: Viruses Are Master Architects of Biomolecular Condensates. *Trends Biochem. Sci.* **2022**, *48*, 229–243. [\[CrossRef\]](#)
24. Geiger, F.; Acker, J.; Papa, G.; Wang, X.; Arter, W.E.; Saar, K.L.; Erkamp, N.A.; Qi, R.; Bravo, J.P.; Strauss, S.; et al. Liquid-Liquid Phase Separation Underpins the Formation of Replication Factories in Rotaviruses. *EMBO J.* **2021**, *40*, e107711. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Ahn, J.H.; Davis, E.S.; Daugird, T.A.; Zhao, S.; Quiroga, I.Y.; Uryu, H.; Li, J.; Storey, A.J.; Tsai, Y.-H.; Keeley, D.P.; et al. Phase Separation Drives Aberrant Chromatin Looping and Cancer Development. *Nature* **2021**, *595*, 591–595. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Zbinden, A.; Pérez-Berlanga, M.; De Rossi, P.; Polymenidou, M. Phase Separation and Neurodegenerative Diseases: A Disturbance in the Force. *Dev. Cell* **2020**, *55*, 45–68. [\[CrossRef\]](#)
27. Elbaum-Garfinkle, S. Matter over Mind: Liquid Phase Separation and Neurodegeneration. *J. Biol. Chem.* **2019**, *294*, 7160–7168. [\[CrossRef\]](#) [\[PubMed\]](#)
28. Boeynaems, S.; Alberti, S.; Fawzi, N.L.; Mittag, T.; Polymenidou, M.; Rousseau, F.; Schymkowitz, J.; Shorter, J.; Wolozin, B.; Van Den Bosch, L.; et al. Protein Phase Separation: A New Phase in Cell Biology. *Trends Cell Biol.* **2018**, *28*, 420–435. [\[CrossRef\]](#)
29. Aguzzi, A.; Altmeppen, M. Phase Separation: Linking Cellular Compartmentalization to Disease. *Trends Cell Biol.* **2016**, *26*, 547–558. [\[CrossRef\]](#) [\[PubMed\]](#)
30. Rowlinson, J.S. Translation of JD van der Waals' "The Thermodynamik Theory of Capillarity under the Hypothesis of a Continuous Variation of Density". *J. Stat. Phys.* **1979**, *20*, 197–200. [\[CrossRef\]](#)
31. Ginzburg, V.V.; Peng, G.; Qiu, F.; Jasnow, D.; Balazs, A.C. Kinetic Model of Phase Separation in Binary Mixtures with Hard Mobile Impurities. *Phys. Rev. E* **1999**, *60 Pt B*, 4352–4359. [\[CrossRef\]](#)
32. Alberti, S.; Saha, S.; Woodruff, J.B.; Franzmann, T.M.; Wang, J.; Hyman, A.A. A User's Guide for Phase Separation Assays with Purified Proteins. *J. Mol. Biol.* **2018**, *430*, 4806–4820. [\[CrossRef\]](#) [\[PubMed\]](#)
33. Himeno, H.; Shimokawa, N.; Komura, S.; Andelman, D.; Hamada, T.; Takagi, M. Charge-Induced Phase Separation in Lipid Membranes. *Soft Matter* **2014**, *10*, 7959–7967. [\[CrossRef\]](#) [\[PubMed\]](#)
34. Laradji, M.; Guo, H.; Grant, M.; Zuckermann, M.J. The Effect of Surfactants on the Dynamics of Phase Separation. *J. Phys. Condens. Matter* **1992**, *4*, 6715. [\[CrossRef\]](#)
35. Choi, J.-M.; Holehouse, A.S.; Pappu, R.V. Physical Principles Underlying the Complex Biology of Intracellular Phase Transitions. *Annu. Rev. Biophys.* **2020**, *49*, 107–133. [\[CrossRef\]](#)
36. De Sancho, D. Phase Separation in Amino Acid Mixtures Is Governed by Composition. *Biophys. J.* **2022**, *121*, 4119–4127. [\[CrossRef\]](#)
37. Choi, J.-M.; Dar, F.; Pappu, R.V. LASSI: A Lattice Model for Simulating Phase Transitions of Multivalent Proteins. *PLoS Comput. Biol.* **2019**, *15*, e1007028. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Gereben, O.; Pusztai, L. Cluster Formation and Percolation in Ethanol-Water Mixtures. *Chem. Phys.* **2017**, *496*, 1–8. [\[CrossRef\]](#)
39. Servis, M.J.; Wu, D.T.; Braley, J.C. Network Analysis and Percolation Transition in Hydrogen Bonded Clusters: Nitric Acid and Water Extracted by Tributyl Phosphate. *Phys. Chem. Chem. Phys.* **2017**, *19*, 11326–11339. [\[CrossRef\]](#)
40. Pothoczki, S.; Pethes, I.; Pusztai, L.; Temleitner, L.; Csókás, D.; Kohara, S.; Ohara, K.; Bakó, I. Hydrogen Bonding and Percolation in Propan-2-Ol—Water Liquid Mixtures: X-Ray Diffraction Experiments and Computer Simulations. *J. Mol. Liq.* **2021**, *329*, 115592. [\[CrossRef\]](#)
41. Alshareedah, I.; Moosa, M.M.; Banerjee, P.R. Programmable Viscoelasticity in Protein-RNA Condensates with Disordered Sticker-Spacer Polypeptides. *bioRxiv* **2021**. [\[CrossRef\]](#)
42. Joseph, J.A.; Reinhardt, A.; Aguirre, A.; Chew, P.Y.; Russell, K.O.; Espinosa, J.R.; Garaizar, A.; Collepardo-Guevara, R. Physics-Driven Coarse-Grained Model for Biomolecular Phase Separation with near-Quantitative Accuracy. *Nat. Comput. Sci.* **2021**, *1*, 732–743. [\[CrossRef\]](#) [\[PubMed\]](#)
43. Deniz, A.A. Percolation Physics and Density Transition Frameworks Converge in Biomolecular Condensation. *Proc. Natl. Acad. Sci. USA* **2022**, *119*, e2210177119. [\[CrossRef\]](#)
44. Choi, J.-M.; Hyman, A.A.; Pappu, R.V. Generalized Models for Bond Percolation Transitions of Associative Polymers. *Phys. Rev. E* **2020**, *102*, 042403. [\[CrossRef\]](#)
45. Jadrach, R.; Schweizer, K.S. Percolation, Phase Separation, and Gelation in Fluids and Mixtures of Spheres and Rods. *J. Chem. Phys.* **2011**, *135*, 234902. [\[CrossRef\]](#) [\[PubMed\]](#)
46. Binder, K. Percolation Effects in the Kinetics of Phase Separation. *Solid State Commun.* **1980**, *34*, 191–194. [\[CrossRef\]](#)
47. Kar, M.; Dar, F.; Welsh, T.J.; Vogel, L.T.; Kühnemuth, R.; Majumdar, A.; Krainer, G.; Franzmann, T.M.; Alberti, S.; Seidel, C.A.M.; et al. Phase-Separating RNA-Binding Proteins Form Heterogeneous Distributions of Clusters in Subsaturated Solutions. *Proc. Natl. Acad. Sci. USA* **2022**, *119*, e2202222119. [\[CrossRef\]](#)
48. Thomsen, C.; Grundevik, P.; Elias, P.; Ståhlberg, A.; Aman, P. A Conserved N-Terminal Motif Is Required for Complex Formation between FUS, EWSR1, TAF15 and Their Oncogenic Fusion Proteins. *FASEB J.* **2013**, *27*, 4965–4974. [\[CrossRef\]](#)
49. Andersson, M.K.; Ståhlberg, A.; Arvidsson, Y.; Olofsson, A.; Semb, H.; Stenman, G.; Nilsson, O.; Aman, P. The Multifunctional FUS, EWS and TAF15 Proto-Oncoproteins Show Cell Type-Specific Expression Patterns and Involvement in Cell Spreading and Stress Response. *BMC Cell Biol.* **2008**, *9*, 37. [\[CrossRef\]](#)

50. McSwiggen, D.T.; Mir, M.; Darzacq, X.; Tjian, R. Evaluating Phase Separation in Live Cells: Diagnosis, Caveats, and Functional Consequences. *Genes Dev.* **2019**, *33*, 1619–1634. [\[CrossRef\]](#)
51. Takamuku, M.; Sugishita, T.; Tamaki, H.; Dong, L.; So, M.; Fujiwara, T.; Matsuki, Y. Evolution of α -Synuclein Conformation Ensemble toward Amyloid Fibril via Liquid-Liquid Phase Separation (LLPS) as Investigated by Dynamic Nuclear Polarization-Enhanced Solid-State MAS NMR. *Neurochem. Int.* **2022**, *157*, 105345. [\[CrossRef\]](#) [\[PubMed\]](#)
52. Murray, D.T.; Kato, M.; Lin, Y.; Thurber, K.R.; Hung, I.; McKnight, S.L.; Tycko, R. Structure of FUS Protein Fibrils and Its Relevance to Self-Assembly and Phase Separation of Low-Complexity Domains. *Cell* **2017**, *171*, 615–627.e16. [\[CrossRef\]](#)
53. Mittag, T.; Pappu, R.V. A Conceptual Framework for Understanding Phase Separation and Addressing Open Questions and Challenges. *Mol. Cell* **2022**, *82*, 2201–2214. [\[CrossRef\]](#) [\[PubMed\]](#)
54. Hardy, J.A.; Higgins, G.A. Alzheimer's Disease: The Amyloid Cascade Hypothesis. *Science* **1992**, *256*, 184–185. [\[CrossRef\]](#) [\[PubMed\]](#)
55. Selkoe, D.J.; Hardy, J. The Amyloid Hypothesis of Alzheimer's Disease at 25 Years. *EMBO Mol. Med.* **2016**, *8*, 595–608. [\[CrossRef\]](#)
56. Hardy, J.; Selkoe, D.J. The Amyloid Hypothesis of Alzheimer's Disease: Progress and Problems on the Road to Therapeutics. *Science* **2002**, *297*, 353–356. [\[CrossRef\]](#)
57. Miller, R.L.; Dhavale, D.D.; O'Shea, J.Y.; Andruska, K.M.; Liu, J.; Franklin, E.E.; Buddhala, C.; Loftin, S.K.; Cirrito, J.R.; Perrin, R.J.; et al. Quantifying Regional α -Synuclein, Amyloid β , and Tau Accumulation in Lewy Body Dementia. *Ann. Clin. Transl. Neurol.* **2022**, *9*, 106–121. [\[CrossRef\]](#)
58. Meade, R.M.; Fairlie, D.P.; Mason, J.M. Alpha-Synuclein Structure and Parkinson's Disease—Lessons and Emerging Principles. *Mol. Neurodegener.* **2019**, *14*, 29. [\[CrossRef\]](#) [\[PubMed\]](#)
59. Benson, M.D.; Buxbaum, J.N.; Eisenberg, D.S.; Merlino, G.; Saraiva, M.J.M.; Sekijima, Y.; Sipe, J.D.; Westermarck, P. Amyloid Nomenclature 2018: Recommendations by the International Society of Amyloidosis (ISA) Nomenclature Committee. *Amyloid* **2018**, *25*, 215–219. [\[CrossRef\]](#)
60. Benson, M.D.; Buxbaum, J.N.; Eisenberg, D.S.; Merlino, G.; Saraiva, M.J.M.; Sekijima, Y.; Sipe, J.D.; Westermarck, P. Amyloid Nomenclature 2020: Update and Recommendations by the International Society of Amyloidosis (ISA) Nomenclature Committee. *Amyloid* **2020**, *27*, 217–222. [\[CrossRef\]](#)
61. Maji, S.K.; Wang, L.; Greenwald, J.; Riek, R. Structure-Activity Relationship of Amyloid Fibrils. *FEBS Lett.* **2009**, *583*, 2610–2617. [\[CrossRef\]](#)
62. Nelson, R.; Eisenberg, D. Recent Atomic Models of Amyloid Fibril Structure. *Curr. Opin. Struct. Biol.* **2006**, *16*, 260–265. [\[CrossRef\]](#)
63. Glabe, C.C. Amyloid Accumulation and Pathogenesis of Alzheimer's Disease: Significance of Monomeric, Oligomeric and Fibrillar A β . In *Alzheimer's Disease: Cellular and Molecular Aspects of Amyloid β* ; Harris, J.R., Fahrenholz, F., Eds.; Springer: Boston, MA, USA, 2005; pp. 167–177. [\[CrossRef\]](#)
64. Roeters, S.J.; Iyer, A.; Pletikapić, G.; Kogan, V.; Subramaniam, V.; Woutersen, S. Evidence for Intramolecular Antiparallel Beta-Sheet Structure in Alpha-Synuclein Fibrils from a Combination of Two-Dimensional Infrared Spectroscopy and Atomic Force Microscopy. *Sci. Rep.* **2017**, *7*, 41051. [\[CrossRef\]](#) [\[PubMed\]](#)
65. Park, S.-B.; Yoon, J.-S.; Jang, S.-M.; Lee, K.-H.; Shin, S.-M. Computational Study on Oligomer Formation of Fibril-Forming Peptide of α -Synuclein. *Bull. Korean Chem. Soc.* **2012**, *33*, 848–854. [\[CrossRef\]](#)
66. Serpell, L.C.; Berriman, J.; Jakes, R.; Goedert, M.; Crowther, R.A. Fiber Diffraction of Synthetic α -Synuclein Filaments Shows Amyloid-like Cross- β Conformation. *Proc. Natl. Acad. Sci. USA* **2000**, *97*, 4897–4902. [\[CrossRef\]](#)
67. Passarella, D.; Goedert, M. Beta-Sheet Assembly of Tau and Neurodegeneration in *Drosophila Melanogaster*. *Neurobiol. Aging* **2018**, *72*, 98–105. [\[CrossRef\]](#) [\[PubMed\]](#)
68. Daebel, V.; Chinnathambi, S.; Biernat, J.; Schwalbe, M.; Habenstein, B.; Loquet, A.; Akoury, E.; Tepper, K.; Müller, H.; Baldus, M.; et al. β -Sheet Core of Tau Paired Helical Filaments Revealed by Solid-State NMR. *J. Am. Chem. Soc.* **2012**, *134*, 13982–13989. [\[CrossRef\]](#)
69. von Bergen, M.; Barghorn, S.; Biernat, J.; Mandelkow, E.-M.; Mandelkow, E. Tau Aggregation Is Driven by a Transition from Random Coil to Beta Sheet Structure. *Biochim. Biophys. Acta* **2005**, *1739*, 158–166. [\[CrossRef\]](#) [\[PubMed\]](#)
70. Spillantini, M.G.; Divane, A.; Goedert, M. Assignment of Human Alpha-Synuclein (SNCA) and Beta-Synuclein (SNCB) Genes to Chromosomes 4q21 and 5q35. *Genomics* **1995**, *27*, 379–381. [\[CrossRef\]](#) [\[PubMed\]](#)
71. Jakes, R.; Spillantini, M.G.; Goedert, M. Identification of Two Distinct Synucleins from Human Brain. *FEBS Lett.* **1994**, *345*, 27–32. [\[CrossRef\]](#)
72. Coelho-Cerqueira, E.; Carmo-Gonçalves, P.; Pinheiro, A.S.; Cortines, J.; Follmer, C. α -Synuclein as an Intrinsically Disordered Monomer—Fact or Artefact? *FEBS J.* **2013**, *280*, 4915–4927. [\[CrossRef\]](#)
73. Hoppe, S.O.; Uzunoğlu, G.; Nussbaum-Krammer, C. α -Synuclein Strains: Does Amyloid Conformation Explain the Heterogeneity of Synucleinopathies? *Biomolecules* **2021**, *11*, 931. [\[CrossRef\]](#)
74. Winner, B.; Jappelli, R.; Maji, S.K.; Desplats, P.A.; Boyer, L.; Aigner, S.; Hetzer, C.; Loher, T.; Vilar, M.; Campioni, S.; et al. In Vivo Demonstration That α -Synuclein Oligomers Are Toxic. *Proc. Natl. Acad. Sci. USA* **2011**, *108*, 4194–4199. [\[CrossRef\]](#)
75. Poudyal, M.; Sakunthala, A.; Mukherjee, S.; Gadhe, L.; Maji, S.K. Phase Separation and Other Forms of α -Synuclein Self-Assemblies. *Essays Biochem.* **2022**, *66*, 987–1000. [\[CrossRef\]](#) [\[PubMed\]](#)

76. Mukherjee, S.; Sakunthala, A.; Gadhe, L.; Poudyal, M.; Sawner, A.S.; Kadu, P.; Maji, S.K. Liquid-Liquid Phase Separation of α -Synuclein: A New Mechanistic Insight for α -Synuclein Aggregation Associated with Parkinson's Disease Pathogenesis. *J. Mol. Biol.* **2022**, *435*, 167713. [[CrossRef](#)] [[PubMed](#)]
77. Huang, S.; Mo, X.; Wang, J.; Ye, X.; Yu, H.; Liu, Y. α -Synuclein Phase Separation and Amyloid Aggregation Are Modulated by C-Terminal Truncations. *FEBS Lett.* **2022**, *596*, 1388–1400. [[CrossRef](#)]
78. Hardenberg, M.C.; Sinnige, T.; Casford, S.; Dada, S.T.; Poudel, C.; Robinson, E.A.; Fuxreiter, M.; Kaminski, C.F.; Kaminski Schierle, G.S.; Nollen, E.A.A.; et al. Observation of an α -Synuclein Liquid Droplet State and Its Maturation into Lewy Body-like Assemblies. *J. Mol. Cell Biol.* **2021**, *13*, 282–294. [[CrossRef](#)] [[PubMed](#)]
79. Ray, S.; Singh, N.; Kumar, R.; Patel, K.; Pandey, S.; Datta, D.; Mahato, J.; Panigrahi, R.; Navalkar, A.; Mehra, S.; et al. α -Synuclein Aggregation Nucleates through Liquid-Liquid Phase Separation. *Nat. Chem.* **2020**, *12*, 705–716. [[CrossRef](#)]
80. Ray, S.; Singh, N.; Patel, K.; Krishnamoorthy, G.; Maji, S.K. FRAP and FRET Investigation of α -Synuclein Fibrillization via Liquid-Liquid Phase Separation In Vitro and in HeLa Cells. In *Protein Aggregation: Methods and Protocols*; Cieplak, A.S., Ed.; Springer: New York, NY, USA, 2023; pp. 395–423. [[CrossRef](#)]
81. Xu, B.; Mo, X.; Chen, J.; Yu, H.; Liu, Y. Myricetin Inhibits α -Synuclein Amyloid Aggregation by Delaying the Liquid-to-Solid Phase Transition. *Chembiochem* **2022**, *23*, e202200216. [[CrossRef](#)]
82. Davies, H.A.; Rigden, D.J.; Phelan, M.M.; Madine, J. Probing Medin Monomer Structure and Its Amyloid Nucleation Using ¹³C-Direct Detection NMR in Combination with Structural Bioinformatics. *Sci. Rep.* **2017**, *7*, 45224. [[CrossRef](#)]
83. Larsson, A.; Söderberg, L.; Westermark, G.T.; Sletten, K.; Engström, U.; Tjernberg, L.O.; Näslund, J.; Westermark, P. Unwinding Fibril Formation of Medin, the Peptide of the Most Common Form of Human Amyloid. *Biochem. Biophys. Res. Commun.* **2007**, *361*, 822–828. [[CrossRef](#)]
84. Peng, S.; Glennert, J.; Westermark, P. Medin-Amyloid: A Recently Characterized Age-Associated Arterial Amyloid Form Affects Mainly Arteries in the Upper Part of the Body. *Amyloid* **2005**, *12*, 96–102. [[CrossRef](#)]
85. Marazuela, P.; Solé, M.; Bonaterra-Pastra, A.; Pizarro, J.; Camacho, J.; Martínez-Sáez, E.; Kuiperij, H.B.; Verbeek, M.M.; de Kort, A.M.; Schreuder, F.H.B.M.; et al. MFG-E8 (LACTADHERIN): A Novel Marker Associated with Cerebral Amyloid Angiopathy. *Acta Neuropathol. Commun.* **2021**, *9*, 154. [[CrossRef](#)]
86. Larsson, A.; Malmström, S.; Westermark, P. Signs of Cross-Seeding: Aortic Medin Amyloid as a Trigger for Protein AA Deposition. *Amyloid* **2011**, *18*, 229–234. [[CrossRef](#)] [[PubMed](#)]
87. Larsson, A.; Peng, S.; Persson, H.; Rosenbloom, J.; Abrams, W.R.; Wassberg, E.; Thelin, S.; Sletten, K.; Gerwins, P.; Westermark, P. Lactadherin Binds to Elastin—A Starting Point for Medin Amyloid Formation? *Amyloid* **2006**, *13*, 78–85. [[CrossRef](#)] [[PubMed](#)]
88. Wagner, J.; Degenhardt, K.; Veit, M.; Louros, N.; Konstantouleas, K.; Skodras, A.; Wild, K.; Liu, P.; Obermüller, U.; Bansal, V.; et al. Medin Co-Aggregates with Vascular Amyloid- β in Alzheimer's Disease. *Nature* **2022**, *612*, 123–131. [[CrossRef](#)]
89. Degenhardt, K.; Wagner, J.; Skodras, A.; Candlish, M.; Koppelman, A.J.; Wild, K.; Maxwell, R.; Rotermund, C.; von Zweydford, F.; Gloeckner, C.J.; et al. Medin Aggregation Causes Cerebrovascular Dysfunction in Aging Wild-Type Mice. *Proc. Natl. Acad. Sci. USA* **2020**, *117*, 23925–23931. [[CrossRef](#)] [[PubMed](#)]
90. Karamanova, N.; Truran, S.; Serrano, G.E.; Beach, T.G.; Madine, J.; Weissig, V.; Davies, H.A.; Veldhuizen, J.; Nikkhah, M.; Hansen, M.; et al. Endothelial Immune Activation by Medin: Potential Role in Cerebrovascular Disease and Reversal by Monosialoganglioside-Containing Nanoliposomes. *J. Am. Heart Assoc.* **2020**, *9*, e014810. [[CrossRef](#)] [[PubMed](#)]
91. Migrino, R.Q.; Karamanova, N.; Truran, S.; Serrano, G.E.; Davies, H.A.; Madine, J.; Beach, T.G. Cerebrovascular Medin Is Associated with Alzheimer's Disease and Vascular Dementia. *Alzheimer's Dement.* **2020**, *12*, e12072. [[CrossRef](#)]
92. Sabha, B.H.; Alzahrani, F.; Almehdar, H.A.; Uversky, V.N.; Redwan, E.M. Disorder in Milk Proteins: Lactadherin Multifunctionality and Structure. *Curr. Protein Pept. Sci.* **2018**, *19*, 983–997. [[CrossRef](#)]
93. Fens, M.H.A.M.; Mastrobattista, E.; de Graaff, A.M.; Flesch, F.M.; Ultee, A.; Rasmussen, J.T.; Molema, G.; Storm, G.; Schiffelers, R.M. Angiogenic Endothelium Shows Lactadherin-Dependent Phagocytosis of Aged Erythrocytes and Apoptotic Cells. *Blood* **2008**, *111*, 4542–4550. [[CrossRef](#)]
94. Silvestre, J.-S.; Théry, C.; Hamard, G.; Boddaert, J.; Aguilar, B.; Delcayre, A.; Houbbron, C.; Tamarat, R.; Blanc-Brude, O.; Heeneman, S.; et al. Lactadherin Promotes VEGF-Dependent Neovascularization. *Nat. Med.* **2005**, *11*, 499–506. [[CrossRef](#)]
95. Bu, H.-F.; Zuo, X.-L.; Wang, X.; Ensslin, M.A.; Koti, V.; Hsueh, W.; Raymond, A.S.; Shur, B.D.; Tan, X.-D. Milk Fat Globule-EGF Factor 8/lactadherin Plays a Crucial Role in Maintenance and Repair of Murine Intestinal Epithelium. *J. Clin. Investig.* **2007**, *117*, 3673–3683. [[CrossRef](#)]
96. Younger, S.; Jang, H.; Davies, H.A.; Niemiec, M.J.; Garcia, J.G.N.; Nussinov, R.; Migrino, R.Q.; Madine, J.; Arce, F.T. Medin Oligomer Membrane Pore Formation: A Potential Mechanism of Vascular Dysfunction. *Biophys. J.* **2020**, *118*, 2769–2782. [[CrossRef](#)] [[PubMed](#)]
97. Migrino, R.Q.; Davies, H.A.; Truran, S.; Karamanova, N.; Franco, D.A.; Beach, T.G.; Serrano, G.E.; Truong, D.; Nikkhah, M.; Madine, J. Amyloidogenic Medin Induces Endothelial Dysfunction and Vascular Inflammation through the Receptor for Advanced Glycation Endproducts. *Cardiovasc. Res.* **2017**, *113*, 1389–1402. [[CrossRef](#)]
98. Arispe, N.; Rojas, E.; Pollard, H.B. Alzheimer Disease Amyloid Beta Protein Forms Calcium Channels in Bilayer Membranes: Blockade by Tromethamine and Aluminum. *Proc. Natl. Acad. Sci. USA* **1993**, *90*, 567–571. [[CrossRef](#)] [[PubMed](#)]
99. Arispe, N.; Pollard, H.B.; Rojas, E. Zn²⁺ Interaction with Alzheimer Amyloid Beta Protein Calcium Channels. *Proc. Natl. Acad. Sci. USA* **1996**, *93*, 1710–1715. [[CrossRef](#)] [[PubMed](#)]

100. Konstantoulea, K.; Louros, N.; Rousseau, F.; Schymkowitz, J. Heterotypic Interactions in Amyloid Function and Disease. *FEBS J.* **2022**, *289*, 2025–2046. [\[CrossRef\]](#)
101. Nag, N.; Tripathi, T. Cross-Seeding with Homologous Sequences Alters Amyloid Aggregation Kinetics and Fibril Structure. *ACS Chem. Neurosci.* **2022**, *13*, 537–539. [\[CrossRef\]](#)
102. Mathieu, C.; Pappu, R.V.; Taylor, J.P. Beyond Aggregation: Pathological Phase Transitions in Neurodegenerative Disease. *Science* **2020**, *370*, 56–60. [\[CrossRef\]](#)
103. St George-Hyslop, P.; Lin, J.Q.; Miyashita, A.; Phillips, E.C.; Qamar, S.; Randle, S.J.; Wang, G. The Physiological and Pathological Biophysics of Phase Separation and Gelation of RNA Binding Proteins in Amyotrophic Lateral Sclerosis and Fronto-Temporal Lobar Degeneration. *Brain Res.* **2018**, *1693 Pt A*, 11–23. [\[CrossRef\]](#)
104. Vanderweyde, T.; Apicco, D.J.; Youmans-Kidder, K.; Ash, P.E.A.; Cook, C.; da Rocha, E.L.; Jansen-West, K.; Frame, A.A.; Citro, A.; Leszyk, J.D.; et al. Interaction of Tau with the RNA-Binding Protein TIA1 Regulates Tau Pathophysiology and Toxicity. *Cell Rep.* **2016**, *15*, 1455–1466. [\[CrossRef\]](#)
105. Gui, X.; Luo, F.; Li, Y.; Zhou, H.; Qin, Z.; Liu, Z.; Gu, J.; Xie, M.; Zhao, K.; Dai, B.; et al. Structural Basis for Reversible Amyloids of hnRNPA1 Elucidates Their Role in Stress Granule Assembly. *Nat. Commun.* **2019**, *10*, 2006. [\[CrossRef\]](#)
106. Ayyadevara, S.; Balasubramaniam, M.; Parcon, P.A.; Barger, S.W.; Griffin, W.S.T.; Alla, R.; Tackett, A.J.; Mackintosh, S.G.; Petricoin, E.; Zhou, W.; et al. Proteins That Mediate Protein Aggregation and Cytotoxicity Distinguish Alzheimer’s Hippocampus from Normal Controls. *Aging Cell* **2016**, *15*, 924–939. [\[CrossRef\]](#) [\[PubMed\]](#)
107. Brunello, C.A.; Yan, X.; Huttunen, H.J. Internalized Tau Sensitizes Cells to Stress by Promoting Formation and Stability of Stress Granules. *Sci. Rep.* **2016**, *6*, 30498. [\[CrossRef\]](#) [\[PubMed\]](#)
108. Maziuk, B.; Ballance, H.I.; Wolozin, B. Dysregulation of RNA Binding Protein Aggregation in Neurodegenerative Disorders. *Front. Mol. Neurosci.* **2017**, *10*, 89. [\[CrossRef\]](#)
109. Bishof, I.; Dammer, E.B.; Duong, D.M.; Kunding, S.R.; Gearing, M.; Lah, J.J.; Levey, A.I.; Seyfried, N.T. RNA-Binding Proteins with Basic-Acidic Dipeptide (BAD) Domains Self-Assemble and Aggregate in Alzheimer’s Disease. *J. Biol. Chem.* **2018**, *293*, 11047–11066. [\[CrossRef\]](#)
110. Kapeli, K.; Martinez, F.J.; Yeo, G.W. Genetic Mutations in RNA-Binding Proteins and Their Roles in ALS. *Hum. Genet.* **2017**, *136*, 1193–1214. [\[CrossRef\]](#)
111. Cookson, M.R. RNA-Binding Proteins Implicated in Neurodegenerative Diseases. *Wiley Interdiscip. Rev. RNA* **2017**, *8*, e1397. [\[CrossRef\]](#) [\[PubMed\]](#)
112. Cascarina, S.M.; Elder, M.R.; Ross, E.D. Atypical Structural Tendencies among Low-Complexity Domains in the Protein Data Bank Proteome. *PLoS Comput. Biol.* **2020**, *16*, e1007487. [\[CrossRef\]](#)
113. Posey, A.E.; Holehouse, A.S.; Pappu, R.V. Phase Separation of Intrinsically Disordered Proteins. *Methods Enzymol.* **2018**, *611*, 1–30. [\[CrossRef\]](#) [\[PubMed\]](#)
114. Uversky, V.N. Intrinsically Disordered Proteins in Overcrowded Milieu: Membrane-Less Organelles, Phase Separation, and Intrinsic Disorder. *Curr. Opin. Struct. Biol.* **2017**, *44*, 18–30. [\[CrossRef\]](#) [\[PubMed\]](#)
115. Hyman, A.A.; Weber, C.A.; Jülicher, F. Liquid-Liquid Phase Separation in Biology. *Annu. Rev. Cell Dev. Biol.* **2014**, *30*, 39–58. [\[CrossRef\]](#)
116. Kragelj, J.; Orand, T.; Delaforge, E.; Tengo, L.; Blackledge, M.; Palencia, A.; Jensen, M.R. Enthalpy-Entropy Compensation in the Promiscuous Interaction of an Intrinsically Disordered Protein with Homologous Protein Partners. *Biomolecules* **2021**, *11*, 1204. [\[CrossRef\]](#) [\[PubMed\]](#)
117. Arbesú, M.; Iruela, G.; Fuentes, H.; Teixeira, J.M.C.; Pons, M. Intramolecular Fuzzy Interactions Involving Intrinsically Disordered Domains. *Front. Mol. Biosci.* **2018**, *5*, 39. [\[CrossRef\]](#) [\[PubMed\]](#)
118. Flock, T.; Weatheritt, R.J.; Latysheva, N.S.; Babu, M.M. Controlling Entropy to Tune the Functions of Intrinsically Disordered Regions. *Curr. Opin. Struct. Biol.* **2014**, *26*, 62–72. [\[CrossRef\]](#)
119. Workman, R.J.; Drake, J.A.; Pettitt, B.M. Chapter 4—Thermodynamic Perspective of Protein Disorder and Phase Separation: Model Systems. In *Structure and Intrinsic Disorder in Enzymology*; Gupta, M.N., Uversky, V.N., Eds.; Academic Press: Cambridge, MA, USA, 2023; pp. 97–126. [\[CrossRef\]](#)
120. Flory, P.J. Thermodynamics of High Polymer Solutions. *J. Chem. Phys.* **1942**, *10*, 51–61. [\[CrossRef\]](#)
121. Huggins, M.L. Some Properties of Solutions of Long-Chain Compounds. *J. Phys. Chem.* **1942**, *46*, 151–158. [\[CrossRef\]](#)
122. Dignon, G.L.; Best, R.B.; Mittal, J. Biomolecular Phase Separation: From Molecular Driving Forces to Macroscopic Properties. *Annu. Rev. Phys. Chem.* **2020**, *71*, 53–75. [\[CrossRef\]](#)
123. Qian, H.; Hopfield, J.J. Entropy-enthalpy Compensation: Perturbation and Relaxation in Thermodynamic Systems. *J. Chem. Phys.* **1996**, *105*, 9292–9298. [\[CrossRef\]](#)
124. Jacobson, K.; Papahadjopoulos, D. Phase Transitions and Phase Separations in Phospholipid Membranes Induced by Changes in Temperature, pH, and Concentration of Bivalent Cations. *Biochemistry* **1975**, *14*, 152–161. [\[CrossRef\]](#) [\[PubMed\]](#)
125. Nott, T.J.; Petsalaki, E.; Farber, P.; Jervis, D.; Fussner, E.; Plochowitz, A.; Craggs, T.D.; Bazett-Jones, D.P.; Pawson, T.; Forman-Kay, J.D.; et al. Phase Transition of a Disordered Nuage Protein Generates Environmentally Responsive Membraneless Organelles. *Mol. Cell* **2015**, *57*, 936–947. [\[CrossRef\]](#)
126. Molliex, A.; Temirov, J.; Lee, J.; Coughlin, M.; Kanagaraj, A.P.; Kim, H.J.; Mittag, T.; Taylor, J.P. Phase Separation by Low Complexity Domains Promotes Stress Granule Assembly and Drives Pathological Fibrillization. *Cell* **2015**, *163*, 123–133. [\[CrossRef\]](#) [\[PubMed\]](#)

127. Riback, J.A.; Katanski, C.D.; Kear-Scott, J.L.; Pilipenko, E.V.; Rojek, A.E.; Sosnick, T.R.; Drummond, D.A. Stress-Triggered Phase Separation Is an Adaptive, Evolutionarily Tuned Response. *Cell* **2017**, *168*, 1028–1040.e19. [[CrossRef](#)] [[PubMed](#)]
128. Cinar, H.; Fetahaj, Z.; Cinar, S.; Vernon, R.M.; Chan, H.S.; Winter, R.H.A. Temperature, Hydrostatic Pressure, and Osmolyte Effects on Liquid-Liquid Phase Separation in Protein Condensates: Physical Chemistry and Biological Implications. *Chemistry* **2019**, *25*, 13049–13069. [[CrossRef](#)]
129. Protter, D.S.W.; Parker, R. Principles and Properties of Stress Granules. *Trends Cell Biol.* **2016**, *26*, 668–679. [[CrossRef](#)] [[PubMed](#)]
130. Tamaki, M.; Kojima, C. pH-Switchable LCST/UCST-Type Thermosensitive Behaviors of Phenylalanine-Modified Zwitterionic Dendrimers. *RSC Adv.* **2020**, *10*, 10452–10460. [[CrossRef](#)]
131. Zhang, H.; Zhang, J.; Dai, W.; Zhao, Y. Facile Synthesis of Thermo-, pH-, CO₂- and Oxidation-Responsive Poly(amido Thioether)s with Tunable LCST and UCST Behaviors. *Polym. Chem.* **2017**, *8*, 5749–5760. [[CrossRef](#)]
132. Jin, X.; Zhou, M.; Chen, S.; Li, D.; Cao, X.; Liu, B. Effects of pH Alterations on Stress- and Aging-Induced Protein Phase Separation. *Cell. Mol. Life Sci.* **2022**, *79*, 380. [[CrossRef](#)]
133. Orij, R.; Brul, S.; Smits, G.J. Intracellular pH Is a Tightly Controlled Signal in Yeast. *Biochim. Biophys. Acta* **2011**, *1810*, 933–944. [[CrossRef](#)]
134. Kroschwald, S.; Munder, M.C.; Maharana, S.; Franzmann, T.M.; Richter, D.; Ruer, M.; Hyman, A.A.; Alberti, S. Different Material States of Pub1 Condensates Define Distinct Modes of Stress Adaptation and Recovery. *Cell Rep.* **2018**, *23*, 3327–3339. [[CrossRef](#)] [[PubMed](#)]
135. Lin, Y.; Fichou, Y.; Longhini, A.P.; Llanes, L.C.; Yin, P.; Bazan, G.C.; Kosik, K.S.; Han, S. Liquid-Liquid Phase Separation of Tau Driven by Hydrophobic Interaction Facilitates Fibrillization of Tau. *J. Mol. Biol.* **2021**, *433*, 166731. [[CrossRef](#)] [[PubMed](#)]
136. Park, S.; Barnes, R.; Lin, Y.; Jeon, B.-J.; Najafi, S.; Delaney, K.T.; Fredrickson, G.H.; Shea, J.-E.; Hwang, D.S.; Han, S. Dehydration Entropy Drives Liquid-Liquid Phase Separation by Molecular Crowding. *Commun. Chem.* **2020**, *3*, 83. [[CrossRef](#)]
137. Elbaum-Garfinkle, S.; Kim, Y.; Szczepaniak, K.; Chen, C.C.-H.; Eckmann, C.R.; Myong, S.; Brangwynne, C.P. The Disordered P Granule Protein LAF-1 Drives Phase Separation into Droplets with Tunable Viscosity and Dynamics. *Proc. Natl. Acad. Sci. USA* **2015**, *112*, 7189–7194. [[CrossRef](#)] [[PubMed](#)]
138. Salis, A.; Ninham, B.W. Models and Mechanisms of Hofmeister Effects in Electrolyte Solutions, and Colloid and Protein Systems Revisited. *Chem. Soc. Rev.* **2014**, *43*, 7358–7377. [[CrossRef](#)]
139. Kalra, A.; Tugcu, N.; Cramer, S.M.; Garde, S. Salting-in and Salting-out of Hydrophobic Solutes in Aqueous Salt Solutions. *J. Phys. Chem. B* **2001**, *105*, 6380–6386. [[CrossRef](#)]
140. Melander, W.; Horváth, C. Salt Effect on Hydrophobic Interactions in Precipitation and Chromatography of Proteins: An Interpretation of the Lyotropic Series. *Arch. Biochem. Biophys.* **1977**, *183*, 200–215. [[CrossRef](#)]
141. Collins, K.D.; Washabaugh, M.W. The Hofmeister Effect and the Behaviour of Water at Interfaces. *Q. Rev. Biophys.* **1985**, *18*, 323–422. [[CrossRef](#)]
142. Friberg, S.E.; Chiu, M. Hydrotropes. *J. Dispers. Sci. Technol.* **1988**, *9*, 443–457. [[CrossRef](#)]
143. Schwierz, N.; Horinek, D.; Netz, R.R. Anionic and Cationic Hofmeister Effects on Hydrophobic and Hydrophilic Surfaces. *Langmuir* **2013**, *29*, 2602–2614. [[CrossRef](#)]
144. Furukawa, K.; Aguirre, C.; So, M.; Sasahara, K.; Miyanoiri, Y.; Sakurai, K.; Yamaguchi, K.; Ikenaka, K.; Mochizuki, H.; Kardos, J.; et al. Isoelectric Point-Amyloid Formation of α -Synuclein Extends the Generality of the Solubility and Supersaturation-Limited Mechanism. *Curr. Res. Struct. Biol.* **2020**, *2*, 35–44. [[CrossRef](#)]
145. Vugmeyster, L.; Clark, M.A.; Falconer, I.B.; Ostrovsky, D.; Gantz, D.; Qiang, W.; Hoatson, G.L. Flexibility and Solvation of Amyloid- β Hydrophobic Core. *J. Biol. Chem.* **2016**, *291*, 18484–18495. [[CrossRef](#)] [[PubMed](#)]
146. Kawai, R.; Chiba, S.; Okuwaki, K.; Kanada, R.; Doi, H.; Ono, M.; Mochizuki, Y.; Okuno, Y. Stabilization Mechanism for a Nonfibrillar Amyloid β Oligomer Based on Formation of a Hydrophobic Core Determined by Dissipative Particle Dynamics. *ACS Chem. Neurosci.* **2020**, *11*, 385–394. [[CrossRef](#)] [[PubMed](#)]
147. Majumdar, A.; Das, D.; Madhu, P.; Avni, A.; Mukhopadhyay, S. Excitation Energy Migration Unveils Fuzzy Interfaces within the Amyloid Architecture. *Biophys. J.* **2020**, *118*, 2621–2626. [[CrossRef](#)]
148. Fernández, A. Time-Resolved Backbone Desolvation and Mutational Hot Spots in Folding Proteins. *Proteins* **2002**, *47*, 447–457. [[CrossRef](#)]
149. Nakagawa, H.; Tamada, T. Hydration and Its Hydrogen Bonding State on a Protein Surface in the Crystalline State as Revealed by Molecular Dynamics Simulation. *Front. Chem.* **2021**, *9*, 738077. [[CrossRef](#)]
150. Bellissent-Funel, M.-C.; Hassanali, A.; Havenith, M.; Henchman, R.; Pohl, P.; Sterpone, F.; van der Spoel, D.; Xu, Y.; Garcia, A.E. Water Determines the Structure and Dynamics of Proteins. *Chem. Rev.* **2016**, *116*, 7673–7697. [[CrossRef](#)] [[PubMed](#)]
151. Levy, Y.; Onuchic, J.N. Water Mediation in Protein Folding and Molecular Recognition. *Annu. Rev. Biophys. Biomol. Struct.* **2006**, *35*, 389–415. [[CrossRef](#)]
152. Chaplin, M. Do We Underestimate the Importance of Water in Cell Biology? *Nat. Rev. Mol. Cell Biol.* **2006**, *7*, 861–866. [[CrossRef](#)]
153. Lum, K.; Chandler, D.; Weeks, J.D. Hydrophobicity at Small and Large Length Scales. *J. Phys. Chem. B* **1999**, *103*, 4570–4577. [[CrossRef](#)]
154. Ahlers, J.; Adams, E.M.; Bader, V.; Pezzotti, S.; Winklhofer, K.F.; Tatzelt, J.; Havenith, M. The Key Role of Solvent in Condensation: Mapping Water in Liquid-Liquid Phase-Separated FUS. *Biophys. J.* **2021**, *120*, 1266–1275. [[CrossRef](#)]

155. Conti Nibali, V.; Pezzotti, S.; Sebastiani, F.; Galimberti, D.R.; Schwaab, G.; Heyden, M.; Gaigeot, M.-P.; Havenith, M. Wrapping Up Hydrophobic Hydration: Locality Matters. *J. Phys. Chem. Lett.* **2020**, *11*, 4809–4816. [\[CrossRef\]](#)
156. Laage, D.; Elsaesser, T.; Hynes, J.T. Water Dynamics in the Hydration Shells of Biomolecules. *Chem. Rev.* **2017**, *117*, 10694–10725. [\[CrossRef\]](#) [\[PubMed\]](#)
157. Fogarty, A.C.; Laage, D. Water Dynamics in Protein Hydration Shells: The Molecular Origins of the Dynamical Perturbation. *J. Phys. Chem. B* **2014**, *118*, 7715–7729. [\[CrossRef\]](#) [\[PubMed\]](#)
158. Liu, Z.; Chan, H.S. Desolvation Is a Likely Origin of Robust Enthalpic Barriers to Protein Folding. *J. Mol. Biol.* **2005**, *349*, 872–889. [\[CrossRef\]](#)
159. Camino, J.D.; Gracia, P.; Cremades, N. The Role of Water in the Primary Nucleation of Protein Amyloid Aggregation. *Biophys. Chem.* **2021**, *269*, 106520. [\[CrossRef\]](#) [\[PubMed\]](#)
160. Castellano, B.M.; Eggers, D.K. Experimental Support for a Desolvation Energy Term in Governing Equations for Binding Equilibria. *J. Phys. Chem. B* **2013**, *117*, 8180–8188. [\[CrossRef\]](#)
161. Kauzmann, W. Some Factors in the Interpretation of Protein Denaturation. *Adv. Protein Chem.* **1959**, *14*, 1–63. [\[CrossRef\]](#)
162. Blokzijl, W.; Engberts, J.B.F.N. Hydrophobic Effects. Opinions and Facts. *Angew. Chem. Int. Ed. Engl.* **1993**, *32*, 1545–1579. [\[CrossRef\]](#)
163. Muller, N. Search for a Realistic View of Hydrophobic Effects. *Acc. Chem. Res.* **1990**, *23*, 23–28. [\[CrossRef\]](#)
164. Hamaker, H.C. The London—Van Der Waals Attraction between Spherical Particles. *Physica* **1937**, *4*, 1058–1072. [\[CrossRef\]](#)
165. Davis, J.G.; Gierszal, K.P.; Wang, P.; Ben-Amotz, D. Water Structural Transformation at Molecular Hydrophobic Interfaces. *Nature* **2012**, *491*, 582–585. [\[CrossRef\]](#) [\[PubMed\]](#)
166. Baldwin, R.L. Dynamic Hydration Shell Restores Kauzmann’s 1959 Explanation of How the Hydrophobic Factor Drives Protein Folding. *Proc. Natl. Acad. Sci. USA* **2014**, *111*, 13052–13056. [\[CrossRef\]](#)
167. Li, J.; Hou, C.; Ma, X.; Guo, S.; Zhang, H.; Shi, L.; Liao, C.; Zheng, B.; Ye, L.; Yang, L.; et al. Entropy-Enthalpy Compensations Fold Proteins in Precise Ways. *Int. J. Mol. Sci.* **2021**, *22*, 9653. [\[CrossRef\]](#)
168. Mancera, R.L. Computer Simulation of the Effect of Salt on the Hydrophobic Effect. *J. Chem. Soc. Faraday Trans.* **1998**, *94*, 3549–3559. [\[CrossRef\]](#)
169. Tuladhar, A.; Dewan, S.; Pezzotti, S.; Brigiano, F.S.; Creazzo, F.; Gaigeot, M.-P.; Borguet, E. Ions Tune Interfacial Water Structure and Modulate Hydrophobic Interactions at Silica Surfaces. *J. Am. Chem. Soc.* **2020**, *142*, 6991–7000. [\[CrossRef\]](#) [\[PubMed\]](#)
170. Arya, S.; Singh, A.K.; Bhasne, K.; Dogra, P.; Datta, A.; Das, P.; Mukhopadhyay, S. Femtosecond Hydration Map of Intrinsically Disordered α -Synuclein. *Biophys. J.* **2018**, *114*, 2540–2551. [\[CrossRef\]](#) [\[PubMed\]](#)
171. Rodriguez, J.A.; Ivanova, M.I.; Sawaya, M.R.; Cascio, D.; Reyes, F.E.; Shi, D.; Sangwan, S.; Guenther, E.L.; Johnson, L.M.; Zhang, M.; et al. Structure of the Toxic Core of α -Synuclein from Invisible Crystals. *Nature* **2015**, *525*, 486–490. [\[CrossRef\]](#)
172. Thirumalai, D.; Reddy, G.; Straub, J.E. Role of Water in Protein Aggregation and Amyloid Polymorphism. *Acc. Chem. Res.* **2012**, *45*, 83–92. [\[CrossRef\]](#)
173. Krone, M.G.; Hua, L.; Soto, P.; Zhou, R.; Berne, B.J.; Shea, J.-E. Role of Water in Mediating the Assembly of Alzheimer Amyloid-Beta A β 16–22 Protofilaments. *J. Am. Chem. Soc.* **2008**, *130*, 11066–11072. [\[CrossRef\]](#)
174. Klement, K.; Wieligmann, K.; Meinhardt, J.; Hortschansky, P.; Richter, W.; Fändrich, M. Effect of Different Salt Ions on the Propensity of Aggregation and on the Structure of Alzheimer’s α 1–40 Amyloid Fibrils. *J. Mol. Biol.* **2007**, *373*, 1321–1333. [\[CrossRef\]](#) [\[PubMed\]](#)
175. van der Lubbe, S.C.C.; Fonseca Guerra, C. The Nature of Hydrogen Bonds: A Delineation of the Role of Different Energy Components on Hydrogen Bond Strengths and Lengths. *Chem. Asian J.* **2019**, *14*, 2760–2769. [\[CrossRef\]](#) [\[PubMed\]](#)
176. Muthachikavil, A.V.; Peng, B.; Kontogeorgis, G.M.; Liang, X. Distinguishing Weak and Strong Hydrogen Bonds in Liquid Water-A Potential of Mean Force-Based Approach. *J. Phys. Chem. B* **2021**, *125*, 7187–7198. [\[CrossRef\]](#) [\[PubMed\]](#)
177. Lin, Y.; Fichou, Y.; Zeng, Z.; Hu, N.Y.; Han, S. Electrostatically Driven Complex Coacervation and Amyloid Aggregation of Tau Are Independent Processes with Overlapping Conditions. *ACS Chem. Neurosci.* **2020**, *11*, 615–627. [\[CrossRef\]](#) [\[PubMed\]](#)
178. Camino, J.D.; Gracia, P.; Chen, S.W.; Sot, J.; de la Arada, I.; Sebastián, V.; Arrondo, J.L.R.; Goñi, F.M.; Dobson, C.M.; Cremades, N. The Extent of Protein Hydration Dictates the Preference for Heterogeneous or Homogeneous Nucleation Generating Either Parallel or Antiparallel β -Sheet α -Synuclein Aggregates. *Chem. Sci.* **2020**, *11*, 11902–11914. [\[CrossRef\]](#) [\[PubMed\]](#)
179. Tarus, B.; Straub, J.E.; Thirumalai, D. Probing the Initial Stage of Aggregation of the A β (10–35)-Protein: Assessing the Propensity for Peptide Dimerization. *J. Mol. Biol.* **2005**, *345*, 1141–1156. [\[CrossRef\]](#)
180. Esler, W.P.; Felix, A.M.; Stimson, E.R.; Lachenmann, M.J.; Ghilardi, J.R.; Lu, Y.A.; Vinters, H.V.; Mantyh, P.W.; Lee, J.P.; Maggio, J.E. Activation Barriers to Structural Transition Determine Deposition Rates of Alzheimer’s Disease a Beta Amyloid. *J. Struct. Biol.* **2000**, *130*, 174–183. [\[CrossRef\]](#)
181. Chang, L.; Ernst, T.; Poland, R.E.; Jenden, D.J. In Vivo Proton Magnetic Resonance Spectroscopy of the Normal Aging Human Brain. *Life Sci.* **1996**, *58*, 2049–2056. [\[CrossRef\]](#)
182. Naber, D.; Korte, U.; Krack, K. Content of Water-Soluble and Total Proteins in the Aging Human Brain. *Exp. Gerontol.* **1979**, *14*, 59–63. [\[CrossRef\]](#)
183. Chong, S.-H.; Ham, S. Dynamics of Hydration Water Plays a Key Role in Determining the Binding Thermodynamics of Protein Complexes. *Sci. Rep.* **2017**, *7*, 8744. [\[CrossRef\]](#)

184. Ribeiro, S.S.; Samanta, N.; Ebbinghaus, S.; Marcos, J.C. The Synergic Effect of Water and Biomolecules in Intracellular Phase Separation. *Nat. Rev. Chem.* **2019**, *3*, 552–561. [\[CrossRef\]](#)
185. Wang, T.; Jo, H.; DeGrado, W.F.; Hong, M. Water Distribution, Dynamics, and Interactions with Alzheimer's β -Amyloid Fibrils Investigated by Solid-State NMR. *J. Am. Chem. Soc.* **2017**, *139*, 6242–6252. [\[CrossRef\]](#) [\[PubMed\]](#)
186. Schwierz, N.; Frost, C.V.; Geissler, P.L.; Zacharias, M. Dynamics of Seeded A β 40-Fibril Growth from Atomistic Molecular Dynamics Simulations: Kinetic Trapping and Reduced Water Mobility in the Locking Step. *J. Am. Chem. Soc.* **2016**, *138*, 527–539. [\[CrossRef\]](#) [\[PubMed\]](#)
187. van der Spoel, D.; van Maaren, P.J.; Larsson, P.; Timneanu, N. Thermodynamics of Hydrogen Bonding in Hydrophilic and Hydrophobic Media. *J. Phys. Chem. B* **2006**, *110*, 4393–4398. [\[CrossRef\]](#) [\[PubMed\]](#)
188. Arunan, E.; Desiraju, G.R.; Klein, R.A.; Sadlej, J.; Scheiner, S.; Alkorta, I.; Clary, D.C.; Crabtree, R.H.; Dannenberg, J.J.; Hobza, P.; et al. Definition of the hydrogen bond (IUPAC Recommendations 2011). *J. Macromol. Sci. Part A Pure Appl. Chem.* **2011**, *83*, 1637–1641. [\[CrossRef\]](#)
189. Dong, K.; Zhang, S.; Wang, J. Understanding the Hydrogen Bonds in Ionic Liquids and Their Roles in Properties and Reactions. *Chem. Commun.* **2016**, *52*, 6744–6764. [\[CrossRef\]](#) [\[PubMed\]](#)
190. Jiang, K.; Liu, L.; Liu, X.; Zhang, X.; Zhang, S. Insight into the Relationship between Viscosity and Hydrogen Bond of a Series of Imidazolium Ionic Liquids: A Molecular Dynamics and Density Functional Theory Study. *Ind. Eng. Chem. Res.* **2019**, *58*, 18848–18854. [\[CrossRef\]](#)
191. Ma, Y.; Liu, Y.; Su, H.; Wang, L.; Zhang, J. Relationship between Hydrogen Bond and Viscosity for a Series of Pyridinium Ionic Liquids: Molecular Dynamics and Quantum Chemistry. *J. Mol. Liq.* **2018**, *255*, 176–184. [\[CrossRef\]](#)
192. Goertz, M.P.; Houston, J.E.; Zhu, X.-Y. Hydrophilicity and the Viscosity of Interfacial Water. *Langmuir* **2007**, *23*, 5491–5497. [\[CrossRef\]](#)
193. Luzar, A.; Chandler, D. Hydrogen-Bond Kinetics in Liquid Water. *Nature* **1996**, *379*, 55–57. [\[CrossRef\]](#)
194. Beal, C. The Viscosity of Air, Water, Natural Gas, Crude Oil and Its Associated Gases at Oil Field Temperatures and Pressures. *Trans. AIME* **1946**, *165*, 94–115. [\[CrossRef\]](#)
195. Hinrikus, H.; Bachmann, M.; Lass, J. Understanding Physical Mechanism of Low-Level Microwave Radiation Effect. *Int. J. Radiat. Biol.* **2018**, *94*, 877–882. [\[CrossRef\]](#)
196. Zong, D.; Hu, H.; Duan, Y.; Sun, Y. Viscosity of Water under Electric Field: Anisotropy Induced by Redistribution of Hydrogen Bonds. *J. Phys. Chem. B* **2016**, *120*, 4818–4827. [\[CrossRef\]](#) [\[PubMed\]](#)
197. Tatoń, G. The Influence of Electromagnetic Wave Originating from WiFi Router on Water Viscosity. *Prz. Elektrotech.* **2018**, *1*, 280–282. [\[CrossRef\]](#)
198. Hinrikus, H.; Lass, J.; Karai, D.; Pilt, K.; Bachmann, M. Microwave Effect on Diffusion: A Possible Mechanism for Non-Thermal Effect. *Electromagn. Biol. Med.* **2015**, *34*, 327–333. [\[CrossRef\]](#)
199. Ghauri, S.A.; Ansari, M.S. Increase of Water Viscosity under the Influence of Magnetic Field. *J. Appl. Phys.* **2006**, *100*, 066101. [\[CrossRef\]](#)
200. Read, F.H. *Electromagnetic Radiation*; John Wiley and Sons: Chichester, UK; New York, NY, USA, 1980.
201. Sellers, P.J.; Berry, J.A.; Collatz, G.J.; Field, C.B.; Hall, F.G. Canopy Reflectance, Photosynthesis, and Transpiration. III. A Reanalysis Using Improved Leaf Models and a New Canopy Integration Scheme. *Remote Sens. Environ.* **1992**, *42*, 187–216. [\[CrossRef\]](#)
202. Sellers, P.J. Canopy Reflectance, Photosynthesis, and Transpiration, II. The Role of Biophysics in the Linearity of Their Interdependence. *Remote Sens. Environ.* **1987**, *21*, 143–183. [\[CrossRef\]](#)
203. Gates, D.M.; Tantraporn, W. The Reflectivity of Deciduous Trees and Herbaceous Plants in the Infrared to 25 Microns. *Science* **1952**, *115*, 613–616. [\[CrossRef\]](#) [\[PubMed\]](#)
204. Baldocchi, D.D.; Ryu, Y.; Dechant, B.; Eichelmann, E.; Hemes, K.; Ma, S.; Sanchez, C.R.; Shortt, R.; Szutu, D.; Valach, A.; et al. Outgoing Near-infrared Radiation from Vegetation Scales with Canopy Photosynthesis across a Spectrum of Function, Structure, Physiological Capacity, and Weather. *J. Geophys. Res. Biogeosci.* **2020**, *125*, e2019JG005534. [\[CrossRef\]](#)
205. Aitken, W.W.; Lombard, J.; Wang, K.; Toro, M.; Byrne, M.; Nardi, M.I.; Kardys, J.; Parrish, A.; Dong, C.; Szapocznik, J.; et al. Relationship of Neighborhood Greenness to Alzheimer's Disease and Non-Alzheimer's Dementia among 249,405 U.S. Medicare Beneficiaries. *J. Alzheimer's Dis.* **2021**, *81*, 597–606. [\[CrossRef\]](#) [\[PubMed\]](#)
206. Veleva, B.I.; Caljouw, M.A.A.; van der Steen, J.T.; Mertens, B.J.A.; Chel, V.G.M.; Numans, M.E. The Effect of Ultraviolet B Irradiation Compared with Oral Vitamin D Supplementation on the Well-Being of Nursing Home Residents with Dementia: A Randomized Controlled Trial. *Int. J. Environ. Res. Public Health* **2020**, *17*, 1684. [\[CrossRef\]](#) [\[PubMed\]](#)
207. Purushothuman, S.; Johnstone, D.M.; Nandasena, C.; van Eersel, J.; Ittner, L.M.; Mitrofanis, J.; Stone, J. Near Infrared Light Mitigates Cerebellar Pathology in Transgenic Mouse Models of Dementia. *Neurosci. Lett.* **2015**, *591*, 155–159. [\[CrossRef\]](#)
208. Purushothuman, S.; Johnstone, D.M.; Nandasena, C.; Mitrofanis, J.; Stone, J. Photobiomodulation with near Infrared Light Mitigates Alzheimer's Disease-Related Pathology in Cerebral Cortex—Evidence from Two Transgenic Mouse Models. *Alzheimer's Res. Ther.* **2014**, *6*, 2. [\[CrossRef\]](#)
209. Comerota, M.M.; Krishnan, B.; Taglialatela, G. Near Infrared Light Decreases Synaptic Vulnerability to Amyloid Beta Oligomers. *Sci. Rep.* **2017**, *7*, 15012. [\[CrossRef\]](#)

210. Comerota, M.M.; Tumurbaatar, B.; Krishnan, B.; Kaye, R.; Taglialatela, G. Near Infrared Light Treatment Reduces Synaptic Levels of Toxic Tau Oligomers in Two Transgenic Mouse Models of Human Tauopathies. *Mol. Neurobiol.* **2019**, *56*, 3341–3355. [\[CrossRef\]](#) [\[PubMed\]](#)
211. Stepanov, Y.V.; Golovynska, I.; Zhang, R.; Golovynskyi, S.; Stepanova, L.I.; Gorbach, O.; Dovbynychuk, T.; Garmanchuk, L.V.; Ohulchanskyi, T.Y.; Qu, J. Near-Infrared Light Reduces β -Amyloid-Stimulated Microglial Toxicity and Enhances Survival of Neurons: Mechanisms of Light Therapy for Alzheimer's Disease. *Alzheimer's Res. Ther.* **2022**, *14*, 84. [\[CrossRef\]](#) [\[PubMed\]](#)
212. Chao, L.L. Effects of Home Photobiomodulation Treatments on Cognitive and Behavioral Function, Cerebral Perfusion, and Resting-State Functional Connectivity in Patients with Dementia: A Pilot Trial. *Photobiomodulation Photomed. Laser Surg.* **2019**, *37*, 133–141. [\[CrossRef\]](#)
213. Berman, M.H.; Halper, J.P.; Nichols, T.W.; Jarrett, H.; Lundy, A.; Huang, J.H. Photobiomodulation with Near Infrared Light Helmet in a Pilot, Placebo Controlled Clinical Trial in Dementia Patients Testing Memory and Cognition. *J. Neurol. Neurosci.* **2017**, *8*, 176. [\[CrossRef\]](#)
214. Nizamutdinov, D.; Qi, X.; Berman, M.H.; Dougal, G.; Dayawansa, S.; Wu, E.; Yi, S.S.; Stevens, A.B.; Huang, J.H. Transcranial Near Infrared Light Stimulations Improve Cognition in Patients with Dementia. *Aging Dis.* **2021**, *12*, 954–963. [\[CrossRef\]](#)
215. Dougal, G.; Ennaceur, A.; Chazot, P.L. Effect of Transcranial Near-Infrared Light 1068 Nm Upon Memory Performance in Aging Healthy Individuals: A Pilot Study. *Photobiomodulation Photomed. Laser Surg.* **2021**, *39*, 654–660. [\[CrossRef\]](#)
216. Han, M.; Wang, Q.; Wang, X.; Zeng, Y.; Huang, Y.; Meng, Q.; Zhang, J.; Wei, X. Near Infra-Red Light Treatment of Alzheimer's Disease. *J. Innov. Opt. Health Sci.* **2018**, *11*, 1750012. [\[CrossRef\]](#)
217. Li, Q.; Peng, J.; Luo, Y.; Zhou, J.; Li, T.; Cao, L.; Peng, S.; Zuo, Z.; Wang, Z. Far Infrared Light Irradiation Enhances A β Clearance via Increased Exocytotic Microglial ATP and Ameliorates Cognitive Deficit in Alzheimer's Disease-like Mice. *J. Neuroinflamm.* **2022**, *19*, 145. [\[CrossRef\]](#) [\[PubMed\]](#)
218. Tan, D.-X.; Reiter, R.J.; Zimmerman, S.; Hardeland, R. Melatonin: Both a Messenger of Darkness and a Participant in the Cellular Actions of Non-Visible Solar Radiation of Near Infrared Light. *Biology* **2023**, *12*, 89. [\[CrossRef\]](#)
219. Zimmerman, S.; Reiter, R.J. Melatonin and the Optics of the Human Body. *Melatonin Res.* **2019**, *2*, 138–160. [\[CrossRef\]](#)
220. Chang, S.-Y.; Lee, M.Y.; Chung, P.-S.; Kim, S.; Choi, B.; Suh, M.-W.; Rhee, C.-K.; Jung, J.Y. Enhanced Mitochondrial Membrane Potential and ATP Synthesis by Photobiomodulation Increases Viability of the Auditory Cell Line after Gentamicin-Induced Intrinsic Apoptosis. *Sci. Rep.* **2019**, *9*, 19248. [\[CrossRef\]](#)
221. Hamblin, M.R. Mechanisms and Applications of the Anti-Inflammatory Effects of Photobiomodulation. *AIMS Biophys.* **2017**, *4*, 337–361. [\[CrossRef\]](#)
222. Karu, T. Mitochondrial Mechanisms of Photobiomodulation in Context of New Data about Multiple Roles of ATP. *Photomed. Laser Surg.* **2010**, *28*, 159–160. [\[CrossRef\]](#) [\[PubMed\]](#)
223. Coon, S.L.; Klein, D.C. Evolution of Arylalkylamine N-Acetyltransferase: Emergence and Divergence. *Mol. Cell. Endocrinol.* **2006**, *252*, 2–10. [\[CrossRef\]](#)
224. Lee, K.; Choi, G.-H.; Back, K. Functional Characterization of Serotonin N-Acetyltransferase in Archaeon Thermoplasma Volcanium. *Antioxidants* **2022**, *11*, 596. [\[CrossRef\]](#)
225. Kang, K.; Lee, K.; Park, S.; Byeon, Y.; Back, K. Molecular Cloning of Rice Serotonin N-Acetyltransferase, the Penultimate Gene in Plant Melatonin Biosynthesis. *J. Pineal Res.* **2013**, *55*, 7–13. [\[CrossRef\]](#)
226. Yoshizawa, T.; Nozawa, R.-S.; Jia, T.Z.; Saio, T.; Mori, E. Biological Phase Separation: Cell Biology Meets Biophysics. *Biophys. Rev.* **2020**, *12*, 519–539. [\[CrossRef\]](#) [\[PubMed\]](#)
227. King, J.T.; Shakya, A. Phase Separation of DNA: From Past to Present. *Biophys. J.* **2021**, *120*, 1139–1149. [\[CrossRef\]](#)
228. Dion, W.; Ballance, H.; Lee, J.; Pan, Y.; Irfan, S.; Edwards, C.; Sun, M.; Zhang, J.; Zhang, X.; Liu, S.; et al. Four-Dimensional Nuclear Speckle Phase Separation Dynamics Regulate Proteostasis. *Sci. Adv.* **2022**, *8*, eabl4150. [\[CrossRef\]](#)
229. Poudyal, R.R.; Pir Cakmak, F.; Keating, C.D.; Bevilacqua, P.C. Physical Principles and Extant Biology Reveal Roles for RNA-Containing Membraneless Compartments in Origins of Life Chemistry. *Biochemistry* **2018**, *57*, 2509–2519. [\[CrossRef\]](#)
230. Loh, D.; Reiter, R.J. Melatonin: Regulation of Biomolecular Condensates in Neurodegenerative Disorders. *Antioxidants* **2021**, *10*, 1483. [\[CrossRef\]](#) [\[PubMed\]](#)
231. Chu, X.-Y.; Xu, Y.-Y.; Tong, X.-Y.; Wang, G.; Zhang, H.-Y. The Legend of ATP: From Origin of Life to Precision Medicine. *Metabolites* **2022**, *12*, 461. [\[CrossRef\]](#)
232. Franzmann, T.M.; Alberti, S. Protein Phase Separation as a Stress Survival Strategy. *Cold Spring Harb. Perspect. Biol.* **2019**, *11*, a034058. [\[CrossRef\]](#)
233. Lerner, A.B.; Case, J.D.; Takahashi, Y.; Lee, T.H.; Mori, W. Isolation of melatonin, the pineal gland factor that lightens melanocytes. *J. Am. Chem. Soc.* **1958**, *80*, 2587. [\[CrossRef\]](#)
234. Loh, D.; Reiter, R.J. Melatonin and Phase Separation: Potential Interactions and Significance. *Melatonin Res.* **2022**, *5*, 186–191. [\[CrossRef\]](#)
235. Loh, D.; Reiter, R.J. Melatonin: Regulation of Prion Protein Phase Separation in Cancer Multidrug Resistance. *Molecules* **2022**, *27*, 705. [\[CrossRef\]](#)
236. Loh, D.; Reiter, R.J. Melatonin: Regulation of Viral Phase Separation and Epitranscriptomics in Post-Acute Sequelae of COVID-19. *Int. J. Mol. Sci.* **2022**, *23*, 8122. [\[CrossRef\]](#) [\[PubMed\]](#)

237. Li, C.; Akinc, M.; Wiench, J.; Pruski, M.; Schilling, C.H. Relationship between Water Mobility and Viscosity of Nanometric Alumina Suspensions. *J. Am. Ceram. Soc.* **2005**, *88*, 2762–2768. [[CrossRef](#)]
238. Ye, H.; Zhang, H.; Zhang, Z.; Zheng, Y. Size and Temperature Effects on the Viscosity of Water inside Carbon Nanotubes. *Nanoscale Res. Lett.* **2011**, *6*, 87. [[CrossRef](#)] [[PubMed](#)]
239. Shaat, M.; Zheng, Y. Fluidity and Phase Transitions of Water in Hydrophobic and Hydrophilic Nanotubes. *Sci. Rep.* **2019**, *9*, 5689. [[CrossRef](#)] [[PubMed](#)]
240. Chaban, V.V.; Prezhdo, V.V.; Prezhdo, O.V. Confinement by Carbon Nanotubes Drastically Alters the Boiling and Critical Behavior of Water Droplets. *ACS Nano* **2012**, *6*, 2766–2773. [[CrossRef](#)]
241. Striolo, A. Water Self-Diffusion through Narrow Oxygenated Carbon Nanotubes. *Nanotechnology* **2007**, *18*, 475704. [[CrossRef](#)]
242. Babu, J.S.; Sathian, S.P. The Role of Activation Energy and Reduced Viscosity on the Enhancement of Water Flow through Carbon Nanotubes. *J. Chem. Phys.* **2011**, *134*, 194509. [[CrossRef](#)]
243. Zhang, D.; Yang, X.; Jiang, W.; Jin, L.; Gao, Y.; Wang, Z. Pauli Repulsion Enhances Mobility of Ultraconfined Water. *ACS Nano* **2021**, *15*, 2490–2496. [[CrossRef](#)]
244. Najafi, S.; Lin, Y.; Longhini, A.P.; Zhang, X.; Delaney, K.T.; Kosik, K.S.; Fredrickson, G.H.; Shea, J.-E.; Han, S. Liquid-Liquid Phase Separation of Tau by Self and Complex Coacervation. *Protein Sci.* **2021**, *30*, 1393–1407. [[CrossRef](#)]
245. Jawerth, L.; Fischer-Friedrich, E.; Saha, S.; Wang, J.; Franzmann, T.; Zhang, X.; Sachweh, J.; Ruer, M.; Ijavi, M.; Saha, S.; et al. Protein Condensates as Aging Maxwell Fluids. *Science* **2020**, *370*, 1317–1323. [[CrossRef](#)] [[PubMed](#)]
246. Tejedor, A.R.; Sanchez-Burgos, I.; Estevez-Espinosa, M.; Garaizar, A.; Collepardo-Guevara, R.; Ramirez, J.; Espinosa, J.R. Protein Structural Transitions Critically Transform the Network Connectivity and Viscoelasticity of RNA-Binding Protein Condensates but RNA Can Prevent It. *Nat. Commun.* **2022**, *13*, 5717. [[CrossRef](#)] [[PubMed](#)]
247. Chen, H.-C.; Mai, F.-D.; Yang, K.-H.; Chen, L.-Y.; Yang, C.-P.; Liu, Y.-C. Quantitative Evaluation on Activated Property-Tunable Bulk Liquid Water with Reduced Hydrogen Bonds Using Deconvoluted Raman Spectroscopy. *Anal. Chem.* **2015**, *87*, 808–815. [[CrossRef](#)] [[PubMed](#)]
248. Yu, S.-H.; Chang, C.-C.; Mai, F.-D.; Yang, C.-P.; Liu, Y.-C. Plasmon-Activated Water Can Form Stronger Intermolecular Hydrogen Bonding with Water-Soluble Alcohols and Dissolve More Hydrophobic Solutes. *Chem. Eng. J.* **2022**, *427*, 131949. [[CrossRef](#)]
249. Zhang, J.; Yan, X.; Tian, Y.; Li, W.; Wang, H.; Li, Q.; Li, Y.; Li, Z.; Wu, T. Synthesis of a New Water-Soluble Melatonin Derivative with Low Toxicity and a Strong Effect on Sleep Aid. *ACS Omega* **2020**, *5*, 6494–6499. [[CrossRef](#)]
250. Renn, T.-Y.; Yang, C.-P.; Wu, U.-I.; Chen, L.-Y.; Mai, F.-D.; Tikhonova, M.A.; Amstislavskaya, T.G.; Liao, W.-C.; Lin, C.-T.; Liu, Y.-C.; et al. Water Composed of Reduced Hydrogen Bonds Activated by Localized Surface Plasmon Resonance Effectively Enhances Anti-Viral and Anti-Oxidative Activities of Melatonin. *Chem. Eng. J.* **2022**, *427*, 131626. [[CrossRef](#)]
251. Rodrigues, A.C.C.; de M. Camargo, L.T.F.; Lopes, Y.F.; Sallum, L.O.; Napolitano, H.B.; Camargo, A.J. Aqueous Solvation Study of Melatonin Using Ab Initio Molecular Dynamics. *J. Mol. Liq.* **2021**, *343*, 117451. [[CrossRef](#)]
252. Florio, G.M.; Zwier, T.S. Solvation of a Flexible Biomolecule in the Gas Phase: The Ultraviolet and Infrared Spectroscopy of Melatonin–Water Clusters. *J. Phys. Chem. A* **2003**, *107*, 974–983. [[CrossRef](#)]
253. Ritwiset, A.; Khajonrit, J.; Kongsuk, S.; Maensiri, S. Molecular Insight on the Formation Structure and Dynamics of Melatonin in an Aqueous Solution and at the Water-Air Interface: A Molecular Dynamics Study. *J. Mol. Graph. Model.* **2021**, *108*, 107983. [[CrossRef](#)] [[PubMed](#)]
254. Chang, H.-M.; Wu, U.-I.; Lin, T.-B.; Lan, C.-T.; Chien, W.-C.; Huang, W.-L.; Shieh, J.-Y. Total Sleep Deprivation Inhibits the Neuronal Nitric Oxide Synthase and Cytochrome Oxidase Reactivities in the Nodose Ganglion of Adult Rats. *J. Anat.* **2006**, *209*, 239–250. [[CrossRef](#)]
255. Cheng, C.-H.; Lin, K.-J.; Hong, C.-T.; Wu, D.; Chang, H.-M.; Liu, C.-H.; Hsiao, I.-T.; Yang, C.-P.; Liu, Y.-C.; Hu, C.-J. Plasmon-Activated Water Reduces Amyloid Burden and Improves Memory in Animals with Alzheimer’s Disease. *Sci. Rep.* **2019**, *9*, 13252. [[CrossRef](#)]
256. Chen, H.-C.; Cheng, C.-Y.; Chen, L.-Y.; Chang, C.-C.; Yang, C.-P.; Mai, F.-D.; Liao, W.-C.; Chang, H.-M.; Liu, Y.-C. Plasmon-Activated Water Effectively Relieves Hepatic Oxidative Damage Resulting from Chronic Sleep Deprivation. *RSC Adv.* **2018**, *8*, 9618–9626. [[CrossRef](#)]
257. Mitroka, S.; Zimmeck, S.; Troya, D.; Tanko, J.M. How Solvent Modulates Hydroxyl Radical Reactivity in Hydrogen Atom Abstractions. *J. Am. Chem. Soc.* **2010**, *132*, 2907–2913. [[CrossRef](#)]
258. Vassilev, P.; Louwerse, M.J.; Baerends, E.J. Hydroxyl Radical and Hydroxide Ion in Liquid Water: A Comparative Electron Density Functional Theory Study. *J. Phys. Chem. B* **2005**, *109*, 23605–23610. [[CrossRef](#)]
259. Wong, H.-S.; Dighe, P.A.; Mezera, V.; Monternier, P.-A.; Brand, M.D. Production of Superoxide and Hydrogen Peroxide from Specific Mitochondrial Sites under Different Bioenergetic Conditions. *J. Biol. Chem.* **2017**, *292*, 16804–16809. [[CrossRef](#)] [[PubMed](#)]
260. Giulivi, C.; Boveris, A.; Cadenas, E. Hydroxyl Radical Generation during Mitochondrial Electron Transfer and the Formation of 8-Hydroxydesoxyguanosine in Mitochondrial DNA. *Arch. Biochem. Biophys.* **1995**, *316*, 909–916. [[CrossRef](#)] [[PubMed](#)]
261. Aleardi, A.M.; Benard, G.; Augereau, O.; Malgat, M.; Talbot, J.C.; Mazat, J.P.; Letellier, T.; Dachary-Prigent, J.; Solaini, G.C.; Rossignol, R. Gradual Alteration of Mitochondrial Structure and Function by Beta-Amyloids: Importance of Membrane Viscosity Changes, Energy Deprivation, Reactive Oxygen Species Production, and Cytochrome c Release. *J. Bioenerg. Biomembr.* **2005**, *37*, 207–225. [[CrossRef](#)] [[PubMed](#)]
262. Gough, D.R.; Cotter, T.G. Hydrogen Peroxide: A Jekyll and Hyde Signalling Molecule. *Cell Death Dis.* **2011**, *2*, e213. [[CrossRef](#)]

263. Ren, M.; Deng, B.; Zhou, K.; Kong, X.; Wang, J.-Y.; Lin, W. Single Fluorescent Probe for Dual-Imaging Viscosity and H₂O₂ in Mitochondria with Different Fluorescence Signals in Living Cells. *Anal. Chem.* **2017**, *89*, 552–555. [\[CrossRef\]](#)
264. Zheng, A.; Liu, H.; Gao, X.; Xu, K.; Tang, B. A Mitochondrial-Targeting Near-Infrared Fluorescent Probe for Revealing the Effects of Hydrogen Peroxide And Heavy Metal Ions on Viscosity. *Anal. Chem.* **2021**, *93*, 9244–9249. [\[CrossRef\]](#)
265. Li, S.; Wang, P.; Feng, W.; Xiang, Y.; Dou, K.; Liu, Z. Simultaneous Imaging of Mitochondrial Viscosity and Hydrogen Peroxide in Alzheimer's Disease by a Single near-Infrared Fluorescent Probe with a Large Stokes Shift. *Chem. Commun.* **2020**, *56*, 1050–1053. [\[CrossRef\]](#) [\[PubMed\]](#)
266. Tan, D.X.; Manchester, L.C.; Reiter, R.J.; Plummer, B.F. Cyclic 3-Hydroxymelatonin: A Melatonin Metabolite Generated as a Result of Hydroxyl Radical Scavenging. *Biol. Signals Recept.* **1999**, *8*, 70–74. [\[CrossRef\]](#)
267. Galano, A. On the Direct Scavenging Activity of Melatonin towards Hydroxyl and a Series of Peroxyl Radicals. *Phys. Chem. Chem. Phys.* **2011**, *13*, 7178–7188. [\[CrossRef\]](#) [\[PubMed\]](#)
268. Bandyopadhyay, D.; Biswas, K.; Bandyopadhyay, U.; Reiter, R.J.; Banerjee, R.K. Melatonin Protects against Stress-Induced Gastric Lesions by Scavenging the Hydroxyl Radical. *J. Pineal Res.* **2000**, *29*, 143–151. [\[CrossRef\]](#)
269. Reiter, R.J.; Tan, D.X.; Osuna, C.; Gitto, E. Actions of Melatonin in the Reduction of Oxidative Stress. A Review. *J. Biomed. Sci.* **2000**, *7*, 444–458. [\[CrossRef\]](#) [\[PubMed\]](#)
270. Galano, A.; Reiter, R.J. Melatonin and Its Metabolites vs Oxidative Stress: From Individual Actions to Collective Protection. *J. Pineal Res.* **2018**, *65*, e12514. [\[CrossRef\]](#)
271. Purushothaman, A.; Sheeja, A.A.; Janardanan, D. Hydroxyl Radical Scavenging Activity of Melatonin and Its Related Indolamines. *Free Radic. Res.* **2020**, *54*, 373–383. [\[CrossRef\]](#) [\[PubMed\]](#)
272. Persson, L.B.; Ambati, V.S.; Brandman, O. Cellular Control of Viscosity Counters Changes in Temperature and Energy Availability. *Cell* **2020**, *183*, 1572–1585.e16. [\[CrossRef\]](#)
273. Scalettar, B.A.; Abney, J.R.; Hackenbrock, C.R. Dynamics, Structure, and Function Are Coupled in the Mitochondrial Matrix. *Proc. Natl. Acad. Sci. USA* **1991**, *88*, 8057–8061. [\[CrossRef\]](#)
274. Hao, X.; Zhan, J.; Geng, C.; Lin, W. Discriminating Normal and Inflammatory Mice Models by Viscosity Changes with a Two-Photon Fluorescent Probe. *Spectrochim. Acta Part A Mol. Biomol. Spectrosc.* **2023**, *284*, 121807. [\[CrossRef\]](#)
275. Guan, X.; Hong, J.; Li, Q.; Feng, G. High-Fidelity Imaging Probe for Lysosomes and Selective Visualization of Cancer Cells and Tissues. *Sens. Actuators B Chem.* **2022**, *369*, 132325. [\[CrossRef\]](#)
276. Song, H.; Zhang, W.; Zhang, Y.; Yin, C.; Huo, F. Viscosity Activated NIR Fluorescent Probe for Visualizing Mitochondrial Viscosity Dynamic and Fatty Liver Mice. *Chem. Eng. J.* **2022**, *445*, 136448. [\[CrossRef\]](#)
277. Li, S.; Huo, F.; Yin, C. NIR Fluorescent Probe for Dual-Response Viscosity and Hydrogen Sulfide and Its Application in Parkinson's Disease Model. *Dyes Pigments* **2022**, *197*, 109825. [\[CrossRef\]](#)
278. Pan, W.; Han, L.; Cao, X.; Shen, S.; Pang, X.; Zhu, Y. Dual-Response near-Infrared Fluorescent Probe for Detecting Cyanide and Mitochondrial Viscosity and Its Application in Bioimaging. *Food Chem.* **2022**, *407*, 135163. [\[CrossRef\]](#) [\[PubMed\]](#)
279. Li, S.; Huo, F.; Wen, Y.; Yin, C. A Dual-Response NIR Probe Reveals Positive Correlation between Biothiols and Viscosity under Cellular Stress Change. *Chem. Commun.* **2022**, *58*, 4881–4884. [\[CrossRef\]](#) [\[PubMed\]](#)
280. Zhang, Y.; Zhou, Q.; Bu, Y.; Xu, T.; Zhu, X.; Zhang, J.; Yu, Z.; Wang, L.; Zhong, F.; Zhou, H. Real-Time Imaging Mitochondrial Viscosity Dynamic during Mitophagy Mediated by Photodynamic Therapy. *Anal. Chim. Acta* **2021**, *1178*, 338847. [\[CrossRef\]](#)
281. Zhou, Y.; Liu, Z.; Qiao, G.; Tang, B.; Li, P. Visualization of Endoplasmic Reticulum Viscosity in the Liver of Mice with Nonalcoholic Fatty Liver Disease by a near-Infrared Fluorescence Probe. *Chin. Chem. Lett.* **2021**, *32*, 3641–3645. [\[CrossRef\]](#)
282. Liu, Y.; Ma, Y.; Gao, W.; Ma, S.; Lin, W. Construction of a Fluorescent Probe with Large Stokes Shift and Deep Red Emission for Sensing of the Viscosity in Hyperglycemic Mice. *Dyes Pigments* **2021**, *195*, 109674. [\[CrossRef\]](#)
283. Xiao, H.; Li, P.; Tang, B. Small Molecular Fluorescent Probes for Imaging of Viscosity in Living Biosystems. *Chemistry* **2021**, *27*, 6880–6898. [\[CrossRef\]](#)
284. Shen, W.; Wang, P.; Xie, Z.; Zhou, H.; Hu, Y.; Fu, M.; Zhu, Q. A Bifunctional Probe Reveals Increased Viscosity and Hydrogen Sulfide in Zebra Fish Model of Parkinson's Disease. *Talanta* **2021**, *234*, 122621. [\[CrossRef\]](#)
285. Fang, Z.; Su, Z.; Qin, W.; Li, H.; Fang, B.; Du, W.; Wu, Q.; Peng, B.; Li, P.; Yu, H.; et al. Two-Photon Dual-Channel Fluorogenic Probe for in Situ Imaging the Mitochondrial H₂S/viscosity in the Brain of Drosophila Parkinson's Disease Model. *Chin. Chem. Lett.* **2020**, *31*, 2903–2908. [\[CrossRef\]](#)
286. Park, S.J.; Shin, B.K.; Lee, H.W.; Song, J.M.; Je, J.T.; Kim, H.M. Asymmetric Cyanine as a Far-Red Fluorescence Probe for Mitochondrial Viscosity. *Dyes Pigments* **2020**, *174*, 108080. [\[CrossRef\]](#)
287. Yang, J.; Guo, Y.; Pistolozzi, M.; Yan, J. Research Progress of Multi-Functional Fluorescent Probes for Alzheimer's Disease Monitoring. *Dyes Pigments* **2021**, *193*, 109466. [\[CrossRef\]](#)
288. Zhu, L.; Fu, M.; Yin, B.; Wang, L.; Chen, Y.; Zhu, Q. A Red-Emitting Fluorescent Probe for Mitochondria-Target Microviscosity in Living Cells and Blood Viscosity Detection in Hyperglycemia Mice. *Dyes Pigments* **2020**, *172*, 107859. [\[CrossRef\]](#)
289. Tan, H.-Y.; Qiu, Y.-T.; Sun, H.; Yan, J.-W.; Zhang, L. A Lysosome-Targeting Dual-Functional Fluorescent Probe for Imaging Intracellular Viscosity and Beta-Amyloid. *Chem. Commun.* **2019**, *55*, 2688–2691. [\[CrossRef\]](#)
290. Li, H.; Xin, C.; Zhang, G.; Han, X.; Qin, W.; Zhang, C.-W.; Yu, C.; Jing, S.; Li, L.; Huang, W. A Mitochondria-Targeted Two-Photon Fluorogenic Probe for the Dual-Imaging of Viscosity and H₂O₂ Levels in Parkinson's Disease Models. *J. Mater. Chem. B Mater. Biol. Med.* **2019**, *7*, 4243–4251. [\[CrossRef\]](#)

291. Yin, J.; Peng, M.; Lin, W. Visualization of Mitochondrial Viscosity in Inflammation, Fatty Liver, and Cancer Living Mice by a Robust Fluorescent Probe. *Anal. Chem.* **2019**, *91*, 8415–8421. [[CrossRef](#)] [[PubMed](#)]
292. Ren, M.; Zhou, K.; Wang, L.; Liu, K.; Lin, W. Construction of a Ratiometric Two-Photon Fluorescent Probe to Monitor the Changes of Mitochondrial Viscosity. *Sens. Actuators B Chem.* **2018**, *262*, 452–459. [[CrossRef](#)]
293. Jiang, N.; Fan, J.; Zhang, S.; Wu, T.; Wang, J.; Gao, P.; Qu, J.; Zhou, F.; Peng, X. Dual Mode Monitoring Probe for Mitochondrial Viscosity in Single Cell. *Sens. Actuators B Chem.* **2014**, *190*, 685–693. [[CrossRef](#)]
294. Aras, S.; Tek, I.; Varli, M.; Yalcin, A.; Cengiz, O.K.; Atmis, V.; Atli, T. Plasma Viscosity: Is a Biomarker for the Differential Diagnosis of Alzheimer's Disease and Vascular Dementia? *Am. J. Alzheimer's Dis. Other Dement.* **2013**, *28*, 62–68. [[CrossRef](#)] [[PubMed](#)]
295. Tang, Y.; Peng, J.; Zhang, Q.; Song, S.; Lin, W. A New NIR Emission Mitochondrial Targetable Fluorescent Probe and Its Application in Detecting Viscosity Changes in Mouse Liver and Kidney Injury. *Talanta* **2022**, *249*, 123647. [[CrossRef](#)]
296. Yu, F.-T.; Huang, Z.; Yang, J.-X.; Yang, L.-M.; Xu, X.-Y.; Huang, J.-Y.; Kong, L. Two Quinoline-Based Two-Photon Fluorescent Probes for Imaging of Viscosity in Subcellular Organelles of Living HeLa Cells. *Spectrochim. Acta Part A Mol. Biomol. Spectrosc.* **2022**, *283*, 121769. [[CrossRef](#)]
297. Wei, Y.-F.; Weng, X.-F.; Sha, X.-L.; Sun, R.; Xu, Y.-J.; Ge, J.-F. Simultaneous Imaging of Lysosomal and Mitochondrial Viscosity under Different Conditions Using a NIR Probe. *Sens. Actuators B Chem.* **2021**, *326*, 128954. [[CrossRef](#)]
298. Yan, F.; Sun, X.; Ma, T.; Zhang, Y.; Jiang, Y.; Wang, R.; Ma, C.; Wei, J.; Chen, L.; Cui, Y. A Viscosity-Dependent Carbon Dots with Anti-VEGF Properties for Monitoring and Promoting Apoptosis in Cancerous Cell. *Chem. Eng. J.* **2021**, *407*, 127801. [[CrossRef](#)]
299. Wu, Y.; Shu, W.; Zeng, C.; Guo, B.; Shi, J.; Jing, J.; Zhang, X. A Mitochondria Targetable and Viscosity Sensitive Fluorescent Probe and Its Applications for Distinguishing Cancerous Cells. *Dyes Pigments* **2019**, *168*, 134–139. [[CrossRef](#)]
300. Yang, Z.; He, Y.; Lee, J.-H.; Park, N.; Suh, M.; Chae, W.-S.; Cao, J.; Peng, X.; Jung, H.; Kang, C.; et al. A Self-Calibrating Bipartite Viscosity Sensor for Mitochondria. *J. Am. Chem. Soc.* **2013**, *135*, 9181–9185. [[CrossRef](#)]
301. Liang, L.-F.; Da, X.; Chen, T.-S.; Pei, Y.-H. Nucleoplasmic viscosity of living cells investigated by fluorescence correlation spectroscopy. *Guang Pu Xue Yu Guang Pu Fen Xi* **2009**, *29*, 459–462. [[PubMed](#)]
302. Zhang, S.; Zhang, Y.; Zhao, L.; Xu, L.; Han, H.; Huang, Y.; Fei, Q.; Sun, Y.; Ma, P.; Song, D. A Novel Water-Soluble near-Infrared Fluorescent Probe for Monitoring Mitochondrial Viscosity. *Talanta* **2021**, *233*, 122592. [[CrossRef](#)]
303. Chernyak, B.V.; Izyumov, D.S.; Lyamzaev, K.G.; Pashkovskaya, A.A.; Pletjushkina, O.Y.; Antonenko, Y.N.; Sakharov, D.V.; Wirtz, K.W.A.; Skulachev, V.P. Production of Reactive Oxygen Species in Mitochondria of HeLa Cells under Oxidative Stress. *Biochim. Biophys. Acta* **2006**, *1757*, 525–534. [[CrossRef](#)]
304. Sun, W.; Shi, Y.-D.; Ding, A.-X.; Tan, Z.-L.; Chen, H.; Liu, R.; Wang, R.; Lu, Z.-L. Imaging Viscosity and Peroxynitrite by a Mitochondria-Targeting Two-Photon Ratiometric Fluorescent Probe. *Sens. Actuators B Chem.* **2018**, *276*, 238–246. [[CrossRef](#)]
305. Xu, T.; Pagadala, V.; Mueller, D.M. Understanding Structure, Function, and Mutations in the Mitochondrial ATP Synthase. *Microb. Cell Factories* **2015**, *2*, 105–125. [[CrossRef](#)] [[PubMed](#)]
306. Nakamoto, R.K.; Scanlon, J.A.B.; Al-Shawi, M.K. The Rotary Mechanism of the ATP Synthase. *Arch. Biochem. Biophys.* **2008**, *476*, 43–50. [[CrossRef](#)]
307. Stock, D.; Gibbons, C.; Arechaga, I.; Leslie, A.G.; Walker, J.E. The Rotary Mechanism of ATP Synthase. *Curr. Opin. Struct. Biol.* **2000**, *10*, 672–679. [[CrossRef](#)]
308. Usukura, E.; Suzuki, T.; Furuike, S.; Soga, N.; Saita, E.-I.; Hisabori, T.; Kinoshita, K., Jr.; Yoshida, M. Torque Generation and Utilization in Motor Enzyme F₀F₁-ATP Synthase: Half-Torque F₁ with Short-Sized Pushrod Helix and Reduced ATP Synthesis by Half-Torque F₀F₁. *J. Biol. Chem.* **2012**, *287*, 1884–1891. [[CrossRef](#)]
309. Novichkova, N.S.; Malyan, A.N. The Effect of the Viscosity of a Trehalose Solution on ATP Hydrolysis by Chloroplast F₁-ATPase. *Biophysics* **2019**, *64*, 853–857. [[CrossRef](#)]
310. Nakanishi-Matsui, M.; Kashiwagi, S.; Hosokawa, H.; Cipriano, D.J.; Dunn, S.D.; Wada, Y.; Futai, M. Stochastic High-Speed Rotation of Escherichia Coli ATP Synthase F₁ Sector: The Epsilon Subunit-Sensitive Rotation. *J. Biol. Chem.* **2006**, *281*, 4126–4131. [[CrossRef](#)]
311. Capaldi, R.A.; Aggeler, R. Mechanism of the F₁(F₀)-Type ATP Synthase, a Biological Rotary Motor. *Trends Biochem. Sci.* **2002**, *27*, 154–160. [[CrossRef](#)]
312. Martin, J.L.; Ishmukhametov, R.; Hornung, T.; Ahmad, Z.; Frasch, W.D. Anatomy of F₁-ATPase Powered Rotation. *Proc. Natl. Acad. Sci. USA* **2014**, *111*, 3715–3720. [[CrossRef](#)]
313. Spetzler, D.; Ishmukhametov, R.; Hornung, T.; Day, L.J.; Martin, J.; Frasch, W.D. Single Molecule Measurements of F₁-ATPase Reveal an Interdependence between the Power Stroke and the Dwell Duration. *Biochemistry* **2009**, *48*, 7979–7985. [[CrossRef](#)] [[PubMed](#)]
314. Omote, H.; Sambonmatsu, N.; Saito, K.; Sambongi, Y.; Iwamoto-Kihara, A.; Yanagida, T.; Wada, Y.; Futai, M. The γ -Subunit Rotation and Torque Generation in F₁-ATPase from Wild-Type or Uncoupled Mutant *Escherichia coli*. *Proc. Natl. Acad. Sci. USA* **1999**, *96*, 7780–7784. [[CrossRef](#)]
315. Watanabe, R.; Hayashi, K.; Ueno, H.; Noji, H. Catalysis-Enhancement via Rotary Fluctuation of F₁-ATPase. *Biophys. J.* **2013**, *105*, 2385–2391. [[CrossRef](#)] [[PubMed](#)]
316. Begum, R.; Powner, M.B.; Hudson, N.; Hogg, C.; Jeffery, G. Treatment with 670 Nm Light up Regulates Cytochrome C Oxidase Expression and Reduces Inflammation in an Age-Related Macular Degeneration Model. *PLoS ONE* **2013**, *8*, e57828. [[CrossRef](#)]

317. Ahamed Basha, A.; Mathangi, D.C.; Shyamala, R. Effect of LED Photobiomodulation on Fluorescent Light Induced Changes in Cellular ATPases and Cytochrome c Oxidase Activity in Wistar Rat. *Lasers Med. Sci.* **2016**, *31*, 1803–1809. [[CrossRef](#)]
318. Gkotsi, D.; Begum, R.; Salt, T.; Lascaratos, G.; Hogg, C.; Chau, K.-Y.; Schapira, A.H.V.; Jeffery, G. Recharging Mitochondrial Batteries in Old Eyes. Near Infra-Red Increases ATP. *Exp. Eye Res.* **2014**, *122*, 50–53. [[CrossRef](#)] [[PubMed](#)]
319. Wong-Riley, M.T.T.; Liang, H.L.; Eells, J.T.; Chance, B.; Henry, M.M.; Buchmann, E.; Kane, M.; Whelan, H.T. Photobiomodulation Directly Benefits Primary Neurons Functionally Inactivated by Toxins: Role of cytochrome c oxidase *. *J. Biol. Chem.* **2005**, *280*, 4761–4771. [[CrossRef](#)] [[PubMed](#)]
320. Li, Y.; Park, J.-S.; Deng, J.-H.; Bai, Y. Cytochrome c Oxidase Subunit IV Is Essential for Assembly and Respiratory Function of the Enzyme Complex. *J. Bioenerg. Biomembr.* **2006**, *38*, 283–291. [[CrossRef](#)] [[PubMed](#)]
321. Sedláč, E.; Fabian, M.; Robinson, N.C.; Musatov, A. Ferricytochrome c Protects Mitochondrial Cytochrome c Oxidase against Hydrogen Peroxide-Induced Oxidative Damage. *Free Radic. Biol. Med.* **2010**, *49*, 1574–1581. [[CrossRef](#)] [[PubMed](#)]
322. Musatov, A.; Robinson, N.C. Susceptibility of Mitochondrial Electron-Transport Complexes to Oxidative Damage. Focus on Cytochrome c Oxidase. *Free Radic. Res.* **2012**, *46*, 1313–1326. [[CrossRef](#)]
323. Choksi, K.B.; Nuss, J.E.; Boylston, W.H.; Rabek, J.P.; Papaconstantinou, J. Age-Related Increases in Oxidatively Damaged Proteins of Mouse Kidney Mitochondrial Electron Transport Chain Complexes. *Free Radic. Biol. Med.* **2007**, *43*, 1423–1438. [[CrossRef](#)] [[PubMed](#)]
324. Zhang, Y.; Marcillat, O.; Giulivi, C.; Ernster, L.; Davies, K.J. The Oxidative Inactivation of Mitochondrial Electron Transport Chain Components and ATPase. *J. Biol. Chem.* **1990**, *265*, 16330–16336. [[CrossRef](#)]
325. Chen, J.; Schenker, S.; Frosto, T.A.; Henderson, G.I. Inhibition of Cytochrome c Oxidase Activity by 4-Hydroxynonenal (HNE). Role of HNE Adduct Formation with the Enzyme Subunits. *Biochim. Biophys. Acta* **1998**, *1380*, 336–344. [[CrossRef](#)] [[PubMed](#)]
326. Karu, T.I.; Pyatibrat, L.V.; Kolyakov, S.F.; Afanasyeva, N.I. Absorption Measurements of a Cell Monolayer Relevant to Phototherapy: Reduction of Cytochrome c Oxidase under near IR Radiation. *J. Photochem. Photobiol. B* **2005**, *81*, 98–106. [[CrossRef](#)]
327. Sommer, A.P.; Haddad, M.K.; Fecht, H.-J. Light Effect on Water Viscosity: Implication for ATP Biosynthesis. *Sci. Rep.* **2015**, *5*, 12029. [[CrossRef](#)] [[PubMed](#)]
328. Sommer, A.P. Mitochondrial Cytochrome c Oxidase Is Not the Primary Acceptor for near Infrared Light-It Is Mitochondrial Bound Water: The Principles of Low-Level Light Therapy. *Ann. Transl. Med.* **2019**, *7* (Suppl. 1), S13. [[CrossRef](#)] [[PubMed](#)]
329. Sommer, A.P.; Schemmer, P.; Pavláth, A.E.; Försterling, H.-D.; Mester, Á.R.; Trelles, M.A. Quantum Biology in Low Level Light Therapy: Death of a Dogma. *Ann. Transl. Med.* **2020**, *8*, 440. [[CrossRef](#)] [[PubMed](#)]
330. Hasinoff, B.B.; Davey, J.P. The Kinetics of the Aerobic Oxidation of Ferrocycytochrome c by Cytochrome c Oxidase in Solvents of Increased Viscosity Are Partially Diffusion Controlled. *Biochim. Biophys. Acta* **1987**, *892*, 1–9. [[CrossRef](#)] [[PubMed](#)]
331. Feng, C.; Kedia, R.V.; Hazzard, J.T.; Hurley, J.K.; Tollin, G.; Enemark, J.H. Effect of Solution Viscosity on Intramolecular Electron Transfer in Sulfite Oxidase. *Biochemistry* **2002**, *41*, 5816–5821. [[CrossRef](#)]
332. Passarella, S.; Ostuni, A.; Atlante, A.; Quagliariello, E. Increase in the ADP/ATP Exchange in Rat Liver Mitochondria Irradiated in Vitro by Helium-Neon Laser. *Biochem. Biophys. Res. Commun.* **1988**, *156*, 978–986. [[CrossRef](#)]
333. Karu, T.; Pyatibrat, L.; Kalendo, G. Irradiation with He Ne Laser Increases ATP Level in Cells Cultivated in Vitro. *J. Photochem. Photobiol. B* **1995**, *27*, 219–223. [[CrossRef](#)]
334. Sommer, A.P.; Bieschke, J.; Friedrich, R.P.; Zhu, D.; Wanker, E.E.; Fecht, H.J.; Mereles, D.; Hunstein, W. 670 Nm Laser Light and EGCG Complementarily Reduce Amyloid- β Aggregates in Human Neuroblastoma Cells: Basis for Treatment of Alzheimer's Disease? *Photomed. Laser Surg.* **2012**, *30*, 54–60. [[CrossRef](#)]
335. Castellano-González, G.; Pichaud, N.; Ballard, J.W.O.; Bessede, A.; Marcal, H.; Guillemin, G.J. Epigallocatechin-3-Gallate Induces Oxidative Phosphorylation by Activating Cytochrome c Oxidase in Human Cultured Neurons and Astrocytes. *Oncotarget* **2016**, *7*, 7426–7440. [[CrossRef](#)]
336. Reiter, R.J.; Rosales-Corral, S.; Tan, D.X.; Jou, M.J.; Galano, A.; Xu, B. Melatonin as a Mitochondria-Targeted Antioxidant: One of Evolution's Best Ideas. *Cell. Mol. Life Sci.* **2017**, *74*, 3863–3881. [[CrossRef](#)] [[PubMed](#)]
337. Petrosillo, G.; De Benedictis, V.; Ruggiero, F.M.; Paradies, G. Decline in Cytochrome c Oxidase Activity in Rat-Brain Mitochondria with Aging. Role of Peroxidized Cardiolipin and Beneficial Effect of Melatonin. *J. Bioenerg. Biomembr.* **2013**, *45*, 431–440. [[CrossRef](#)] [[PubMed](#)]
338. Agil, A.; Chayah, M.; Visiedo, L.; Navarro-Alarcon, M.; Rodríguez Ferrer, J.M.; Tassi, M.; Reiter, R.J.; Fernández-Vázquez, G. Melatonin Improves Mitochondrial Dynamics and Function in the Kidney of Zucker Diabetic Fatty Rats. *J. Clin. Med. Res.* **2020**, *9*, 2916. [[CrossRef](#)] [[PubMed](#)]
339. Fernández Vázquez, G.; Reiter, R.J.; Agil, A. Melatonin Increases Brown Adipose Tissue Mass and Function in Zucker Diabetic Fatty Rats: Implications for Obesity Control. *J. Pineal Res.* **2018**, *64*, e12472. [[CrossRef](#)]
340. Randi, E.B.; Zuhra, K.; Pecze, L.; Panagaki, T.; Szabo, C. Physiological Concentrations of Cyanide Stimulate Mitochondrial Complex IV and Enhance Cellular Bioenergetics. *Proc. Natl. Acad. Sci. USA* **2021**, *118*, e2026245118. [[CrossRef](#)]
341. Pacher, P. Cyanide Emerges as an Endogenous Mammalian Gasotransmitter. *Proc. Natl. Acad. Sci. USA* **2021**, *118*, e2108040118. [[CrossRef](#)]
342. Zuhra, K.; Szabo, C. The Two Faces of Cyanide: An Environmental Toxin and a Potential Novel Mammalian Gasotransmitter. *FEBS J.* **2022**, *289*, 2481–2515. [[CrossRef](#)]
343. Nelson, L. Acute Cyanide Toxicity: Mechanisms and Manifestations. *J. Emerg. Nurs.* **2006**, *32* (Suppl. 4), S8–S11. [[CrossRef](#)]

344. Leavesley, H.B.; Li, L.; Prabhakaran, K.; Borowitz, J.L.; Isom, G.E. Interaction of Cyanide and Nitric Oxide with Cytochrome c Oxidase: Implications for Acute Cyanide Toxicity. *Toxicol. Sci.* **2008**, *101*, 101–111. [\[CrossRef\]](#)
345. Gunasekar, P.G.; Borowitz, J.L.; Isom, G.E. Cyanide-Induced Generation of Oxidative Species: Involvement of Nitric Oxide Synthase and Cyclooxygenase-2. *J. Pharmacol. Exp. Ther.* **1998**, *285*, 236–241. [\[PubMed\]](#)
346. Zhang, W.; Xu, K.; Yue, L.; Shao, Z.; Feng, Y.; Fang, M. Two-Dimensional Carbazole-Based Derivatives as Versatile Chemosensors for Colorimetric Detection of Cyanide and Two-Photon Fluorescence Imaging of Viscosity in Vitro. *Dyes Pigments* **2017**, *137*, 560–568. [\[CrossRef\]](#)
347. Martín, M.; Macías, M.; León, J.; Escames, G.; Khaldy, H.; Acuña-Castroviejo, D. Melatonin Increases the Activity of the Oxidative Phosphorylation Enzymes and the Production of ATP in Rat Brain and Liver Mitochondria. *Int. J. Biochem. Cell Biol.* **2002**, *34*, 348–357. [\[CrossRef\]](#)
348. Martín, M.; Macías, M.; Escames, G.; Reiter, R.J.; Agapito, M.T.; Ortiz, G.G.; Acuña-Castroviejo, D. Melatonin-Induced Increased Activity of the Respiratory Chain Complexes I and IV Can Prevent Mitochondrial Damage Induced by Ruthenium Red in Vivo. *J. Pineal Res.* **2000**, *28*, 242–248. [\[CrossRef\]](#)
349. Jiang, G.-B.; Zheng, X.; Yao, J.-H.; Han, B.-J.; Li, W.; Wang, J.; Huang, H.-L.; Liu, Y.-J. Ruthenium(II) Polypyridyl Complexes Induce BEL-7402 Cell Apoptosis by ROS-Mediated Mitochondrial Pathway. *J. Inorg. Biochem.* **2014**, *141*, 170–179. [\[CrossRef\]](#) [\[PubMed\]](#)
350. Li, W.; Jiang, G.-B.; Yao, J.-H.; Wang, X.-Z.; Wang, J.; Han, B.-J.; Xie, Y.-Y.; Lin, G.-J.; Huang, H.-L.; Liu, Y.-J. Ruthenium(II) Complexes: DNA-Binding, Cytotoxicity, Apoptosis, Cellular Localization, Cell Cycle Arrest, Reactive Oxygen Species, Mitochondrial Membrane Potential and Western Blot Analysis. *J. Photochem. Photobiol. B* **2014**, *140*, 94–104. [\[CrossRef\]](#)
351. Li, Y.; Zhang, J.; Wan, J.; Liu, A.; Sun, J. Melatonin Regulates A β Production/clearance Balance and A β Neurotoxicity: A Potential Therapeutic Molecule for Alzheimer's Disease. *Biomed. Pharmacother.* **2020**, *132*, 110887. [\[CrossRef\]](#) [\[PubMed\]](#)
352. Cardinali, D.P. Melatonin: Clinical Perspectives in Neurodegeneration. *Front. Endocrinol.* **2019**, *10*, 480. [\[CrossRef\]](#)
353. Vincent, B. Protective Roles of Melatonin against the Amyloid-Dependent Development of Alzheimer's Disease: A Critical Review. *Pharmacol. Res.* **2018**, *134*, 223–237. [\[CrossRef\]](#)
354. Wang, J.-Z.; Wang, Z.-F. Role of Melatonin in Alzheimer-like Neurodegeneration. *Acta Pharmacol. Sin.* **2006**, *27*, 41–49. [\[CrossRef\]](#)
355. Srinivasan, V.; Pandi-Perumal, S.R.; Maestroni, G.J.; Esquifino, A.I.; Hardeland, R.; Cardinali, D.P. Role of Melatonin in Neurodegenerative Diseases. *Neurotox. Res.* **2005**, *7*, 293–318. [\[CrossRef\]](#) [\[PubMed\]](#)
356. Ono, K.; Mochizuki, H.; Ikeda, T.; Nihira, T.; Takasaki, J.-I.; Teplow, D.B.; Yamada, M. Effect of Melatonin on α -Synuclein Self-Assembly and Cytotoxicity. *Neurobiol. Aging* **2012**, *33*, 2172–2185. [\[CrossRef\]](#)
357. Lin, A.M.Y.; Fang, S.F.; Chao, P.L.; Yang, C.H. Melatonin Attenuates Arsenite-Induced Apoptosis in Rat Brain: Involvement of Mitochondrial and Endoplasmic Reticulum Pathways and Aggregation of Alpha-Synuclein. *J. Pineal Res.* **2007**, *43*, 163–171. [\[CrossRef\]](#)
358. Pappolla, M.; Bozner, P.; Soto, C.; Shao, H.; Robakis, N.K.; Zagorski, M.; Frangione, B.; Ghiso, J. Inhibition of Alzheimer Beta-Fibrillogenesis by Melatonin. *J. Biol. Chem.* **1998**, *273*, 7185–7188. [\[CrossRef\]](#) [\[PubMed\]](#)
359. Skribanek, Z.; Balásperi, L.; Mák, M. Interaction between Synthetic Amyloid-Beta-Peptide (1-40) and Its Aggregation Inhibitors Studied by Electrospray Ionization Mass Spectrometry. *J. Mass Spectrom.* **2001**, *36*, 1226–1229. [\[CrossRef\]](#) [\[PubMed\]](#)
360. Poeggeler, B.; Miravalle, L.; Zagorski, M.G.; Wisniewski, T.; Chyan, Y.J.; Zhang, Y.; Shao, H.; Bryant-Thomas, T.; Vidal, R.; Frangione, B.; et al. Melatonin Reverses the Profibrillogenic Activity of Apolipoprotein E4 on the Alzheimer Amyloid Abeta Peptide. *Biochemistry* **2001**, *40*, 14995–15001. [\[CrossRef\]](#)
361. Pappolla, M.A.; Matsubara, E.; Vidal, R.; Pacheco-Quinto, J.; Poeggeler, B.; Zagorski, M.; Sambamurti, K. Melatonin Treatment Enhances A β Lymphatic Clearance in a Transgenic Mouse Model of Amyloidosis. *Curr. Alzheimer Res.* **2018**, *15*, 637–642. [\[CrossRef\]](#)
362. Matsubara, E.; Bryant-Thomas, T.; Pacheco Quinto, J.; Henry, T.L.; Poeggeler, B.; Herbert, D.; Cruz-Sanchez, F.; Chyan, Y.-J.; Smith, M.A.; Perry, G.; et al. Melatonin Increases Survival and Inhibits Oxidative and Amyloid Pathology in a Transgenic Model of Alzheimer's Disease. *J. Neurochem.* **2003**, *85*, 1101–1108. [\[CrossRef\]](#)
363. Quinn, J.; Kulhanek, D.; Nowlin, J.; Jones, R.; Praticò, D.; Rokach, J.; Stackman, R. Chronic Melatonin Therapy Fails to Alter Amyloid Burden or Oxidative Damage in Old Tg2576 Mice: Implications for Clinical Trials. *Brain Res.* **2005**, *1037*, 209–213. [\[CrossRef\]](#)
364. Lahiri, D.K.; Chen, D.; Ge, Y.-W.; Bondy, S.C.; Sharman, E.H. Dietary Supplementation with Melatonin Reduces Levels of Amyloid Beta-Peptides in the Murine Cerebral Cortex. *J. Pineal Res.* **2004**, *36*, 224–231. [\[CrossRef\]](#)
365. Ali, T.; Kim, M.O. Melatonin Ameliorates Amyloid Beta-Induced Memory Deficits, Tau Hyperphosphorylation and Neurodegeneration via PI3/Akt/GSK3 β Pathway in the Mouse Hippocampus. *J. Pineal Res.* **2015**, *59*, 47–59. [\[CrossRef\]](#) [\[PubMed\]](#)
366. Luengo, E.; Buendia, I.; Fernández-Mendivil, C.; Trigo-Alonso, P.; Negredo, P.; Michalska, P.; Hernández-García, B.; Sánchez-Ramos, C.; Bernal, J.A.; Ikezu, T.; et al. Pharmacological Doses of Melatonin Impede Cognitive Decline in Tau-Related Alzheimer Models, Once Tauopathy Is Initiated, by Restoring the Autophagic Flux. *J. Pineal Res.* **2019**, *67*, e12578. [\[CrossRef\]](#)
367. Balmik, A.A.; Das, R.; Dangi, A.; Gorantla, N.V.; Marelli, U.K.; Chinnathambi, S. Melatonin Interacts with Repeat Domain of Tau to Mediate Disaggregation of Paired Helical Filaments. *Biochim. Biophys. Acta Gen. Subj.* **2020**, *1864*, 129467. [\[CrossRef\]](#)
368. Das, R.; Balmik, A.A.; Chinnathambi, S. Effect of Melatonin on Tau Aggregation and Tau-Mediated Cell Surface Morphology. *Int. J. Biol. Macromol.* **2020**, *152*, 30–39. [\[CrossRef\]](#) [\[PubMed\]](#)

369. Kumar, S.; Nussinov, R. Salt Bridge Stability in Monomeric Proteins. *J. Mol. Biol.* **1999**, *293*, 1241–1255. [[CrossRef](#)]
370. Xu, D.; Tsai, C.J.; Nussinov, R. Hydrogen Bonds and Salt Bridges across Protein-Protein Interfaces. *Protein Eng.* **1997**, *10*, 999–1012. [[CrossRef](#)]
371. Musafia, B.; Buchner, V.; Arad, D. Complex Salt Bridges in Proteins: Statistical Analysis of Structure and Function. *J. Mol. Biol.* **1995**, *254*, 761–770. [[CrossRef](#)] [[PubMed](#)]
372. Tarus, B.; Straub, J.E.; Thirumalai, D. Dynamics of Asp23-Lys28 Salt-Bridge Formation in Abeta10-35 Monomers. *J. Am. Chem. Soc.* **2006**, *128*, 16159–16168. [[CrossRef](#)]
373. Zwier, T.S. The spectroscopy of solvation in hydrogen-bonded aromatic clusters. *Annu. Rev. Phys. Chem.* **1996**, *47*, 205–241. [[CrossRef](#)]
374. Zhu, L.; Gong, Y.; Lju, H.; Sun, G.; Zhang, Q.; Qian, Z. Mechanisms of Melatonin Binding and Destabilizing the Protofilament and Filament of Tau R3–R4 Domains Revealed by Molecular Dynamics Simulation. *Phys. Chem. Chem. Phys.* **2021**, *23*, 20615–20626. [[CrossRef](#)]
375. Köpke, E.; Tung, Y.C.; Shaikh, S.; Alonso, A.C.; Iqbal, K.; Grundke-Iqbal, I. Microtubule-Associated Protein Tau. Abnormal Phosphorylation of a Non-Paired Helical Filament Pool in Alzheimer Disease. *J. Biol. Chem.* **1993**, *268*, 24374–24384. [[CrossRef](#)] [[PubMed](#)]
376. Wang, J.Z.; Gong, C.X.; Zaidi, T.; Grundke-Iqbal, I.; Iqbal, K. Dephosphorylation of Alzheimer Paired Helical Filaments by Protein Phosphatase-2A and -2B. *J. Biol. Chem.* **1995**, *270*, 4854–4860. [[CrossRef](#)] [[PubMed](#)]
377. Wang, J.Z.; Grundke-Iqbal, I.; Iqbal, K. Restoration of Biological Activity of Alzheimer Abnormally Phosphorylated Tau by Dephosphorylation with Protein Phosphatase-2A, -2B and -1. *Brain Res. Mol. Brain Res.* **1996**, *38*, 200–208. [[CrossRef](#)] [[PubMed](#)]
378. Poppek, D.; Keck, S.; Ermak, G.; Jung, T.; Stolzing, A.; Ullrich, O.; Davies, K.J.A.; Grune, T. Phosphorylation Inhibits Turnover of the Tau Protein by the Proteasome: Influence of RCAN1 and Oxidative Stress. *Biochem. J.* **2006**, *400*, 511–520. [[CrossRef](#)]
379. Bloom, G.S. Amyloid- β and Tau: The Trigger and Bullet in Alzheimer Disease Pathogenesis. *JAMA Neurol.* **2014**, *71*, 505–508. [[CrossRef](#)]
380. Cline, E.N.; Bicca, M.A.; Viola, K.L.; Klein, W.L. The Amyloid- β Oligomer Hypothesis: Beginning of the Third Decade. *J. Alzheimer's Dis.* **2018**, *64*, S567–S610. [[CrossRef](#)]
381. Kametani, F.; Hasegawa, M. Reconsideration of Amyloid Hypothesis and Tau Hypothesis in Alzheimer's Disease. *Front. Neurosci.* **2018**, *12*, 25. [[CrossRef](#)]
382. Chen, X.-Q.; Mobley, W.C. Alzheimer Disease Pathogenesis: Insights From Molecular and Cellular Biology Studies of Oligomeric A β and Tau Species. *Front. Neurosci.* **2019**, *13*, 659. [[CrossRef](#)]
383. Chen, X.; Chen, M.; Schafer, N.P.; Wolynes, P.G. Exploring the Interplay between Fibrillization and Amorphous Aggregation Channels on the Energy Landscapes of Tau Repeat Isoforms. *Proc. Natl. Acad. Sci. USA* **2020**, *117*, 4125–4130. [[CrossRef](#)] [[PubMed](#)]
384. Ruben, G.C.; Ciardelli, T.L.; Grundke-Iqbal, I.; Iqbal, K. Alzheimer Disease Hyperphosphorylated Tau Aggregates Hydrophobically. *Synapse* **1997**, *27*, 208–229. [[CrossRef](#)]
385. Wegmann, S.; Eftekharzadeh, B.; Tepper, K.; Zoltowska, K.M.; Bennett, R.E.; Dujardin, S.; Laskowski, P.R.; MacKenzie, D.; Kamath, T.; Commins, C.; et al. Tau Protein Liquid-Liquid Phase Separation Can Initiate Tau Aggregation. *EMBO J.* **2018**, *37*, e98049. [[CrossRef](#)] [[PubMed](#)]
386. Mandl, I.; Grauer, A.; Neuberger, C. Solubilization of Insoluble Matter in Nature; I. The Part Played by Salts of Adenosinetriphosphate. *Biochim. Biophys. Acta* **1952**, *8*, 654–663. [[CrossRef](#)] [[PubMed](#)]
387. Hayes, M.H.; Peuchen, E.H.; Dovichi, N.J.; Weeks, D.L. Dual Roles for ATP in the Regulation of Phase Separated Protein Aggregates in Xenopus Oocyte Nucleoli. *Life* **2018**, *7*, e35224. [[CrossRef](#)]
388. Patel, A.; Malinowska, L.; Saha, S.; Wang, J.; Alberti, S.; Krishnan, Y.; Hyman, A.A. ATP as a Biological Hydrotrope. *Science* **2017**, *356*, 753–756. [[CrossRef](#)]
389. Kurisaki, I.; Tanaka, S. ATP Converts A β 42 Oligomer into Off-Pathway Species by Making Contact with Its Backbone Atoms Using Hydrophobic Adenosine. *J. Phys. Chem. B* **2019**, *123*, 9922–9933. [[CrossRef](#)] [[PubMed](#)]
390. Mehringer, J.; Do, T.-M.; Touraud, D.; Hohenschutz, M.; Khoshshima, A.; Horinek, D.; Kunz, W. Hofmeister versus Neuberger: Is ATP Really a Biological Hydrotrope? *Cell Rep. Phys. Sci.* **2021**, *2*, 100343. [[CrossRef](#)]
391. Zhang, C.; Rissman, R.A.; Feng, J. Characterization of ATP Alternations in an Alzheimer's Disease Transgenic Mouse Model. *J. Alzheimer's Dis.* **2015**, *44*, 375–378. [[CrossRef](#)]
392. Pal, S.; Paul, S. ATP Controls the Aggregation of A β 16-22 Peptides. *J. Phys. Chem. B* **2020**, *124*, 210–223. [[CrossRef](#)]
393. Oro, J. Mechanism of Synthesis of Adenine from Hydrogen Cyanide under Possible Primitive Earth Conditions. *Nature* **1961**, *191*, 1193–1194. [[CrossRef](#)]
394. Newby, A.C. Adenosine and the Concept of “retaliatory Metabolites”. *Trends Biochem. Sci.* **1984**, *9*, 42–44. [[CrossRef](#)]
395. Cate, J.H.; Gooding, A.R.; Podell, E.; Zhou, K.; Golden, B.L.; Szewczak, A.A.; Kundrot, C.E.; Cech, T.R.; Doudna, J.A. RNA Tertiary Structure Mediation by Adenosine Platforms. *Science* **1996**, *273*, 1696–1699. [[CrossRef](#)]
396. Ren, C.-L.; Shan, Y.; Zhang, P.; Ding, H.-M.; Ma, Y.-Q. Uncovering the Molecular Mechanism for Dual Effect of ATP on Phase Separation in FUS Solution. *Sci. Adv.* **2022**, *8*, eabo7885. [[CrossRef](#)]
397. Kang, J.; Lim, L.; Song, J. ATP Enhances at Low Concentrations but Dissolves at High Concentrations Liquid-Liquid Phase Separation (LLPS) of ALS/FTD-Causing FUS. *Biochem. Biophys. Res. Commun.* **2018**, *504*, 545–551. [[CrossRef](#)]

398. Henninger, J.E.; Oksuz, O.; Shrinivas, K.; Sagi, I.; LeRoy, G.; Zheng, M.M.; Andrews, J.O.; Zamudio, A.V.; Lazaris, C.; Hannett, N.M.; et al. RNA-Mediated Feedback Control of Transcriptional Condensates. *Cell* **2021**, *184*, 207–225.e24. [\[CrossRef\]](#) [\[PubMed\]](#)
399. Maharana, S.; Wang, J.; Papadopoulos, D.K.; Richter, D.; Pozniakovsky, A.; Poser, I.; Bickle, M.; Rizk, S.; Guillén-Boixet, J.; Franzmann, T.M.; et al. RNA Buffers the Phase Separation Behavior of Prion-like RNA Binding Proteins. *Science* **2018**, *360*, 918–921. [\[CrossRef\]](#) [\[PubMed\]](#)
400. Aida, H.; Shigeta, Y.; Harada, R. The Role of ATP in Solubilizing RNA-Binding Protein Fused in Sarcoma. *Proteins* **2022**, *90*, 1606–1612. [\[CrossRef\]](#)
401. Coskuner, O.; Murray, I.V.J. Adenosine Triphosphate (ATP) Reduces Amyloid- β Protein Misfolding in Vitro. *J. Alzheimer's Dis.* **2014**, *41*, 561–574. [\[CrossRef\]](#) [\[PubMed\]](#)
402. Cohen, T.; Frydman-Marom, A.; Rechter, M.; Gazit, E. Inhibition of Amyloid Fibril Formation and Cytotoxicity by Hydroxyindole Derivatives. *Biochemistry* **2006**, *45*, 4727–4735. [\[CrossRef\]](#) [\[PubMed\]](#)
403. Morshedi, D.; Rezaei-Ghaleh, N.; Ebrahim-Habibi, A.; Ahmadian, S.; Nemat-Gorgani, M. Inhibition of Amyloid Fibrillation of Lysozyme by Indole Derivatives—Possible Mechanism of Action. *FEBS J.* **2007**, *274*, 6415–6425. [\[CrossRef\]](#)
404. Swan, I.D. The Inhibition of Hen Egg-White Lysozyme by Imidazole and Indole Derivatives. *J. Mol. Biol.* **1972**, *65*, 59–62. [\[CrossRef\]](#) [\[PubMed\]](#)
405. Di Bella, G.; Mascia, F.; Gualano, L.; Di Bella, L. Melatonin Anticancer Effects: Review. *Int. J. Mol. Sci.* **2013**, *14*, 2410–2430. [\[CrossRef\]](#)
406. Di Bella, G.; Gualano, L.; Di Bella, L. Melatonin with Adenosine Solubilized in Water and Stabilized with Glycine for Oncological Treatment—Technical Preparation, Effectivity and Clinical Findings. *Neuro Endocrinol. Lett.* **2017**, *38*, 465–474.
407. Todisco, M. Effectiveness of a Treatment Based on Melatonin in Five Patients with Systemic Sclerosis. *Am. J. Ther.* **2006**, *13*, 84–87. [\[CrossRef\]](#)
408. Todisco, M.; Rossi, N. Melatonin for Refractory Idiopathic Thrombocytopenic Purpura: A Report of 3 Cases. *Am. J. Ther.* **2002**, *9*, 524–526. [\[CrossRef\]](#) [\[PubMed\]](#)
409. Pattanayak, G.K.; Liao, Y.; Wallace, E.W.J.; Budnik, B.; Drummond, D.A.; Rust, M.J. Daily Cycles of Reversible Protein Condensation in Cyanobacteria. *Cell Rep.* **2020**, *32*, 108032. [\[CrossRef\]](#) [\[PubMed\]](#)
410. Borea, P.A.; Gessi, S.; Merighi, S.; Vincenzi, F.; Varani, K. Pathological Overproduction: The Bad Side of Adenosine. *Br. J. Pharmacol.* **2017**, *174*, 1945–1960. [\[CrossRef\]](#) [\[PubMed\]](#)
411. Zimmermann, H. Extracellular Metabolism of ATP and Other Nucleotides. *Naunyn-Schmiedeberg's Arch. Pharmacol.* **2000**, *362*, 299–309. [\[CrossRef\]](#)
412. Fredholm, B.B. Physiological and Pathophysiological Roles of Adenosine. *Sleep Biol. Rhythms* **2011**, *9*, 24–28. [\[CrossRef\]](#)
413. Chang, C.-P.; Wu, K.-C.; Lin, C.-Y.; Chern, Y. Emerging Roles of Dysregulated Adenosine Homeostasis in Brain Disorders with a Specific Focus on Neurodegenerative Diseases. *J. Biomed. Sci.* **2021**, *28*, 70. [\[CrossRef\]](#)
414. Antonioli, L.; Blandizzi, C.; Pacher, P.; Haskó, G. Immunity, Inflammation and Cancer: A Leading Role for Adenosine. *Nat. Rev. Cancer* **2013**, *13*, 842–857. [\[CrossRef\]](#)
415. Morandi, F.; Horenstein, A.L.; Rizzo, R.; Malavasi, F. The Role of Extracellular Adenosine Generation in the Development of Autoimmune Diseases. *Mediat. Inflamm.* **2018**, *2018*, 7019398. [\[CrossRef\]](#) [\[PubMed\]](#)
416. Blay, J.; White, T.D.; Hoskin, D.W. The Extracellular Fluid of Solid Carcinomas Contains Immunosuppressive Concentrations of Adenosine. *Cancer Res.* **1997**, *57*, 2602–2605.
417. Rudolph, K.A.; Schubert, P. Adenosine and Brain Ischemia. In *Adenosine and Adenine Nucleotides: From Molecular Biology to Integrative Physiology*; Belardinelli, L., Pelleg, A., Eds.; Springer: Boston, MA, USA, 1995; pp. 391–397. [\[CrossRef\]](#)
418. Tan, D.-X.; Hardeland, R. The Reserve/Maximum Capacity of Melatonin's Synthetic Function for the Potential Dimorphism of Melatonin Production and Its Biological Significance in Mammals. *Molecules* **2021**, *26*, 7302. [\[CrossRef\]](#) [\[PubMed\]](#)
419. Löfgren, L.; Pehrsson, S.; Hägglund, G.; Tjellström, H.; Nylander, S. Accurate Measurement of Endogenous Adenosine in Human Blood. *PLoS ONE* **2018**, *13*, e0205707. [\[CrossRef\]](#)
420. Waldhauser, F.; Weiszenbacher, G.; Tatzer, E.; Gisinger, B.; Waldhauser, M.; Schemper, M.; Frisch, H. Alterations in Nocturnal Serum Melatonin Levels in Humans with Growth and Aging. *J. Clin. Endocrinol. Metab.* **1988**, *66*, 648–652. [\[CrossRef\]](#) [\[PubMed\]](#)
421. Kennaway, D.J.; Lushington, K.; Dawson, D.; Lack, L.; van den Heuvel, C.; Rogers, N. Urinary 6-Sulfatoxymelatonin Excretion and Aging: New Results and a Critical Review of the Literature. *J. Pineal Res.* **1999**, *27*, 210–220. [\[CrossRef\]](#)
422. Waldhauser, F.; Kovács, J.; Reiter, E. Age-Related Changes in Melatonin Levels in Humans and Its Potential Consequences for Sleep Disorders. *Exp. Gerontol.* **1998**, *33*, 759–772. [\[CrossRef\]](#)
423. Waldhauser, F.; Weiszenbacher, G.; Frisch, H.; Zeitlhuber, U.; Waldhauser, M.; Wurtman, R.J. Fall in Nocturnal Serum Melatonin during Prepuberty and Pubescence. *Lancet* **1984**, *1*, 362–365. [\[CrossRef\]](#)
424. Burch, J.B.; Reif, J.S.; Yost, M.G.; Keefe, T.J.; Pitrat, C.A. Reduced Excretion of a Melatonin Metabolite in Workers Exposed to 60 Hz Magnetic Fields. *Am. J. Epidemiol.* **1999**, *150*, 27–36. [\[CrossRef\]](#)
425. Prayag, A.S.; Najjar, R.P.; Gronfier, C. Melatonin Suppression Is Exquisitely Sensitive to Light and Primarily Driven by Melanopsin in Humans. *J. Pineal Res.* **2019**, *66*, e12562. [\[CrossRef\]](#)
426. Bojkowski, C.J.; Aldhous, M.E.; English, J.; Franey, C.; Poulton, A.L.; Skene, D.J.; Arendt, J. Suppression of Nocturnal Plasma Melatonin and 6-Sulphatoxymelatonin by Bright and Dim Light in Man. *Horm. Metab. Res.* **1987**, *19*, 437–440. [\[CrossRef\]](#)

427. Tan, D.-X.; Reiter, R.J.; Manchester, L.C.; Yan, M.-T.; El-Sawi, M.; Sainz, R.M.; Mayo, J.C.; Kohen, R.; Allegra, M.; Hardeland, R. Chemical and Physical Properties and Potential Mechanisms: Melatonin as a Broad Spectrum Antioxidant and Free Radical Scavenger. *Curr. Top. Med. Chem.* **2002**, *2*, 181–197. [CrossRef]
428. National Center for Biotechnology Information. *PubChem Compound Summary for CID 5957, Adenosine-5'-Triphosphate*; National Center for Biotechnology Information: Bethesda, MD, USA, 2023. Available online: https://pubchem.ncbi.nlm.nih.gov/compound/Adenosine-5_-triphosphate (accessed on 5 January 2023).
429. National Center for Biotechnology Information. *PubChem Compound Summary for CID 896, Melatonin*; National Center for Biotechnology Information: Bethesda, MD, USA, 2023. Available online: <https://pubchem.ncbi.nlm.nih.gov/compound/Melatonin> (accessed on 7 January 2023).
430. Ishii, M.; Wang, G.; Racchumi, G.; Dyke, J.P.; Iadecola, C. Transgenic Mice Overexpressing Amyloid Precursor Protein Exhibit Early Metabolic Deficits and a Pathologically Low Leptin State Associated with Hypothalamic Dysfunction in Arcuate Neuropeptide Y Neurons. *J. Neurosci.* **2014**, *34*, 9096–9106. [CrossRef]
431. Feng, Z.; Qin, C.; Chang, Y.; Zhang, J.-T. Early Melatonin Supplementation Alleviates Oxidative Stress in a Transgenic Mouse Model of Alzheimer's Disease. *Free. Radic. Biol. Med.* **2006**, *40*, 101–109. [CrossRef] [PubMed]
432. Bachmanov, A.A.; Reed, D.R.; Beauchamp, G.K.; Tordoff, M.G. Food Intake, Water Intake, and Drinking Spout Side Preference of 28 Mouse Strains. *Behav. Genet.* **2002**, *32*, 435–443. [CrossRef] [PubMed]
433. Gould, S.J. Allometry in Primates, with Emphasis on Scaling and the Evolution of the Brain. *Contrib. Primatol.* **1975**, *5*, 244–292. [PubMed]
434. Mirth, C.K.; Frankino, W.A.; Shingleton, A.W. Allometry and Size Control: What Can Studies of Body Size Regulation Teach Us about the Evolution of Morphological Scaling Relationships? *Curr. Opin. Insect Sci.* **2016**, *13*, 93–98. [CrossRef]
435. Galilei, G. *Dialogues Concerning Two New Sciences*, 1637; Crew, H.; De Salvio, A., Translators; Macmillan: New York, NY, USA, 1914.
436. Boxenbaum, H. Interspecies Scaling, Allometry, Physiological Time, and the Ground Plan of Pharmacokinetics. *J. Pharmacokinet. Pharmacodyn.* **1982**, *10*, 201–227. [CrossRef] [PubMed]
437. Boxenbaum, H.; Fertig, J.B. Scaling of Antipyrine Intrinsic Clearance of Unbound Drug in 15 Mammalian Species. *Eur. J. Drug Metab. Pharmacokinet.* **1984**, *9*, 177–183. [CrossRef]
438. Boxenbaum, H. Interspecies Pharmacokinetic Scaling and the Evolutionary-Comparative Paradigm. *Drug Metab. Rev.* **1984**, *15*, 1071–1121. [CrossRef]
439. Ritschel, W.A.; Vachharajani, N.N.; Johnson, R.D.; Hussain, A.S. The Allometric Approach for Interspecies Scaling of Pharmacokinetic Parameters. *Comp. Biochem. Physiol. Part C* **1992**, *103*, 249–253. [CrossRef] [PubMed]
440. Boxenbaum, H.; DiLea, C. First-Time-in-Human Dose Selection: Allometric Thoughts and Perspectives. *J. Clin. Pharmacol.* **1995**, *35*, 957–966. [CrossRef] [PubMed]
441. Hu, T.M.; Hayton, W.L. Allometric Scaling of Xenobiotic Clearance: Uncertainty versus Universality. *AAPS PharmSci* **2001**, *3*, E29. [CrossRef] [PubMed]
442. Gibson, S.; Numa, A. The Importance of Metabolic Rate and the Folly of Body Surface Area Calculations. *Anaesthesia* **2003**, *58*, 50–55. [CrossRef] [PubMed]
443. Savage, V.M.; Gillooly, J.F.; Woodruff, W.H.; West, G.B.; Allen, A.P.; Enquist, B.J.; Brown, J.H. The Predominance of Quarter-Power Scaling in Biology. *Funct. Ecol.* **2004**, *18*, 257–282. [CrossRef]
444. Glazier, D.S. Beyond the “3/4-Power Law”: Variation in the Intra- and Interspecific Scaling of Metabolic Rate in Animals. *Biol. Rev. Camb. Philos. Soc.* **2005**, *80*, 611–662. [CrossRef]
445. West, G.B.; Brown, J.H. The Origin of Allometric Scaling Laws in Biology from Genomes to Ecosystems: Towards a Quantitative Unifying Theory of Biological Structure and Organization. *J. Exp. Biol.* **2005**, *208 Pt 9*, 1575–1592. [CrossRef]
446. Rhomberg, L.R.; Lewandowski, T.A. Methods for Identifying a Default Cross-Species Scaling Factor. *Hum. Ecol. Risk Assess. Int. J.* **2006**, *12*, 1094–1127. [CrossRef]
447. Reagan-Shaw, S.; Nihal, M.; Ahmad, N. Dose Translation from Animal to Human Studies Revisited. *FASEB J.* **2008**, *22*, 659–661. [CrossRef]
448. White, C.R.; Kearney, M.R. Metabolic Scaling in Animals: Methods, Empirical Results, and Theoretical Explanations. *Compr. Physiol.* **2014**, *4*, 231–256. [CrossRef]
449. Glazier, D.S. Effects of Contingency versus Constraints on the Body-Mass Scaling of Metabolic Rate. *Challenges* **2018**, *9*, 4. [CrossRef]
450. Li, X.-X.; He, J.-H. Along the evolution process kleiber's 3/4 law makes way for rubner's surface law: A fractal approach. *Fractals* **2019**, *27*, 1950015. [CrossRef]
451. Dodds, P.S.; Rothman, D.H.; Weitz, J.S. Re-Examination of the “3/4-Law” of Metabolism. *J. Theor. Biol.* **2001**, *209*, 9–27. [CrossRef] [PubMed]
452. White, C.R.; Seymour, R.S. Mammalian Basal Metabolic Rate Is Proportional to Body mass^{2/3}. *Proc. Natl. Acad. Sci. USA* **2003**, *100*, 4046–4049. [CrossRef]
453. White, C.R.; Cassey, P.; Blackburn, T.M. Allometric Exponents Do Not Support a Universal Metabolic Allometry. *Ecology* **2007**, *88*, 315–323. [CrossRef]
454. Banavar, J.R.; Moses, M.E.; Brown, J.H.; Damuth, J.; Rinaldo, A.; Sibly, R.M.; Maritan, A. A General Basis for Quarter-Power Scaling in Animals. *Proc. Natl. Acad. Sci. USA* **2010**, *107*, 15816–15820. [CrossRef]

455. Dreyer, G. The normal basal metabolism in man: And its relation to the size of the body and age, expressed in simple formulæ. *Lancet* **1920**, *196*, 289–291. [\[CrossRef\]](#)
456. Kleiber, M. Body Size and Metabolism. *Hilgardia* **1932**, *6*, 315–353. [\[CrossRef\]](#)
457. Nair, A.B.; Jacob, S. A Simple Practice Guide for Dose Conversion between Animals and Human. *J. Basic Clin. Pharm.* **2016**, *7*, 27–31. [\[CrossRef\]](#) [\[PubMed\]](#)
458. Reading, B.D.; Freeman, B. Simple Formula for the Surface Area of the Body and a Simple Model for Anthropometry. *Clin. Anat.* **2005**, *18*, 126–130. [\[CrossRef\]](#)
459. Cheung, M.C.; Spalding, P.B.; Gutierrez, J.C.; Balkan, W.; Namias, N.; Koniaris, L.G.; Zimmers, T.A. Body Surface Area Prediction in Normal, Hypermuscular, and Obese Mice. *J. Surg. Res.* **2009**, *153*, 326–331. [\[CrossRef\]](#)
460. Gouma, E.; Simos, Y.; Verginadis, I.; Lykoudis, E.; Evangelou, A.; Karkabounas, S. A Simple Procedure for Estimation of Total Body Surface Area and Determination of a New Value of Meeh's Constant in Rats. *Lab. Anim.* **2012**, *46*, 40–45. [\[CrossRef\]](#)
461. Toutain, P.L.; Bousquet-Mélou, A. Bioavailability and Its Assessment. *J. Vet. Pharmacol. Ther.* **2004**, *27*, 455–466. [\[CrossRef\]](#) [\[PubMed\]](#)
462. Waldhauser, F.; Waldhauser, M.; Lieberman, H.R.; Deng, M.H.; Lynch, H.J.; Wurtman, R.J. Bioavailability of Oral Melatonin in Humans. *Neuroendocrinology* **1984**, *39*, 307–313. [\[CrossRef\]](#)
463. Zetner, D.; Andersen, L.P.K.; Alder, R.; Jessen, M.L.; Tolstrup, A.; Rosenberg, J. Pharmacokinetics and Safety of Intravenous, Intravesical, Rectal, Transdermal, and Vaginal Melatonin in Healthy Female Volunteers: A Cross-Over Study. *Pharmacology* **2021**, *106*, 169–176. [\[CrossRef\]](#) [\[PubMed\]](#)
464. Lane, E.A.; Moss, H.B. Pharmacokinetics of Melatonin in Man: First Pass Hepatic Metabolism. *J. Clin. Endocrinol. Metab.* **1985**, *61*, 1214–1216. [\[CrossRef\]](#)
465. DeMuro, R.L.; Nafziger, A.N.; Blask, D.E.; Menhinick, A.M.; Bertino, J.S., Jr. The Absolute Bioavailability of Oral Melatonin. *J. Clin. Pharmacol.* **2000**, *40*, 781–784. [\[CrossRef\]](#) [\[PubMed\]](#)
466. Harpsøe, N.G.; Andersen, L.P.H.; Gögenur, I.; Rosenberg, J. Clinical Pharmacokinetics of Melatonin: A Systematic Review. *Eur. J. Clin. Pharmacol.* **2015**, *71*, 901–909. [\[CrossRef\]](#)
467. Andersen, L.P.H.; Werner, M.U.; Rosenkilde, M.M.; Harpsøe, N.G.; Fuglsang, H.; Rosenberg, J.; Gögenur, I. Pharmacokinetics of Oral and Intravenous Melatonin in Healthy Volunteers. *BMC Pharmacol. Toxicol.* **2016**, *17*, 8. [\[CrossRef\]](#)
468. Di, W.L.; Kadva, A.; Johnston, A.; Silman, R. Variable Bioavailability of Oral Melatonin. *N. Engl. J. Med.* **1997**, *336*, 1028–1029. [\[CrossRef\]](#)
469. Bagci, S.; Horoz, Ö.Ö.; Yildizdas, D.; Reinsberg, J.; Bartmann, P.; Müller, A. Melatonin Status in Pediatric Intensive Care Patients with Sepsis. *Pediatr. Crit. Care Med.* **2012**, *13*, e120–e123. [\[CrossRef\]](#)
470. Tordjman, S.; Anderson, G.M.; Pichard, N.; Charbuy, H.; Touitou, Y. Nocturnal Excretion of 6-Sulphatoxymelatonin in Children and Adolescents with Autistic Disorder. *Biol. Psychiatry* **2005**, *57*, 134–138. [\[CrossRef\]](#) [\[PubMed\]](#)
471. Tordjman, S.; Anderson, G.M.; Bellissant, E.; Botbol, M.; Charbuy, H.; Camus, F.; Gaignic, R.; Kermarrec, S.; Fougere, C.; Cohen, D.; et al. Day and Nighttime Excretion of 6-Sulphatoxymelatonin in Adolescents and Young Adults with Autistic Disorder. *Psychoneuroendocrinology* **2012**, *37*, 1990–1997. [\[CrossRef\]](#)
472. Kennaway, D.J.; Flanagan, D.E.; Moore, V.M.; Cockington, R.A.; Robinson, J.S.; Phillips, D.I. The Impact of Fetal Size and Length of Gestation on 6-Sulphatoxymelatonin Excretion in Adult Life. *J. Pineal Res.* **2001**, *30*, 188–192. [\[CrossRef\]](#)
473. Girotti, L.; Lago, M.; Ianovsky, O.; Carbajales, J.; Elizari, M.V.; Brusco, L.I.; Cardinali, D.P. Low Urinary 6-Sulphatoxymelatonin Levels in Patients with Coronary Artery Disease. *J. Pineal Res.* **2000**, *29*, 138–142. [\[CrossRef\]](#)
474. Yelawaram, K.; McLaughlin, L.G.; Knipe, J.O.; Schabdach, D. Pharmacokinetics and Oral Bioavailability of Exogenous Melatonin in Preclinical Animal Models and Clinical Implications. *J. Pineal Res.* **1997**, *22*, 45–51. [\[CrossRef\]](#) [\[PubMed\]](#)
475. Kopin, I.J.; Pare, C.M.; Axelrod, J.; Weissbach, H. The Fate of Melatonin in Animals. *J. Biol. Chem.* **1961**, *236*, 3072–3075. [\[CrossRef\]](#)
476. Raynaud, F.; Mauviard, F.; Geoffriau, M.; Claustrat, B.; Pevet, P. Plasma 6-Hydroxymelatonin, 6-Sulphatoxymelatonin and Melatonin Kinetics after Melatonin Administration to Rats. *Biol. Signals* **1993**, *2*, 358–366. [\[CrossRef\]](#) [\[PubMed\]](#)
477. Choudhary, S.; Kumar, A.; Saha, N.; Chaudhury, N.K. PK-PD Based Optimal Dose and Time for Orally Administered Supra-Pharmacological Dose of Melatonin to Prevent Radiation Induced Mortality in Mice. *Life Sci.* **2019**, *219*, 31–39. [\[CrossRef\]](#) [\[PubMed\]](#)
478. Galley, H.F.; Allen, L.; Colin, P.J.; Galt, S.P.; Webster, N.R. Dose Assessment of Melatonin in Sepsis (DAMSEL2) Study: Pharmacokinetics of Two Doses of Oral Melatonin in Patients with Sepsis. *J. Pineal Res.* **2022**, *73*, e12830. [\[CrossRef\]](#)
479. Mao, S.; Chen, J.; Wei, Z.; Liu, H.; Bi, D. Intranasal Administration of Melatonin Starch Microspheres. *Int. J. Pharm.* **2004**, *272*, 37–43. [\[CrossRef\]](#)
480. Pripem, A.; Johns, J.R.; Limsitthichaikoon, S.; Limphirath, W.; Mahakunakorn, P.; Johns, N.P. Intranasal Melatonin Nanoniosomes: Pharmacokinetic, Pharmacodynamics and Toxicity Studies. *Ther. Deliv.* **2017**, *8*, 373–390. [\[CrossRef\]](#)
481. Maeda, H.; Ogawa, Y.; Nakayama, H. Inclusion Complexes of Melatonin with Modified Cyclodextrins. *J. Incl. Phenom. Macrocycl. Chem.* **2014**, *78*, 217–224. [\[CrossRef\]](#)
482. Sakellaropoulou, A.; Siamidi, A.; Vlachou, M. Melatonin/Cyclodextrin Inclusion Complexes: A Review. *Molecules* **2022**, *27*, 445. [\[CrossRef\]](#)
483. Barchas, J.; DaCosta, F.; Spector, S. Acute Pharmacology of Melatonin. *Nature* **1967**, *214*, 919–920. [\[CrossRef\]](#)
484. Boutin, J.A.; Jockers, R. Melatonin Controversies, an Update. *J. Pineal Res.* **2021**, *70*, e12702. [\[CrossRef\]](#) [\[PubMed\]](#)

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485. Cai, R.; Yang, H.; He, J.; Zhu, W. The Effects of Magnetic Fields on Water Molecular Hydrogen Bonds. *J. Mol. Struct.* **2009**, *938*, 15–19. [[CrossRef](#)]
486. Menczel Schrire, Z.; Gordon, C.J.; Palmer, J.R.; Murray, J.; Hickie, I.; Rogers, N.L.; Lewis, S.J.; Terpening, Z.; Pye, J.E.; Naismith, S.L.; et al. Actigraphic and Melatonin Alignment in Older Adults with Varying Dementia Risk. *Chronobiol. Int.* **2022**, 1–12. [[CrossRef](#)]
487. Glacet, R.; Reynaud, E.; Robin-Choteau, L.; Reix, N.; Hugueny, L.; Ruppert, E.; Geoffroy, P.A.; Kilic-Huck, Ü.; Comtet, H.; Bourgin, P. A Comparison of Four Methods to Estimate Dim Light Melatonin Onset: A Repeatability and Agreement Study. *Chronobiol. Int.* **2022**, 1–9. [[CrossRef](#)] [[PubMed](#)]
488. Irwin, M.R.; Vitiello, M.V. Implications of Sleep Disturbance and Inflammation for Alzheimer’s Disease Dementia. *Lancet Neurol.* **2019**, *18*, 296–306. [[CrossRef](#)] [[PubMed](#)]

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